TM Forum Specification

Loyalty Management User Guide

TMF658

Maturity Level: General availability (GA)	**Team Approved Date: 18-Feb-2025**
Release Status: Preview	**Approval Status: Team Approved**
Version 5.0.0	**IPR Mode: RAND**

© TM Forum 2025. All Rights Reserved.

NOTICE

Copyright © TM Forum 2025. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

Table of Contents

NOTICE . . . i
Introduction . . . 3
API Standard coverage of Functional Framework . . . 3
Fundamentals . . . 3
Dependencies . . . 3
Loyalty Programme Management Support . . . 3
Program Specification . . . 3
Member onboarding . . . 4
Loyalty Account Management . . . 4
Program onboarding . . . 4
Benefit Partner Management . . . 4
Loyalty Earn operations . . . 4
Direct Loyalty Burn interfaces . . . 4
Loyalty Exection Points . . . 5
Use Cases . . . 6
Introduction . . . 6
Use Case 1 – Rewarding Citizens in a Smart City for use of clean inner-city transportation . . . 6
Use Case 2 - Prepaid subscriber behavioural changes to preserve network capacity . . . 7
Use Case 3 - Loyalty Rewards for optimal refuse removal . . . 7
Use Case 4 – E-commerce integration as part of purchase journeys . . . 8
Use Case 5 – Direct earning integration using the LoyaltyEarn API operation . . . 8
Use Case 6 – Managing a diversity of rewards benefit types . . . 8
Reliance on TMF723 Policy Management PolicyRule resources . . . 9
Example of PolicyRule resource for Use Case 1 . . . 9
Example of PolicyRule resource for Use Case 2 . . . 10
Support of polymorphism and extension patterns . . . 13
RESOURCE MODEL . . . 14
Managed Entity and Task Resource Models . . . 14
LoyaltyProgramProductSpecification resource . . . 14
LoyaltyProgramMember resource . . . 36
LoyaltyProgramProduct resource . . . 53
LoyaltyAccount resource . . . 104
LoyaltyProgramPartner resource . . . 116
LoyaltyEarn resource . . . 122
LoyaltyBurn resource . . . 125
LoyaltyExecutionPoint resource . . . 137
Notification Resource Models . . . 141
LoyaltyProgramProductSpecification . . . 143
LoyaltyProgramMember . . . 148

© TM Forum 2025. All Rights Reserved.

LoyaltyProgramProduct . . . 152
LoyaltyAccount . . . 156
LoyaltyProgramPartner . . . 160
LoyaltyEarn . . . 164
LoyaltyBurn . . . 166
LoyaltyExecutionPoint . . . 169
API OPERATIONS . . . 174
Operations on LoyaltyProgramProductSpecification . . . 174
List or find LoyaltyProgramProductSpecification objects . . . 174
Retrieves a LoyaltyProgramProductSpecification by ID . . . 178
Creates a LoyaltyProgramProductSpecification . . . 181
Updates partially a LoyaltyProgramProductSpecification . . . 187
Deletes a LoyaltyProgramProductSpecification . . . 197
Operations on LoyaltyProgramMember . . . 197
List or find LoyaltyProgramMember objects . . . 198
Retrieves a LoyaltyProgramMember by ID . . . 200
Creates a LoyaltyProgramMember . . . 202
Updates partially a LoyaltyProgramMember . . . 205
Deletes a LoyaltyProgramMember . . . 212
Operations on LoyaltyProgramProduct . . . 212
Retrieves a LoyaltyProgramProduct by ID . . . 212
List or find LoyaltyProgramProduct objects . . . 214
Creates a LoyaltyProgramProduct . . . 217
Updates partially a LoyaltyProgramProduct . . . 222
Deletes a LoyaltyProgramProduct . . . 231
Operations on LoyaltyAccount . . . 231
List or find LoyaltyAccount objects . . . 231
Retrieves a LoyaltyAccount by ID . . . 233
Creates a LoyaltyAccount . . . 234
Updates partially a LoyaltyAccount . . . 237
Deletes a LoyaltyAccount . . . 245
Operations on LoyaltyProgramPartner . . . 245
List or find LoyaltyProgramPartner objects . . . 245
Retrieves a LoyaltyProgramPartner by ID . . . 248
Creates a LoyaltyProgramPartner . . . 251
Updates partially a LoyaltyProgramPartner . . . 254
Deletes a LoyaltyProgramPartner . . . 263
Operations on LoyaltyEarn . . . 264
List or find LoyaltyEarn objects . . . 264
Retrieves a LoyaltyEarn by ID . . . 266
Creates a LoyaltyEarn . . . 267
Operations on LoyaltyBurn . . . 269

© TM Forum 2025. All Rights Reserved.

Retrieves a LoyaltyBurn by ID . . . 269
List or find LoyaltyBurn objects . . . 270
Creates a LoyaltyBurn . . . 274
Operations on LoyaltyExecutionPoint . . . 277
List or find LoyaltyExecutionPoint objects . . . 277
Retrieves a LoyaltyExecutionPoint by ID . . . 280
API NOTIFICATIONS . . . 282
Register listener . . . 282
Unregister listener . . . 283
Publish Event to listener . . . 283
Acknowledgements . . . 285
Release History . . . 285
Contributors to Document . . . 285

© TM Forum 2025. All Rights Reserved.

Introduction

The TMF658 Loyalty Management API delivers an interface for the management of Loyalty Programs

API Standard coverage of Functional Framework

This API provides an interface standard for the management of customer loyalty functions as defined in the Functional Framework (GB1033 Functional Framework Guidebook) with specific reference to 3.8.1 (Customer Loyalty and Retention) at sub-domain level 1.

The main use case is to achieve an API-first approach to end-to-end Loyalty Program Management by:

1. Defining a loyalty management library of reusable resources such as:
    1. Event types/topics recognised for rewards processing.
    2. Conditions in the form of reusable expressions applied to qualifying event types.
    3. Actions resulting from qualifying events in terms of program rule definitions.
2. Defining Loyalty Program Product Specifications as a collection of rules where a rule is comprised of qualifying events, conditions and actions.

Fundamentals

The TMF658 API relies on the asynchronous event pattern. This means that any loyalty program product specification must include one or more PolicyEvent resources that define subscription events/topics in an event-driven ecosystem that represent qualifying events for rewards processing. This implies that the underlying Loyalty Engine will implement event subscriptions in accordance with the LoyaltyProgramProductSpecification PolicyEvent specifications and that these events will be subjected to Loyalty Program Policy Rules.

Dependencies

In addition, the Loyalty API supports referencing of Policy Rules as defined by TMF723 (Policy Management). A Policy Rule binds events to conditions and, ultimately, resulting actions when events processed meet the policy conditions. In the case of Loyalty Program rules, the actions take the form of behavioural rewards such a points / virtual currency rewards, order creation, offer recommendations, voucher issuing or other forms of businessi interactions. It is important for adopters of TMF658 to understand the Policy Management API as the correct operation of the Loyalty Management functions underpinning it relies on the implementation of Loyalty Rules using the rule definitions managed using TMF723.

The TMF658 API supports the following Loyalty Management business functions:

Loyalty Programme Management Support

Program Specification

The API supports agility through continuous extensions to overall Loyalty Programs with the management of multiple LoyaltyProgramProductSpecification resources, each with its bound

© TM Forum 2025. All Rights Reserved.

events/topics and associated Loyalty Program policies defining rewarding business rules. It is envisaged that, with adoption of TMF658, Loyalty Programs can be continuously extended to ensure that the programs aids in objectives such as customer behavioural change programs, retention and customer value proposional improvements.

Member onboarding

Program members are onboarded using the LoyaltyProgramMember resource which is modelled as a specialisation of TMF669 PartyRole. Enrolment is therefore by definition a LoyaltyProgramMember subscribed to a LoyaltyProgramProductSpecification and with an associated LoyaltyAccount to which rewards are accrued to rewards earnings buckets and balances.

Loyalty Account Management

Loyalty Account is a specialization of the Account schema and is provided as a managed entity of TMF658. This enables specifications of accounts that may be dedicated for a Program Member or may be shared between members. Loyalty Rewards points / virtual currency accrual can therefore be shared by multiple members (e.g. Corporate Programs where employees may utilise from a rewards resource pool) or be a dedicated account per member.

Program onboarding

Customers may be enrolled to one or more LoyaltyProgramProductSpecification to entitle customers to the benefits associated with the Loyalty Program.

Benefit Partner Management

Program Benefit Partners may be formally managed as part of Loyalty Programs. Benefit partners can therefore be managed, inclusive of partner benefit catalogues that may be used as part of rewards actions as defined by the TMF723 Policy Management policy definitions.

Loyalty Earn operations

In certain cases, loyalty programs may allow the consumption of a direct earn operation as opposed to event-policy-action driven rewarding based on emitted operational events. Examples of the use of direct Loyalty Earn operations are:

- Partners with authorization to post directly Loyalty Earn events when customers interact with their brand
- Simple Loyalty Programs that do not require a policy engine to determine earnings
- Extarnal policy/pricing engines determining loyalty rewards actions and use the loyaltyEarn operation to record externally determined rewards in the form of virtual currency / points.

Direct Loyalty Burn interfaces

The TMF658 loyaltyBurn interface is used by authorized API consumers as part of e-commerce purchase journeys. Loyalty Rewards can therefore be used as payment

© TM Forum 2025. All Rights Reserved.

instrument for partial or full payment during online or other digitally integrated purchase journeys.

Loyalty Exection Points

The LoyaltyExectionPoint resource represents a rewardable or rewarded point of brand interaction by a Loyalty Program member with the organization running the Loyalty Program. It is an implicit requirement of an underpinning Loyalty Engine to generate accounting records of all Loyalty Points / Virtual Currency earnings and burn transactions. The LoyaltyExecutionPoint resources are retrievable with a GET loyaltyExecutionPoint operations to enable Loyalty Applications to retrieve earnings accrual and usage (burn) records as part of Loyalty Program Management user experience.

© TM Forum 2025. All Rights Reserved.

Use Cases

Introduction

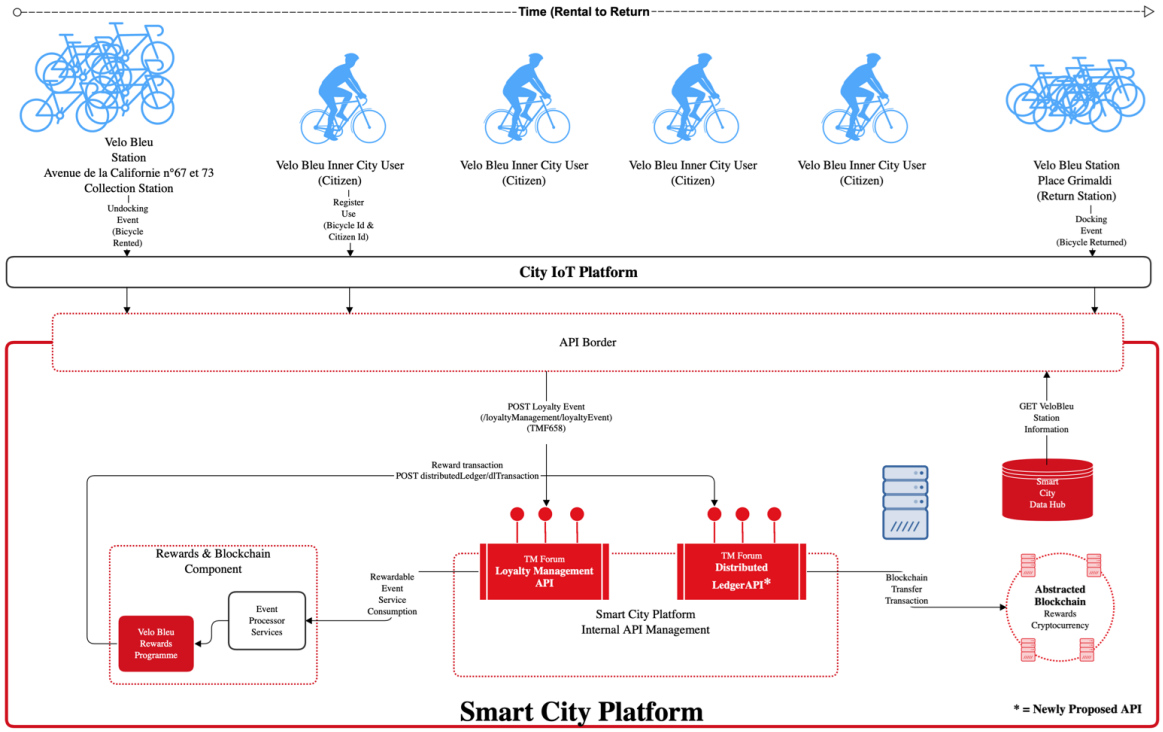
The use cases listed in this section illustrates a variety of uses of the TMF658 API to achieve different Loyalty Program objectives. The examples used throughout this specification are structured to illustrated the first two use cases described below.

Use Case 1 – Rewarding Citizens in a Smart City for use of clean inner-city transportation

The City of Nice identified a rewards use case as part of the Connected Citizen Catalyst project. In this use case, it identified rewarding citizens that park their cars in the various car parks at the edge of the inner city and use green inner-city transportation, e.g the Velo Bleu bicycle transportation system.

The city identified the establishment of a Smart City coalition blockchain and that citizens will be rewarded with a private-ledger cryptocurrency that could then be used to pay for city services and be exchanged for benefits at cities forming part of the Smart City coalition.

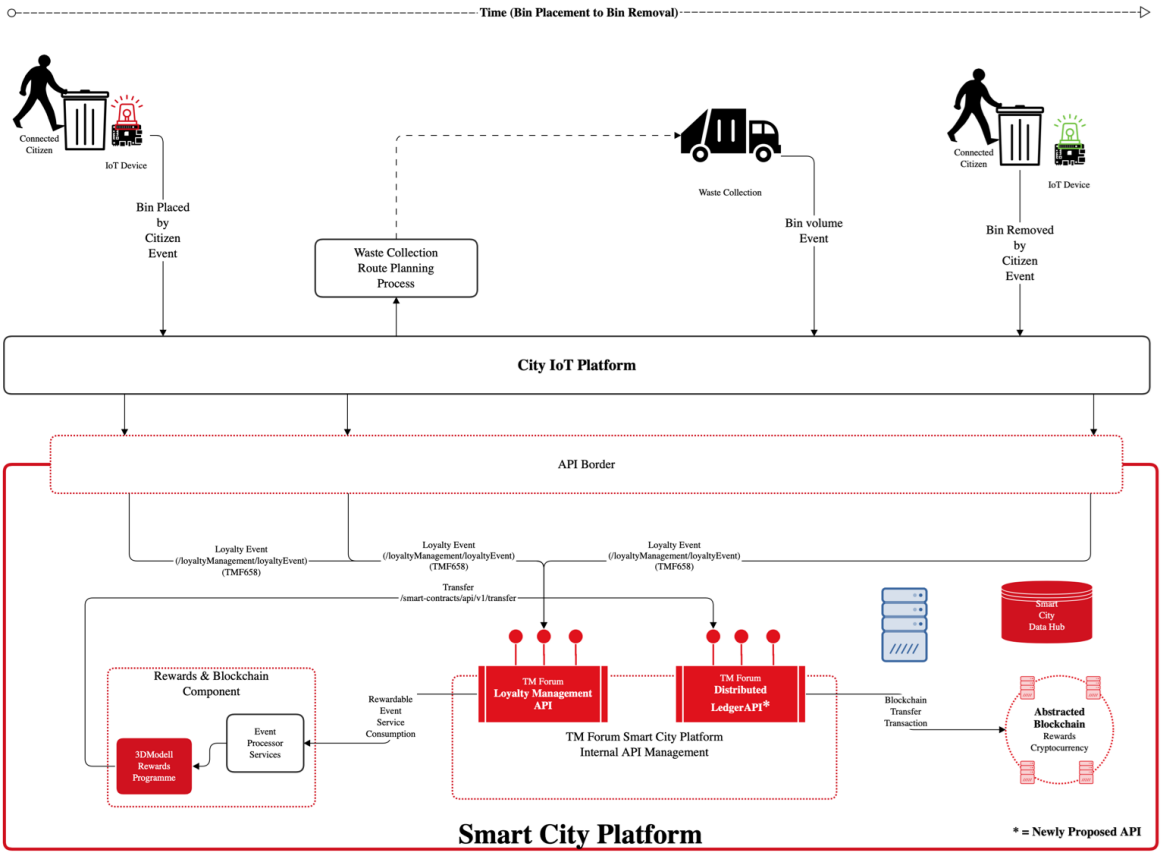
The Rewards Program goal is therefore to increase the use of green inner-city transportation with the benefits of reducing noise pollution, increase quality of inner-city life and to reduce pollution.



© TM Forum 2025. All Rights Reserved.

Use Case 2 - Prepaid subscriber behavioural changes to preserve network capacity

A predominantly prepaid market Mobile Network Operator has determined that their network is overloaded with peak voucher recharge and other top-up mechanisms placing strain on the IN SDP nodes where balance is retained. It was found that peak traffic places high demand on the Account Balance Management Function in the network between 18h00 and 20h00 each weekday.The planning teams were faced with either expanding the ABMF (IN SDP) infrastructure as a costly exercise but worked with Marketing and Customer Retention teams and decided to rather create an incentive program using the TMF658-based Loyalty Engine to move the peak balance top-up load through this program. The Loyalty Program was extended to reward the first 10,000 customers (configurable) recharging from 06h00 to 08h00 each weekday with surprise and delight free data and SMS offers.



Use Case 3 - Loyalty Rewards for optimal refuse removal

The City of Saint Quentin identified a loyalty program use case in which citizens are incentivised to place bins in the street when full instead of scheduled refuse collection days and where IoT events are triggered to report bin placement and bin removal.

This use case is supported with intelligent route planning for refuse removal and with the aim of minimising the periods that bins appear on pavements and therefore contributes to improving the aesthetics of the town.

Citizens are incentivised to remove the bins from the pavement as soon after refuse

© TM Forum 2025. All Rights Reserved.

collection was done. This program is therefore implemented with three rewardable events triggered from IoT devices as follows:

1. A bin placement event - an IoT event is emitted if the citizen places the bin at a measured location
2. A bin volume event - an IoT event is emitted with a refuse volume measurement
3. A bin removal event - an IoT event is emitted when a bin removal event takes place

Use Case 4 – E-commerce integration as part of purchase journeys

The LoyaltyBurn API resource is a form of payment that can be incorporated as part of customer online purchase journeys for partial or full payment for services and goods using loyalty currency (virtual or monetary). This capability will typically be used by the Loyalty Program owner organisation for purchases by its own customer base, but could also be extended to benefit and other partners to enable Loyalty Program Members to use loyalty earnings towards payment for loyalty program partner products and services.

Use Case 5 – Direct earning integration using the LoyaltyEarn API operation

While the API standard is primarily designed to support a rules and policy engine approach that processes event streams for rewarding, API consumers with access to the LoyaltyEarn API operation can use this interface to honour agreements for direct loyalty earnings.

The API consumers using this operation will typically be trusted internal systems or benefit Partners with which explicit earnings agreements exist.

Use Case 6 – Managing a diversity of rewards benefit types

For qualifying events received by an underlying Loyalty Engine based on the TMF658 resource model, a variety of Loyalty Action types can result from qualifying events. These are as follows:

- **LoyaltyEarn**: The action resulting from a qualifying event is that the customer earns loyalty points/ virtual currency.
- **LoyaltyBurn**: The action resulting from a qualifying event is that the customer loyalty points/virtual currency balance is reduced.
- **CustomerOrder**: An order is placed on behalf of the Loyalty Program Member
- **BusinessInteraction**: This action type represents a specific business interaction that could be a fulfilment operation with a benefit partner. These interactions could be simple, atomic transactions or could be orchestrated, multi-step events involved to fulfil the intended benefit.
- **VoucherIssue**: A voucher is issued to the Loyalty Program Member for use.
- **Offer**: An offer is created and communicated with the Loyalty
- **TierChange**: The qualifying event results in a Loyalty Program tier update within a Tiered loyalty program context.

© TM Forum 2025. All Rights Reserved.

Reliance on TMF723 Policy Management PolicyRule resources

This User Guide focuses on example payloads to support Use Cases 1 and 2. These LoyaltyProgramProductSpecification resources to represent the two programs refer to PolicyRule resources that **must** be defined in TMF723 as resources.

Example of PolicyRule resource for Use Case 1

This example provides an illustration of a PolicyRule incorporated by reference in the LoyaltyProgramProductSpecification for Use Case 1.


```
{
    "@type": "PolicyRule",
    "id": "cityOfNiceRewardsRule1",
    "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/cityOfNiceRewardsRule1",
    "name": "LoyaltyProgramProductSpecification1Rule1",
    "version": "1.0.0",
    "description": "Verifies that it is a VeloBleu cycle docking or undocking event
and evaluates this against the conditions and rewards actions",
    "lifecycleState": "active",
    "isComposite": true,
    "policyDomain": [
        {
            "@type": "PolicyDomainRef",
            "@baseType": "EntityRef",
            "@referredType": "PolicyDomain",
            "id": "LoyaltyProgramRewards",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyDomain/LoyaltyProgramRewards",
        }
    ],
    "combiningAlgorithm": "PermitUnlessDeny",
    "policyRuleRelationship": [
        {
            "@type": "PolicyRuleRelationship",
            "policyRule": {
                "@type": "PolicyRuleRef",
                "@baseType": "EntityRef",
                "id": "policyRuleQualifyingUndockEvent",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/policyRuleQualifyingUndockEvent",
                "@referredType": "PolicyRule"
            }
        },
        {
            "policyRule": {
                "@type": "PolicyRuleRef",
                "@baseType": "EntityRef",
                "id": "policyRuleQualifyingUndockEvent",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/policyRuleQualifyingDockEvent",
                "@referredType": "PolicyRule"
            }
        }
    ],
    "policyEventRelationship": [
        {
            "@type": "PolicyEventRelationship",
```


© TM Forum 2025. All Rights Reserved.

```
            "priority": 10,
            "policyEvent": {
                "@type": "PolicyEventRef",
                "@baseType": "EntityRef",
                "id": "veloBleuDockingEvent",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyEvent/veloBleuDockingEvent",
                "name": "Velo Bleu Docking Event",
                "@referredType": "VeloBleuDockingEvent"
            }
        },
        {
            "@type": "PolicyEventRelationship",
            "priority": 10,
            "policyEvent": {
                "@type": "PolicyEventRef",
                "@baseType": "EntityRef",
                "id": "veloBleuUndockingEvent",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyEvent/veloBleuUndockingEvent",
                "name": "Velo Bleu Undocking Event",
                "@referredType": "VeloBleuUndockingEvent"
            }
        }
    ],
    "policyActionRelationship": [
        {
            "@type": "PolicyActionRelationship",
            "priority": 10,
            "policyAction": {
                "@type": "PolicyActionRef",
                "@baseType": "EntityRef",
                "id": "policyActionRewardPoints",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyAction/policyActionRewardPoints",
                "name": "Reward Velo Bleu Usage",
                "@referredType": "PolicyAction"
            }
        }
    ],
    "priority": 0
}
```

Example of PolicyRule resource for Use Case 2

This example provides an illustration of a PolicyRule incorporated by reference in the LoyaltyProgramProductSpecification for Use Case 2.

```
{
    "@type": "PolicyRule",
    "id": "voucherRechargeBehaviourRule",
    "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/voucherRechargeBehaviourRule",
    "name": "VoucherRechargeBehaviourRule",
    "version": "1.0.0",
    "description": "Verifies that it is a voucher recharge operation between 06h00
and 09h00 and on a weekday and if met reward with the rewards action.",
    "lifecycleState": "active",
    "isComposite": true,
```

© TM Forum 2025. All Rights Reserved.

```
    "policyDomain": [
        {
            "@type": "PolicyDomainRef",
            "@baseType": "EntityRef",
            "@referredType": "PolicyDomain",
            "id": "LoyaltyProgramRewards",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyDomain/LoyaltyProgramRewards",
        }
    ],
    "combiningAlgorithm": "PermitUnlessDeny",
    "policyRuleRelationship": [
        {
            "@type": "PolicyRuleRelationship",
            "policyRule": {
                "@type": "PolicyRuleRef",
                "@baseType": "EntityRef",
                "id": "policyRuleTimeWindow",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/policyRuleTimeWindow",
                "@referredType": "PolicyRule"
            }
        },
        {
            "policyRule": {
                "@type": "PolicyRuleRef",
                "@baseType": "EntityRef",
                "id": "policyRuleIsWeekDay",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/policyRuleIsWeekDay",
                "@referredType": "PolicyRule"
            }
        }
    ],
    "policyEventRelationship": [
        {
            "@type": "PolicyEventRelationship",
            "priority": 10,
            "policyEvent": {
                "@type": "PolicyEventRef",
                "@baseType": "EntityRef",
                "id": "voucherRechargeEvent",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyEvent/voucherRechargeEvent",
                "name": "Voucher Recharge Event",
                "@referredType": "VoucherRechargeEvent"
            }
        }
    ],
    "actionExecutionStrategy": "DoAll",
    "sequencedActions": "Mandatory",
    "policyActionRelationship": [
        {
            "@type": "PolicyActionRelationship",
            "priority": 10,
            "policyAction": {
                "@type": "PolicyActionRef",
                "@baseType": "EntityRef",
                "id": "policyRewardSurpriseAndDelightOffer",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyAction/policyRewardSurpriseAndDelightOffer",
                "name": "Surprise and Delight Offer",
                "@referredType": "PolicyAction"
```


© TM Forum 2025. All Rights Reserved.

```
            }
        }
    ],
    "priority": 0
}
```

© TM Forum 2025. All Rights Reserved.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of LoyaltyAccount where other could be instances of FinancialAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as LoyaltyAccountRef or FinancialAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

© TM Forum 2025. All Rights Reserved.

RESOURCE MODEL

Managed Entity and Task Resource Models

LoyaltyProgramProductSpecification resource

The loyalty program product specification resource is the root entity for product specification management. A loyalty program product specification is a detailed description of a loyalty program made available externally in the form of a LoyaltyProduct to LoyaltyProgramMembers. LoyaltyProgramProductSpecification defines one or more Loyalty Rules (PolicyRule resource) that must be checked to identify the actions to apply.

Resource model

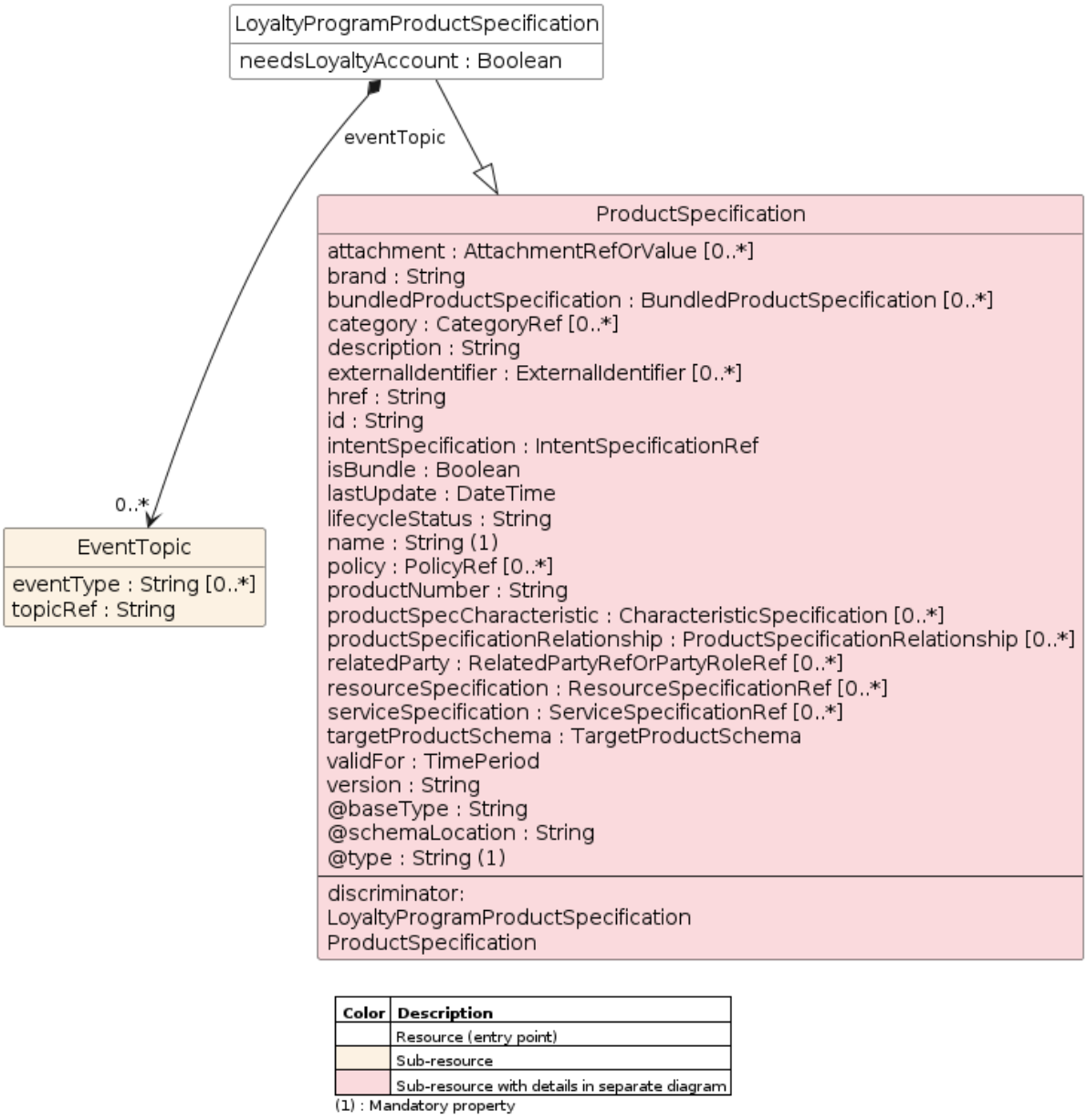


Figure 1 - LoyaltyProgramProductSpecification

© TM Forum 2025. All Rights Reserved.

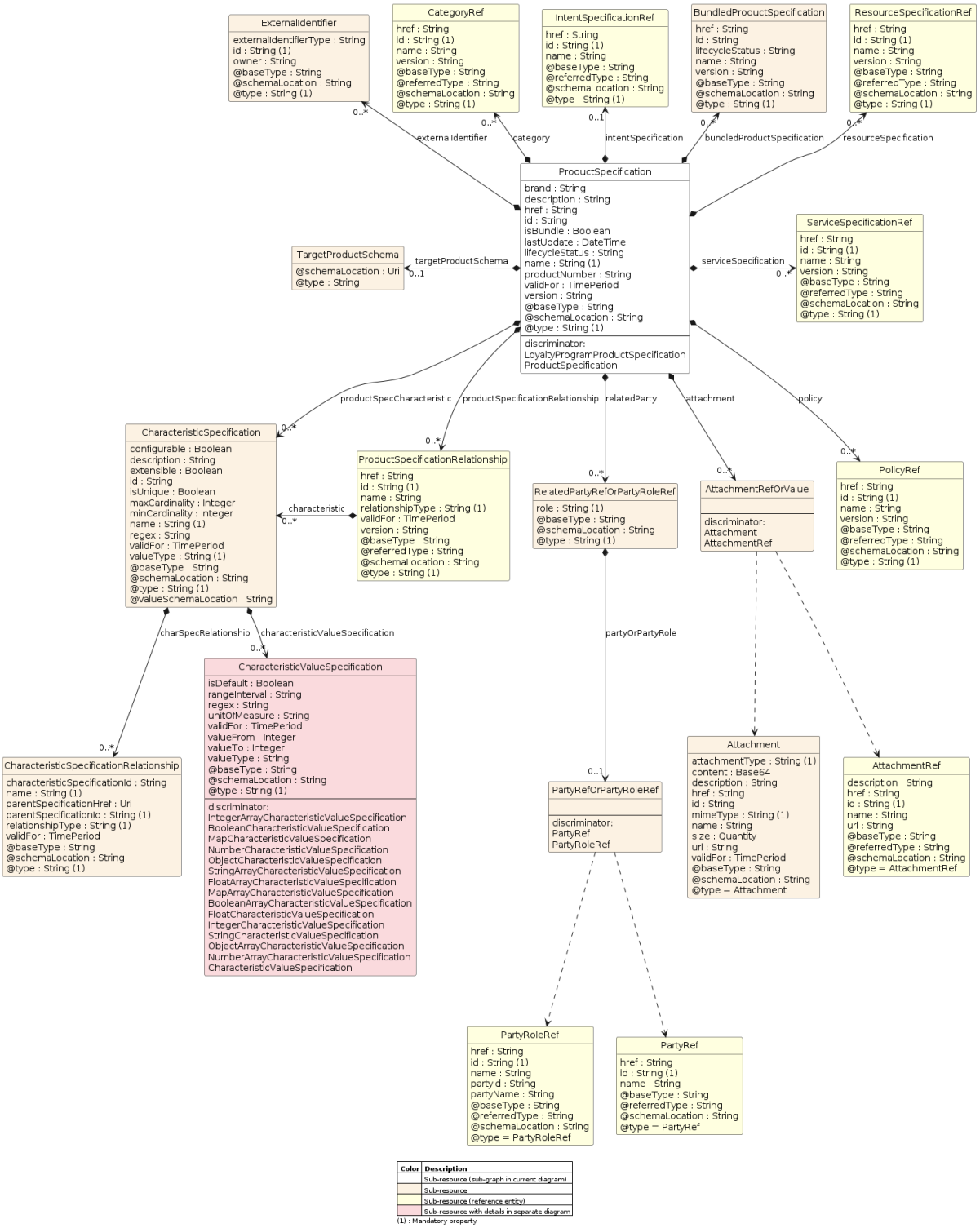


Figure 2 - ProductSpecification

© TM Forum 2025. All Rights Reserved.

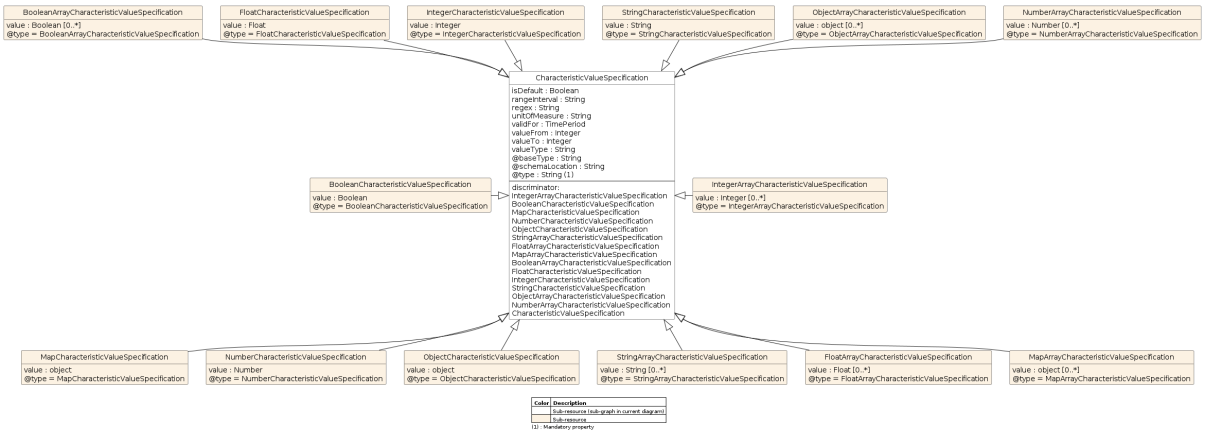


Figure 3 - CharacteristicValueSpecification

Field descriptions

LoyaltyProgramProductSpecification fields

attachment	An AttachmentRefOrValue. Complements the description of an element (for instance a product) through video, pictures…
brand	A String. The manufacturer or trademark of the specification.
bundledProductSpecification	A BundledProductSpecification. A type of ProductSpecification that belongs to a grouping of ProductSpecifications made available to the market. It inherits of all attributes of ProductSpecification.
category	A CategoryRef. The category resource is used to group product specifications in logical containers. Categories can contain other categories and/or other catalog entries.
description	A String. A narrative that explains in detail what the product specification is.
eventTopic	An EventTopic. An event topic subscribed to by the Loyalty Engine as part of the event-driven rewards processing architecture.
externalIdentifier	An ExternalIdentifier. List of external identifieers for the specification, e.g. identifier in source catalog.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. IsBundle determines whether a productSpecification represents a single productSpecification (false), or a bundle of productSpecification (true).
lastUpdate	A DateTime. Date and time of the last update.
lifecycleStatus	A String. Used to indicate the current lifecycle status.
name	A String. Name of the product specification.

© TM Forum 2025. All Rights Reserved.

needsLoyaltyAccount	A Boolean. Indicator if a loyaltyAccount is required for the program as in some instances programs may reward with business interactions, offers only and therefore would not need an account for points or virtual currency balances.
policy	A PolicyRef. The Policy resource represents a policy/rule applied to ProductSpecification.
productNumber	A String. An identification number assigned to uniquely identity the specification.
productSpecCharacteristic	A CharacteristicSpecification. A characteristic quality or distinctive feature of a ProductSpecification. The characteristic can be take on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
productSpecificationRelationship	A ProductSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among product specifications.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A related party defines party or party role linked to a specific entity.
resourceSpecification	A ResourceSpecificationRef. The ResourceSpecification is required to realize a ProductSpecification.
serviceSpecification	A ServiceSpecificationRef. ServiceSpecification(s) required to realize a ProductSpecification.
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Product specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.

© TM Forum 2025. All Rights Reserved.

id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BundledProductSpecification sub-resource fields

href	A String. Reference of the product specification.
id	A String. Unique identifier of the product specification.
lifecycleStatus	A String. Used to indicate the current lifecycle status.
name	A String. Name of the product specification.
version	A String. Version of the product specification.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

CategoryRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the category.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EventTopic sub-resource fields

eventType	A String.
topicRef	A String. The URI of the topic as defined in the Async API. This could be a topic from a Kafka Event Broker or similar.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

© TM Forum 2025. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
characteristicValueSpecification	A CharacteristicValueSpecification. A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on. CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification

© TM Forum 2025. All Rights Reserved.

configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.

© TM Forum 2025. All Rights Reserved.

parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2025. All Rights Reserved.

	CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
value	This property is present in subclasses

BooleanArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristicValueSpecification sub-resource fields

© TM Forum 2025. All Rights Reserved.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.

© TM Forum 2025. All Rights Reserved.

valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2025. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

MapArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MapCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

© TM Forum 2025. All Rights Reserved.

value	An object. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.

© TM Forum 2025. All Rights Reserved.

rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Collection of characteristic values.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

© TM Forum 2025. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2025. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductSpecificationRelationship sub-resource fields

characteristic	A CharacteristicSpecification. A characteristic that refines the relationship. For example, consider the relationship between broadband and TV. For a 4k TV it is important to know the minimum bandwidth to support 4k, so a characteristic Resolution might be defined on the relationship to allow capturing of this in the inventory.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of the relationship, for example override, discount, etc.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the referred product specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

© TM Forum 2025. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.

© TM Forum 2025. All Rights Reserved.

id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Service specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetProductSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target product.
@type	A String. Class type of the target product.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyProgramProductSpecification' resource object.


```
{
    "id": "100",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "My MVNO",
    "description": "Rewards Program to incentivise onboarding with benefits",
    "isBundle": false,
    "productNumber": "LOYALTY-01",
    "category": [
        {
            "id": "1",
            "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/category/1",
            "@type": "CategoryRef",
            "name": "Smart City Loyalty Incentive",
            "@referredType": "Category",
            "version": "1.0"
        }
    ],
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "version": "1.0",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Catalog Vendor",
            "partyOrPartyRole": {
```


© TM Forum 2025. All Rights Reserved.

```
                "id": "501",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/501",
                "@type": "PartyRoleRef",
                "name": "Loyalty Benefit Partner",
                "@referredType": "LoyaltyProgramPartner"
            }
        }
    ],
    "lastUpdate": "2024-05-01T18:15:13.414Z",
    "lifecycleStatus": "Active",
    "name": "Inner City Transportation Incentive Program",
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
```


© TM Forum 2025. All Rights Reserved.

```
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "88987",
            "href": "https://mycsp.com:8080/tmf-
api/policyManagement/v5/policyRule/88987",
            "@type": "PolicyRef",
            "name": "High LSM Rule",
            "@referredType": "PolicyRule"
        }
    ]
}
```


LoyaltyProgramMember resource

An Individual or Organization that opts in to a loyalty program.

Resource model

© TM Forum 2025. All Rights Reserved.

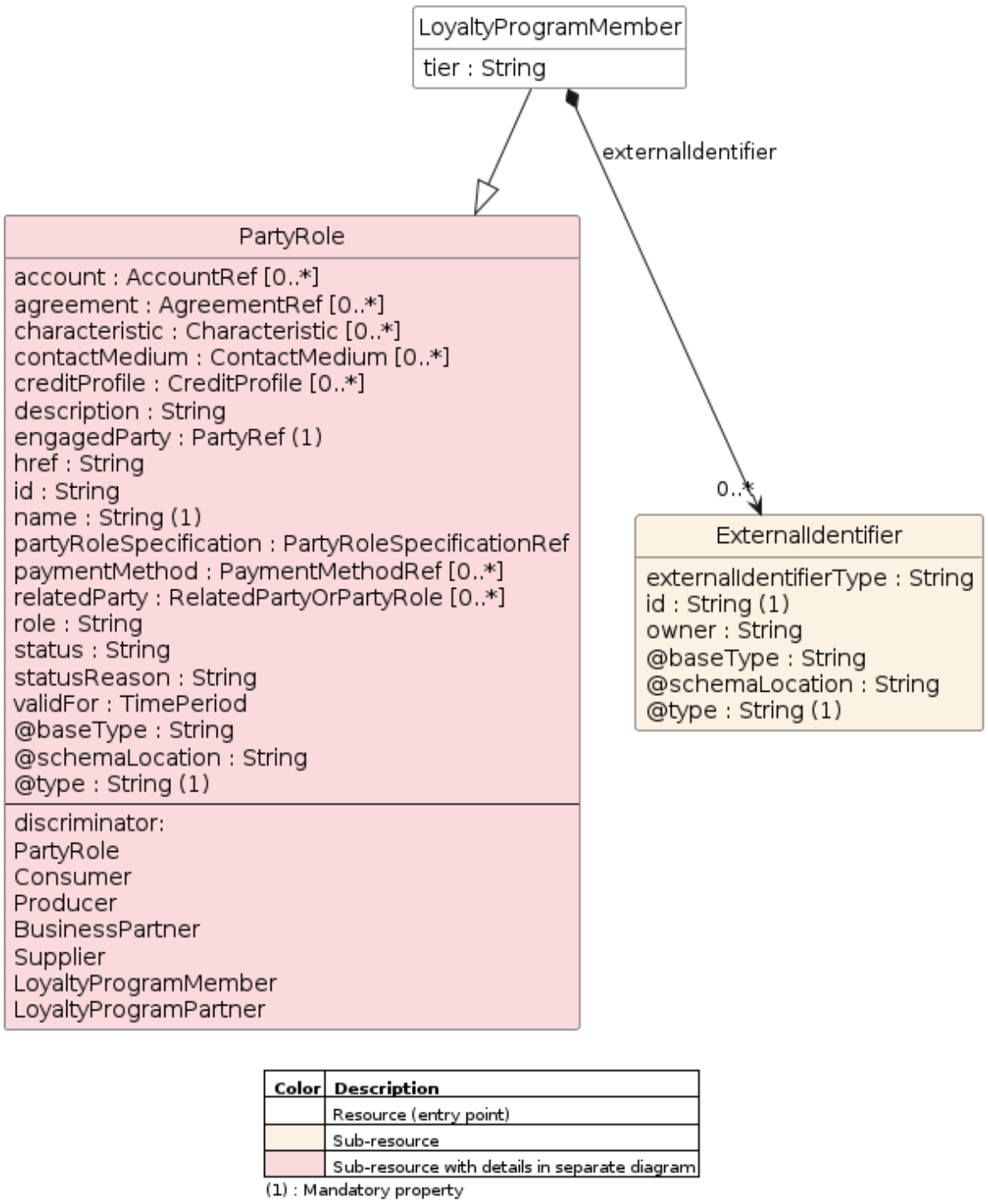


Figure 4 - LoyaltyProgramMember

© TM Forum 2025. All Rights Reserved.

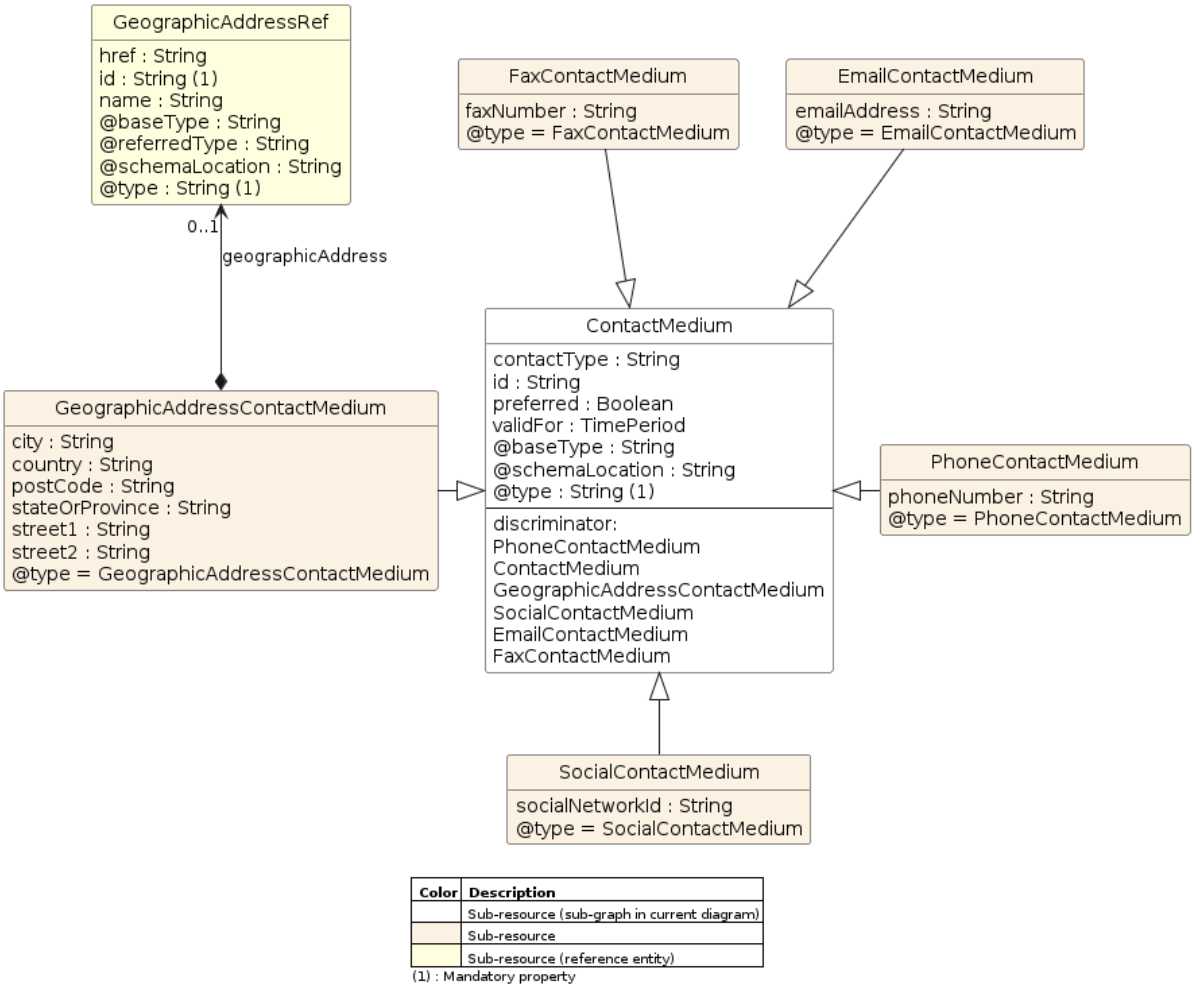


Figure 5 - ContactMedium

© TM Forum 2025. All Rights Reserved.

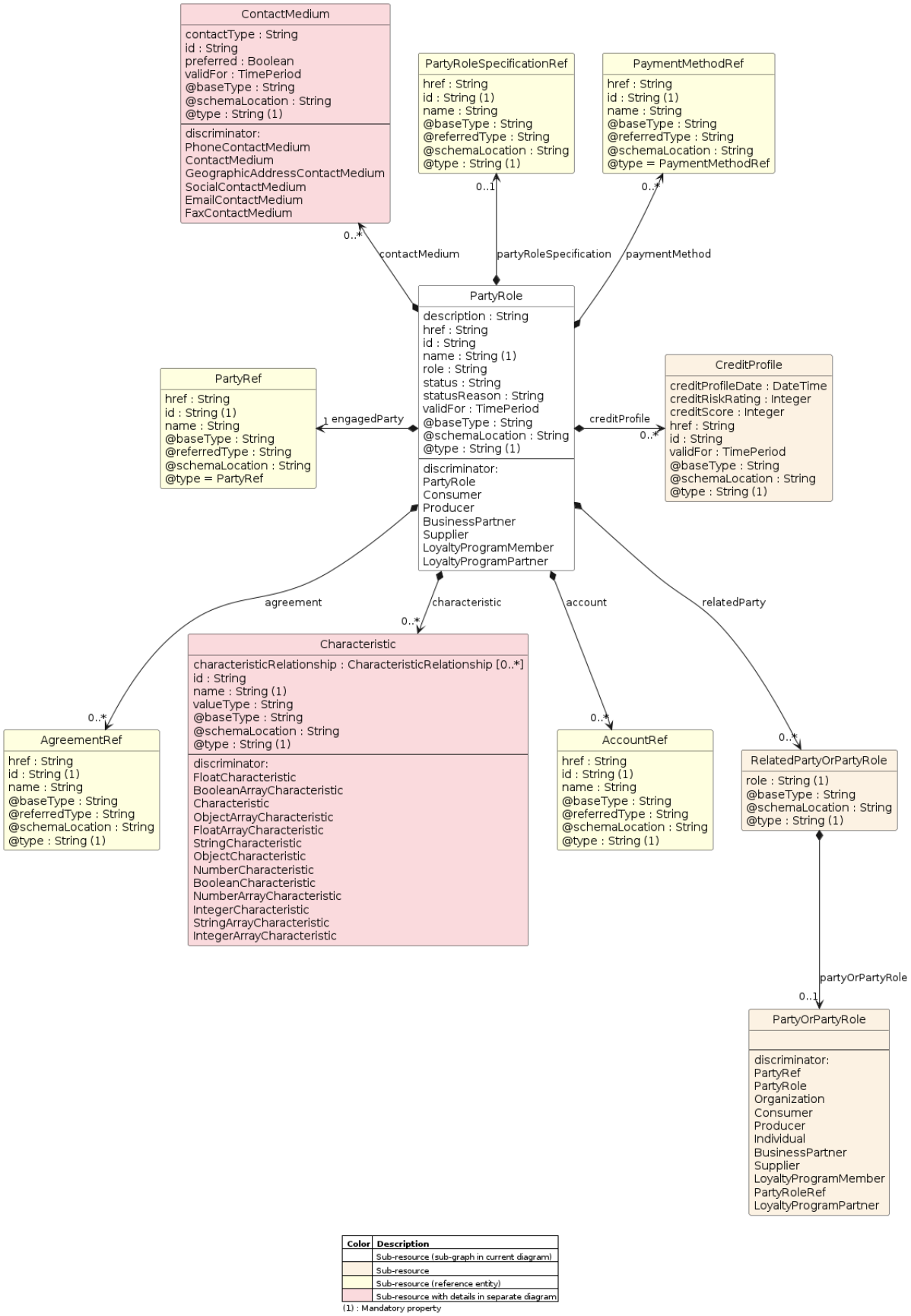


Figure 6 - PartyRole

© TM Forum 2025. All Rights Reserved.

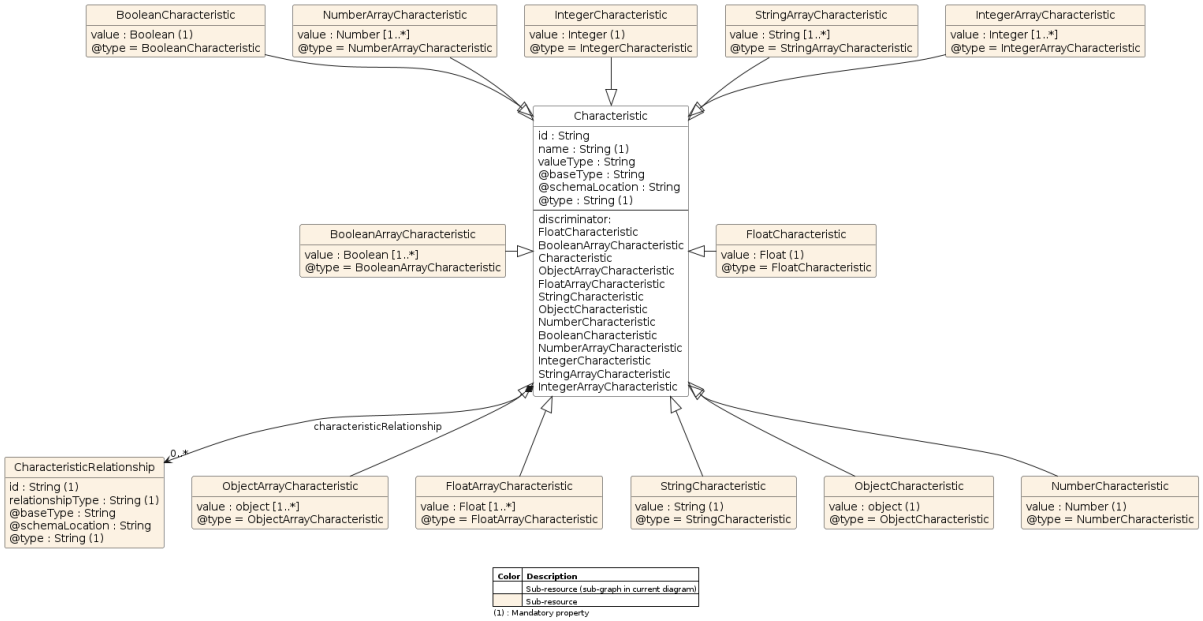


Figure 7 - Characteristic

Field descriptions

LoyaltyProgramMember fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.

© TM Forum 2025. All Rights Reserved.

engagedParty	A PartyRef. A Party reference.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
tier	A String. The member tier based on tiered loyalty program rules.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AccountRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2025. All Rights Reserved.

AgreementRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.

© TM Forum 2025. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. A characteristic which value is an array of Float(s).

© TM Forum 2025. All Rights Reserved.

valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

© TM Forum 2025. All Rights Reserved.

value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.

© TM Forum 2025. All Rights Reserved.

name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.

© TM Forum 2025. All Rights Reserved.

id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
city	This property is present in subclasses
country	This property is present in subclasses
emailAddress	This property is present in subclasses
faxNumber	This property is present in subclasses
geographicAddress	This property is present in subclasses
phoneNumber	This property is present in subclasses
postCode	This property is present in subclasses
socialNetworkId	This property is present in subclasses
stateOrProvince	This property is present in subclasses
street1	This property is present in subclasses

© TM Forum 2025. All Rights Reserved.

street2	This property is present in subclasses

GeographicAddressRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EmailContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
emailAddress	A String. Full email address in standard format.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FaxContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
faxNumber	A String. The fax number of the contact.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2025. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressContactMedium sub-resource fields

city	A String. The city.
contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
country	A String. The country.
geographicAddress	A GeographicAddressRef. Reference to a Geographic Address.
id	A String. Identifier for this contact medium.
postCode	A String. Postcode.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
stateOrProvince	A String. State or province.
street1	A String. Describes the street.
street2	A String. Complementary street description.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhoneContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
phoneNumber	A String. The phone number of the contact.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2025. All Rights Reserved.

SocialContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
socialNetworkId	A String. Identifier as a member of a social network.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CreditProfile sub-resource fields

creditProfileDate	A DateTime. The date the profile was established.
creditRiskRating	An Integer. This is an integer whose value is used to rate the risk.
creditScore	An Integer. A measure of a person or organizations creditworthiness calculated on the basis of a combination of factors such as their income and credit history.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

See [PartyRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

PartyRoleSpecificationRef sub-resource fields

href	A String. Hyperlink reference.

© TM Forum 2025. All Rights Reserved.

id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PaymentMethodRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyOrPartyRole sub-resource fields

partyOrPartyRole	A PartyOrPartyRole.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyProgramMember' resource object.

```
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
```

© TM Forum 2025. All Rights Reserved.

```
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "active",
    "statusReason": "Loyalty Member Enrolment",
```


© TM Forum 2025. All Rights Reserved.

```
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
}
```

LoyaltyProgramProduct resource

An instance of a loyalty program product, linked to a specified loyalty member.

Resource model

© TM Forum 2025. All Rights Reserved.

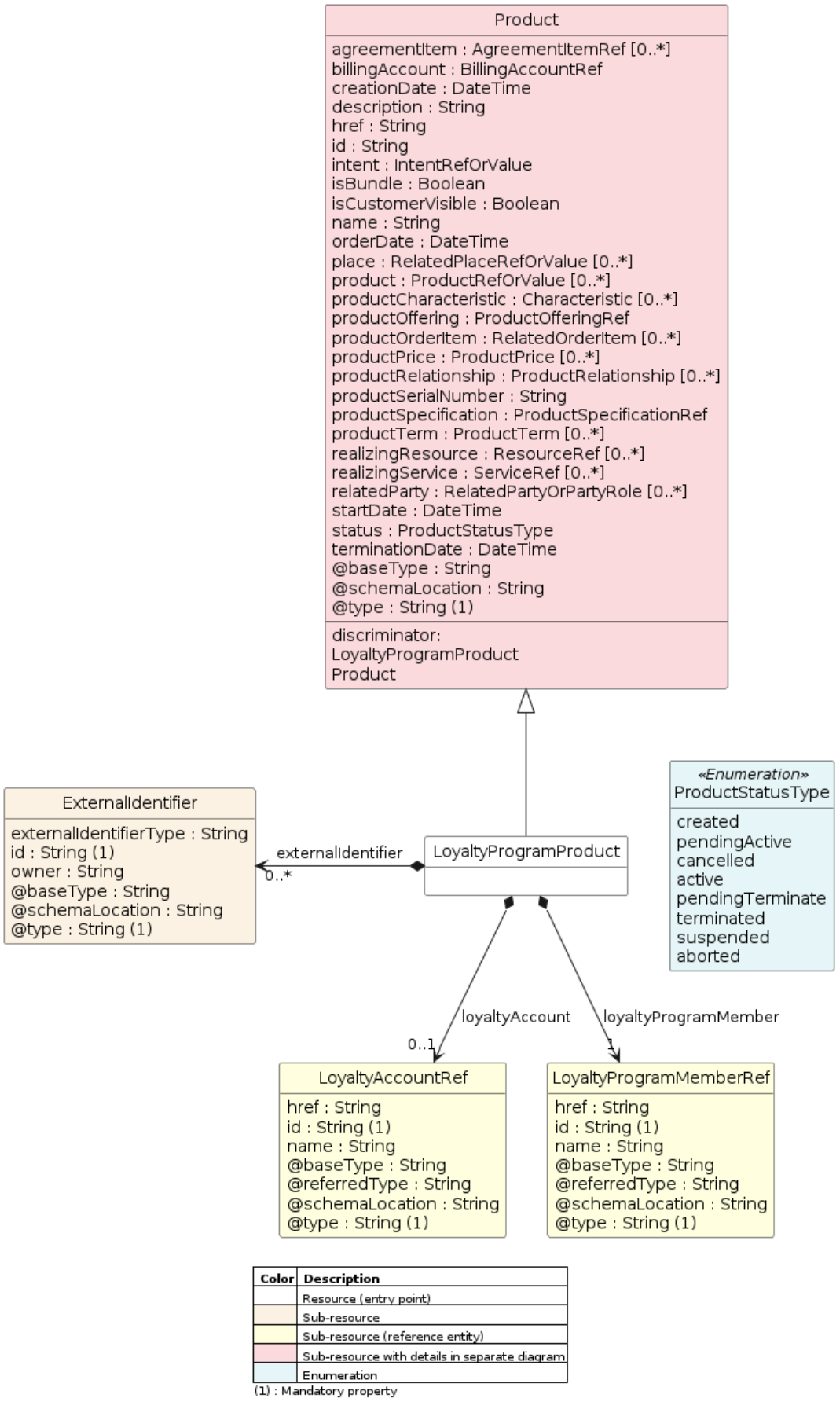


Figure 8 - LoyaltyProgramProduct

© TM Forum 2025. All Rights Reserved.

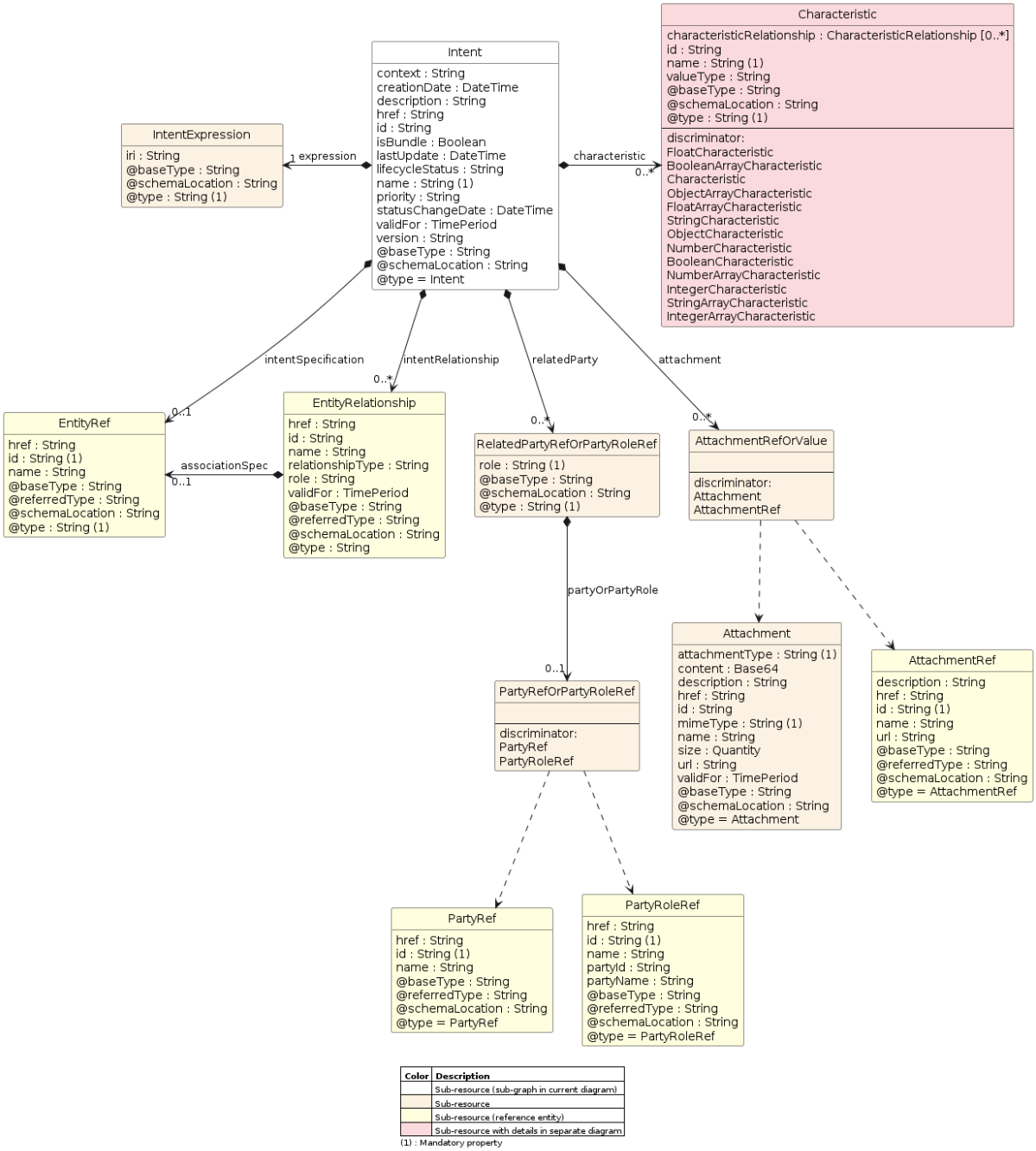


Figure 9 - Intent

See Figure 5 for ContactMedium diagrams.

See Figure 7 for Characteristic diagrams.

© TM Forum 2025. All Rights Reserved.

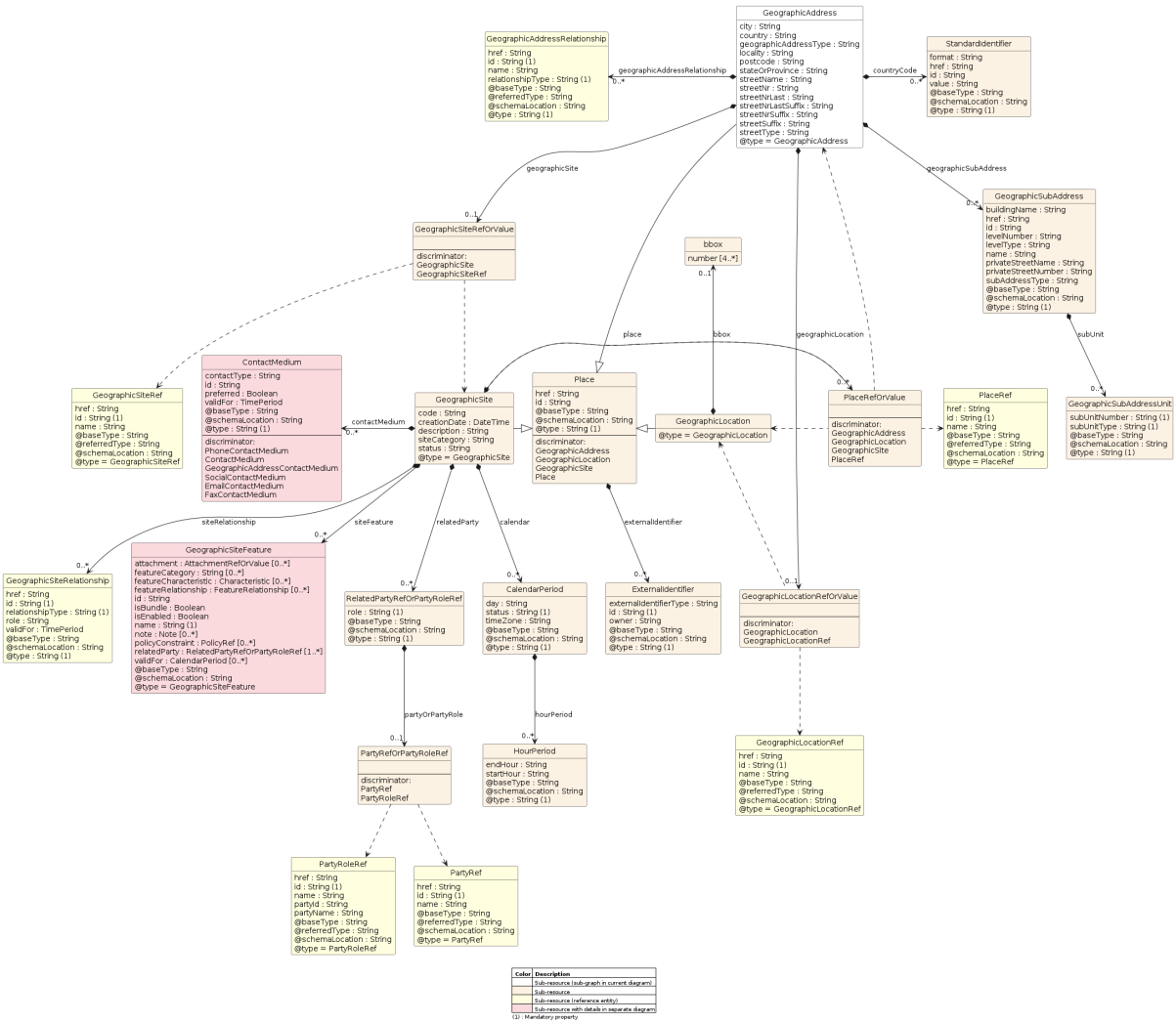


Figure 10 - GeographicAddress

© TM Forum 2025. All Rights Reserved.

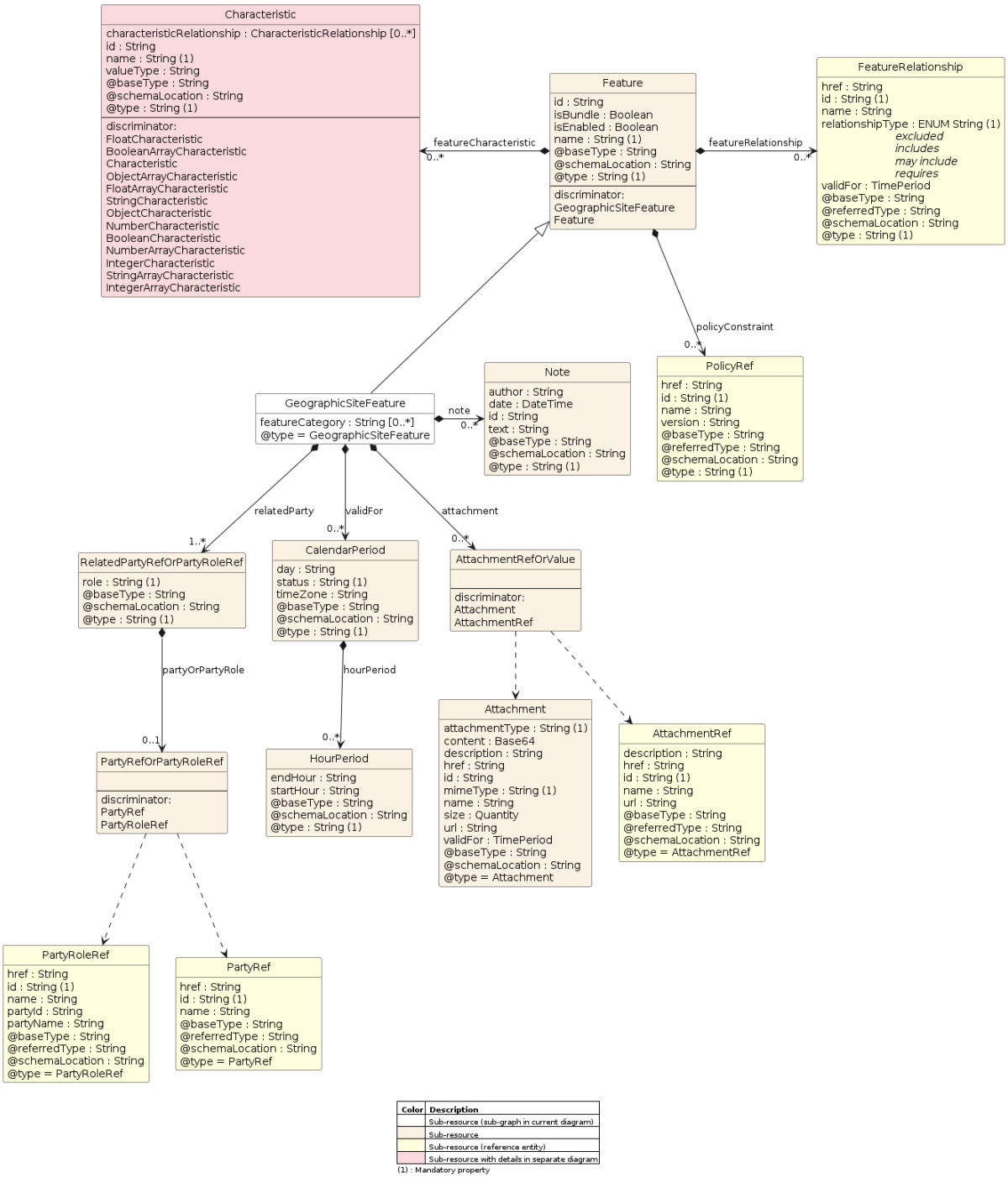


Figure 11 - GeographicSiteFeature

© TM Forum 2025. All Rights Reserved.

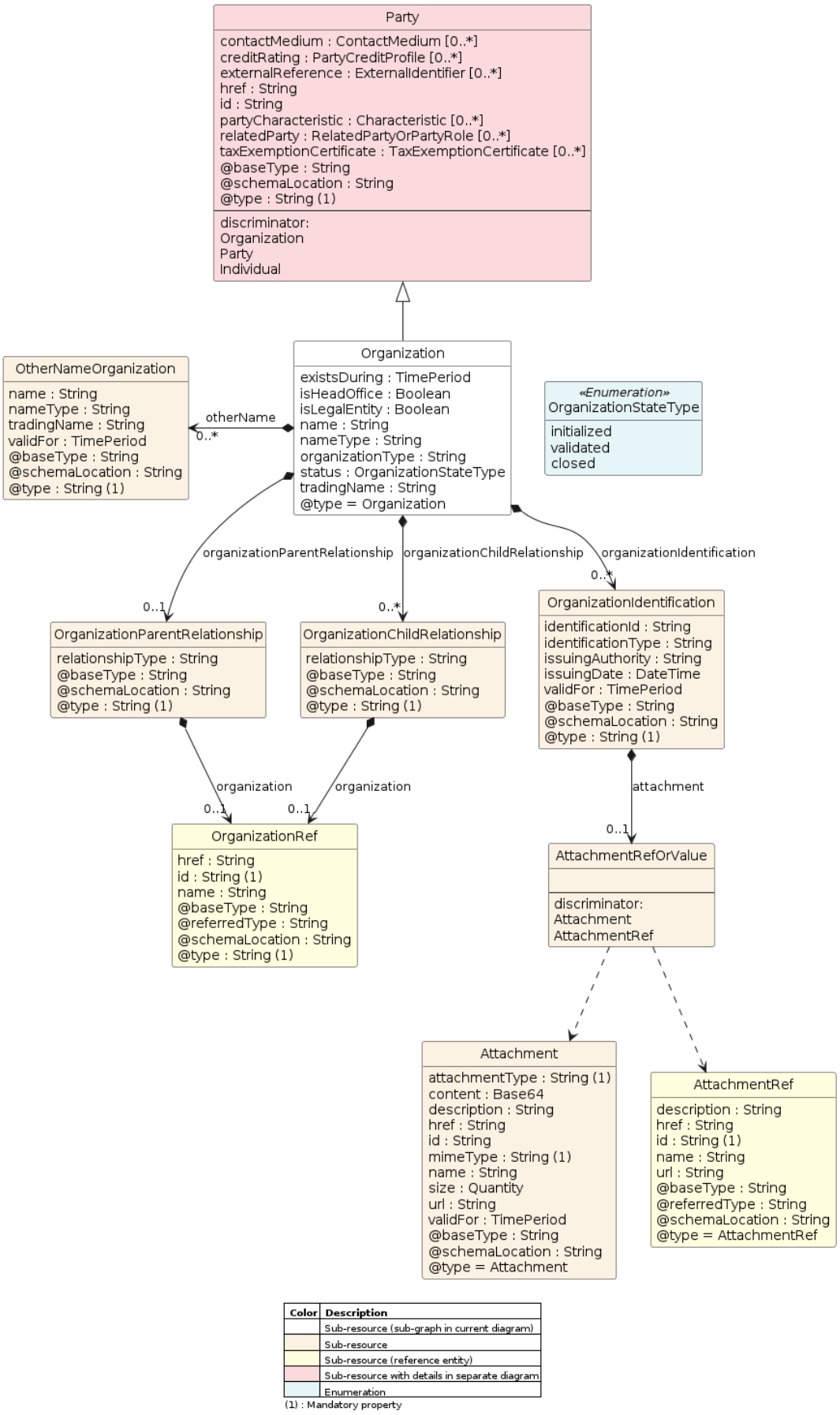


Figure 12 - Organization

© TM Forum 2025. All Rights Reserved.

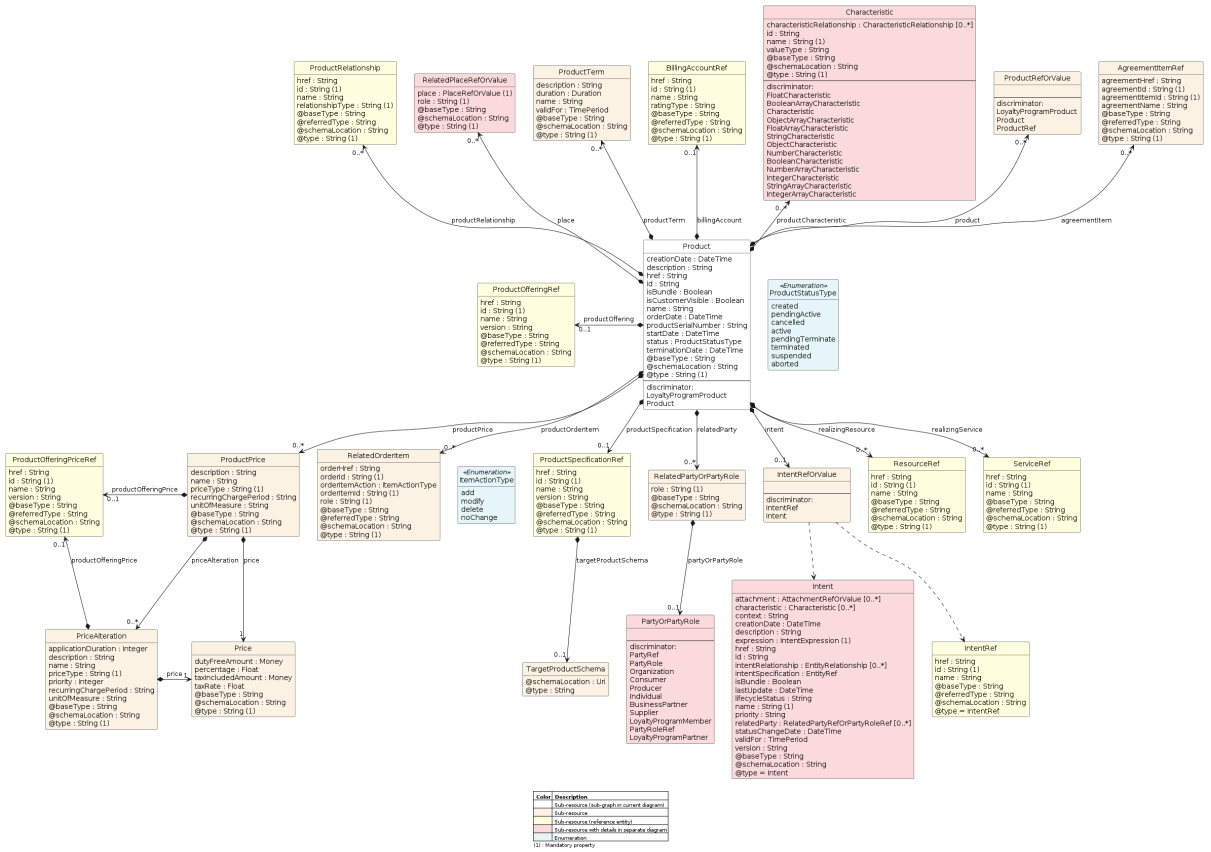


Figure 13 - Product

© TM Forum 2025. All Rights Reserved.

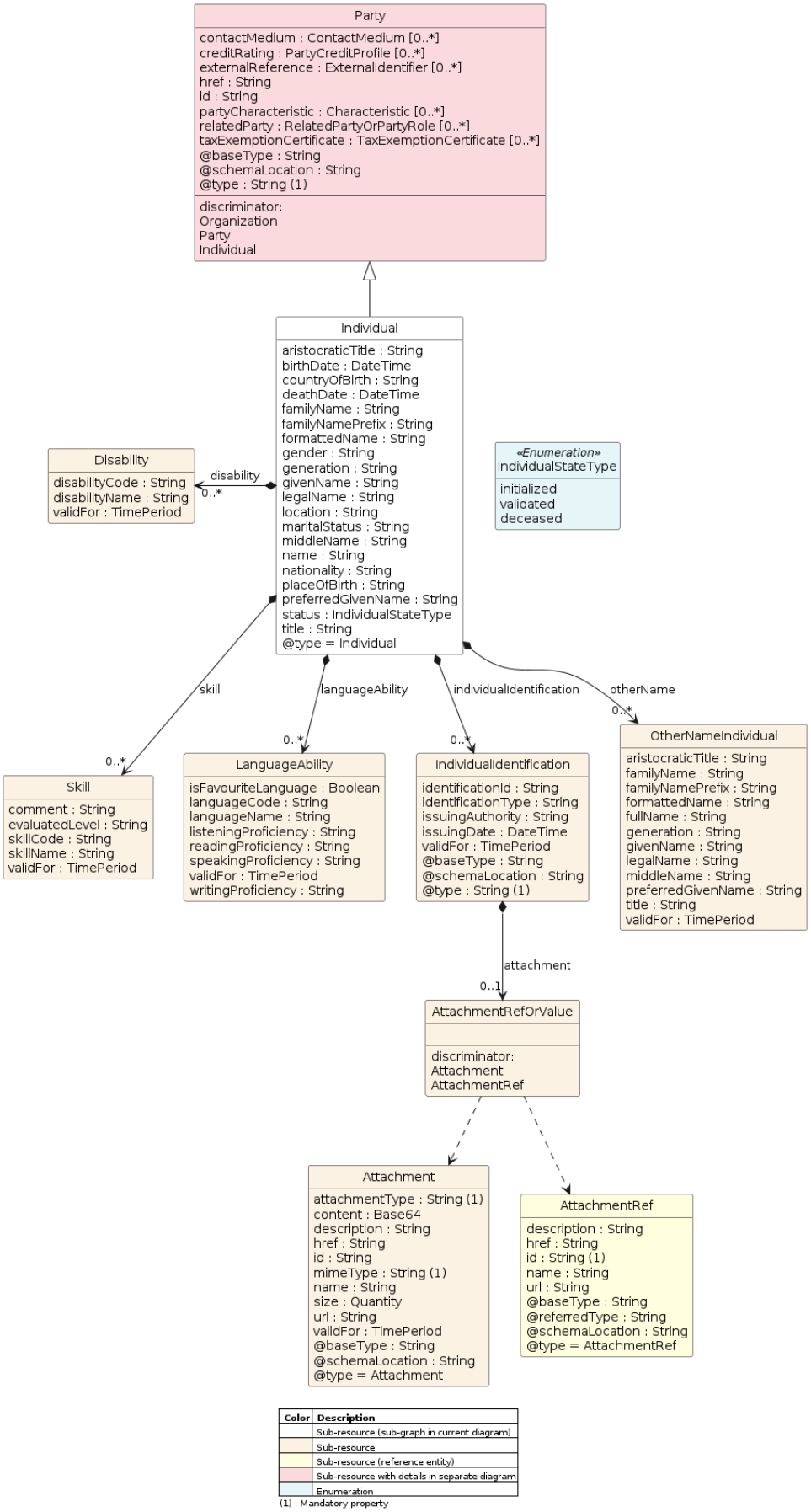


Figure 14 - Individual

© TM Forum 2025. All Rights Reserved.

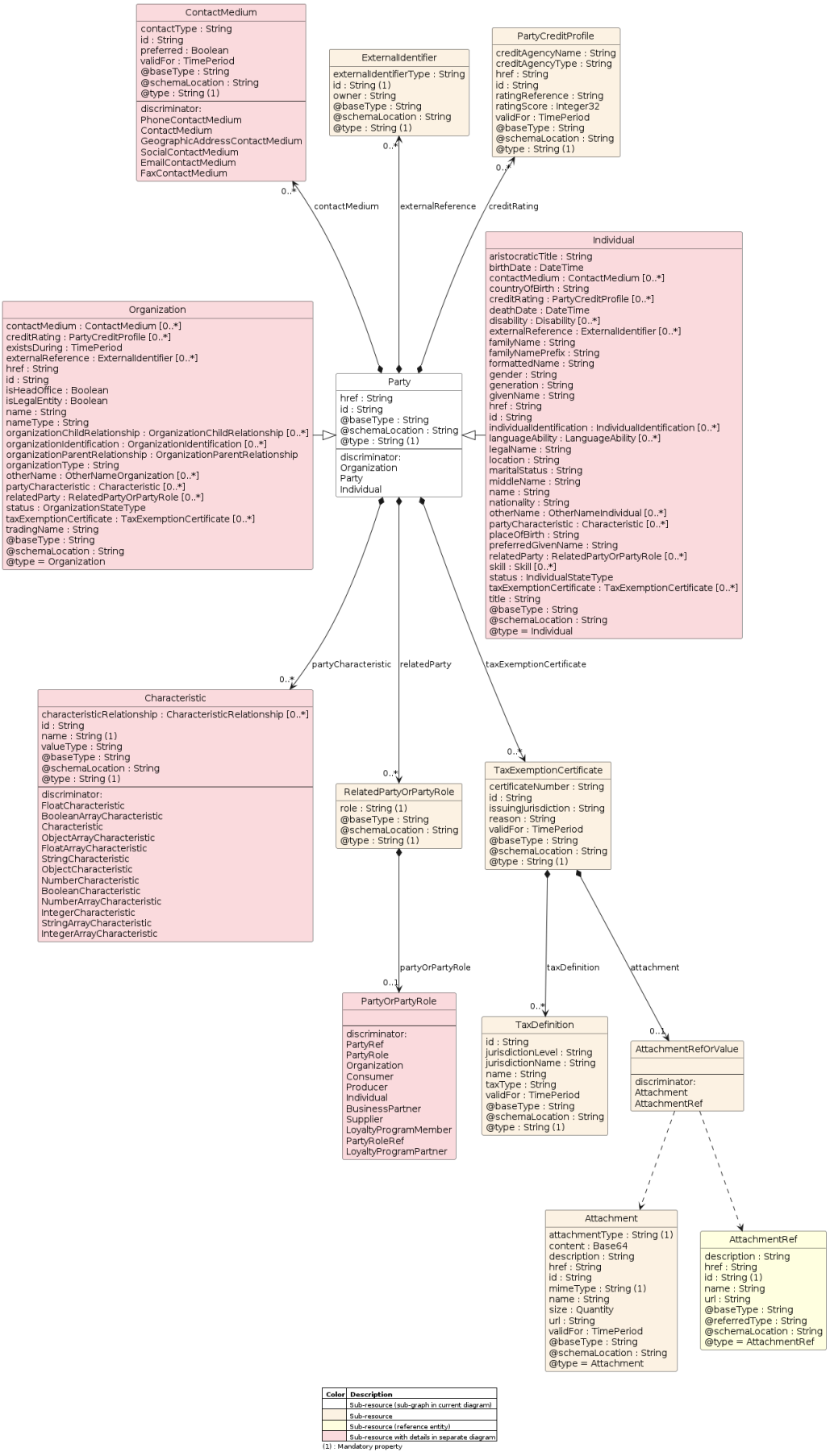


Figure 15 - Party

© TM Forum 2025. All Rights Reserved.

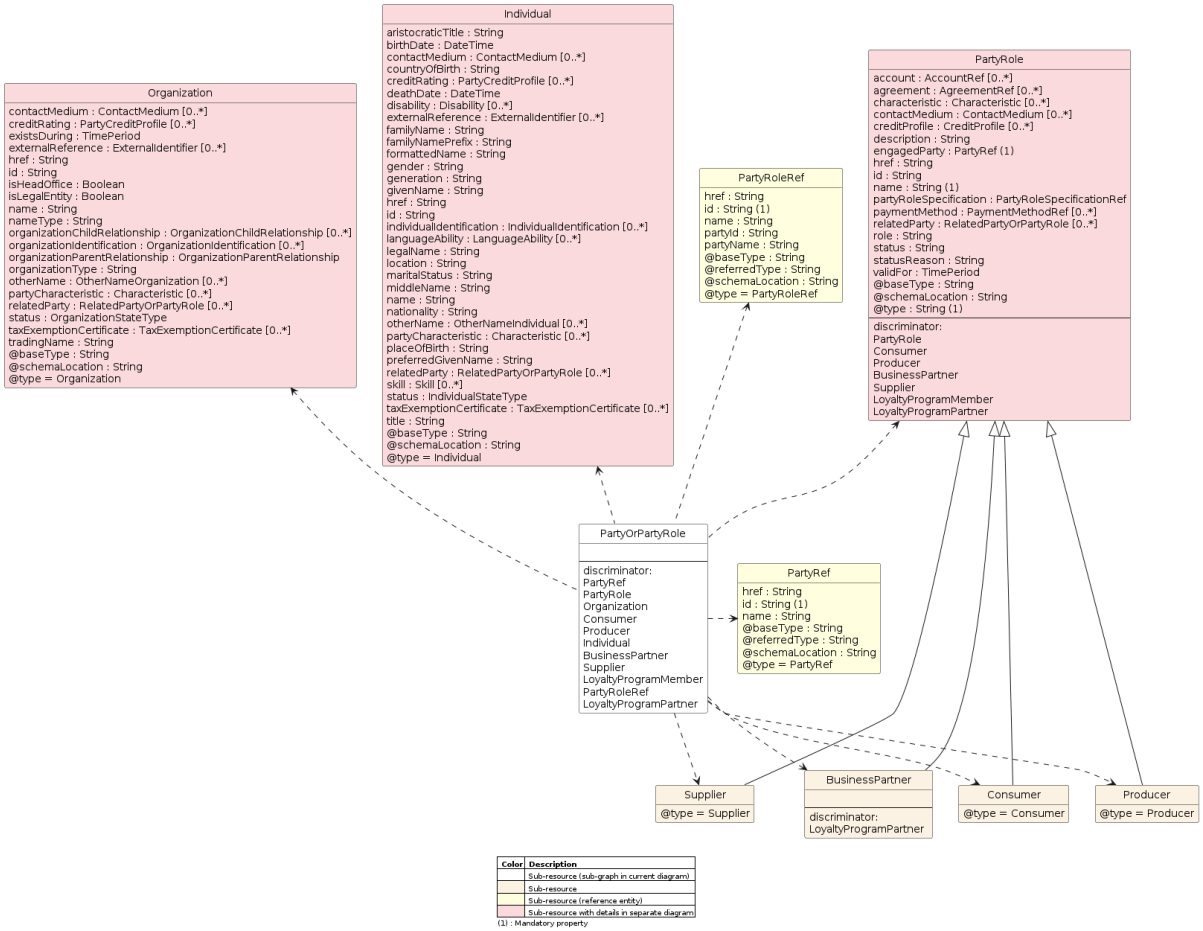


Figure 16 - PartyOrPartyRole

See Figure 6 for PartyRole diagrams.

Field descriptions

LoyaltyProgramProduct fields

agreementItem	An AgreementItemRef. Collection of agreements.
billingAccount	A BillingAccountRef. BillingAccount reference. A BillingAccount is a detailed description of a bill structure.
creationDate	A DateTime. Date and time when the product was created.
description	A String. Is the description of the product. It could be copied from the description of the Product Offering.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRefOrValue. Intent Ref (if Intent already exists) or Value (if Intent be created or its details be presented).

© TM Forum 2025. All Rights Reserved.

isBundle	A Boolean. If true, the product is a ProductBundle which is an instantiation of a BundledProductOffering. If false, the product is a ProductComponent which is an instantiation of a SimpleProductOffering.
isCustomerVisible	A Boolean. If true, the product is visible by the customer.
loyaltyAccount	A LoyaltyAccountRef. Reference to a loyalty account.
loyaltyProgramMember	A LoyaltyProgramMemberRef. Reference to a loyalty program member.
name	A String. Name of the product. It could be the same as the name of the product offering.
orderDate	A DateTime. Is the date when the product was ordered.
place	A RelatedPlaceRefOrValue. Collection of places.
product	A ProductRefOrValue. Collection of products.
productCharacteristic	A Characteristic. Collection of product characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
productOrderItem	A RelatedOrderItem. Collection of product order items.
productPrice	A ProductPrice. Collection of product prices.
productRelationship	A ProductRelationship. Collection of product relationships.
productSerialNumber	A String. Is the serial number for the product. This is typically applicable to tangible products e.g. Broadband Router.
productSpecification	A ProductSpecificationRef. ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog.
productTerm	A ProductTerm. Collection of product terms.
realizingResource	A ResourceRef. Collection of realizing resources.
realizingService	A ServiceRef. Collection of realizing services.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
startDate	A DateTime. Is the date from which the product starts.

© TM Forum 2025. All Rights Reserved.

status	A ProductStatusType. Possible values for the status of the product. ENUMERATED with values: * created * pendingActive * cancelled * active * pendingTerminate * terminated * suspended * aborted
terminationDate	A DateTime. Is the date when the product was terminated.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AgreementItemRef sub-resource fields

agreementHref	A String. Reference of the related entity.
agreementId	A String. Unique identifier of a related entity.
agreementItemId	A String. Id of an item of a Agreement.
agreementName	A String. Name of the related entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillingAccountRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
ratingType	A String. Indicates whether the account follows a specific payment option such as prepaid or postpaid.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

Intent sub-resource fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this intent, such as picture, document, media.
characteristic	A Characteristic. Collection of characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
context	A String. A string used to give a context to the intent.
creationDate	A DateTime. Date and time of the creation of this REST resource.
description	A String. The description of the intent.
expression	An IntentExpression. An Intent expression.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentRelationship	An EntityRelationship. A list of intents related to this intent.
intentSpecification	An EntityRef. Entity reference schema to be use for all entityRef class.
isBundle	A Boolean. IsBundle determines whether an intent represents a single intent (false), or a bundle of intents(true).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this intent.
name	A String. The name of the intent.
priority	A String. Can be used by intent owner to prioritize intents in an intent management system.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Collection of related parties.

© TM Forum 2025. All Rights Reserved.

statusChangeDate	A DateTime. A date time(DateTime). The date that the entity status changed to the current one.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. A field that identifies the specific version of an instance of an intent.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

See [Attachment] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

© TM Forum 2025. All Rights Reserved.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

IntentExpression sub-resource fields

iri	A String. Internationalized Resource Identifier of the intent Expression.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRelationship sub-resource fields

associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String.
id	A String.
name	A String.
relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2025. All Rights Reserved.

@referredType	A String.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String.

EntityRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

See [EntityRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

LoyaltyAccountRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LoyaltyProgramMemberRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRefOrValue sub-resource fields

place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddress sub-resource fields

© TM Forum 2025. All Rights Reserved.

city	A String. City that the address is in.
country	A String. Country that the address is in.
countryCode	A StandardIdentifier. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geographicAddressRelationship	A GeographicAddressRelationship. A list of geographic addresses relationships. Represents a relationship between geographic addresses.
geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSite	A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress. Collection of sub addresses.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.

© TM Forum 2025. All Rights Reserved.

streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.

© TM Forum 2025. All Rights Reserved.

externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

calendar	A CalendarPeriod. Collection of calendar items.
code	A String. A code that may be used for some addressing schemes eg: [ANSI T1.253-1999].
contactMedium	A ContactMedium. Collection of contact information. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Text describing additional information regarding the site.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
place	A PlaceRefOrValue. Collection of place objects.

© TM Forum 2025. All Rights Reserved.

relatedParty	A RelatedPartyRefOrPartyRoleRef. Related parties collection.
siteCategory	A String. Site classification/category.
siteFeature	A GeographicSiteFeature. Collection of site features.
siteRelationship	A GeographicSiteRelationship. Collection of site siteRelationships.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A String. Day where the calendar status applies (e.g.: monday, mon-to-fri, weekdays, weekend, all week, …).
hourPeriod	A HourPeriod. Collection of hour intervals.
status	A String. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information (e.g.: Paris, GMT+1).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

endHour	A String. The time when the status ends applying.
startHour	A String. The time when the status starts applying.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

See [ContactMedium] for details.

GeographicAddressRef sub-resource fields

© TM Forum 2025. All Rights Reserved.

See [GeographicAddressRef] for details.

EmailContactMedium sub-resource fields

See [EmailContactMedium] for details.

FaxContactMedium sub-resource fields

See [FaxContactMedium] for details.

GeographicAddressContactMedium sub-resource fields

See [GeographicAddressContactMedium] for details.

PhoneContactMedium sub-resource fields

See [PhoneContactMedium] for details.

SocialContactMedium sub-resource fields

See [SocialContactMedium] for details.

GeographicSiteFeature sub-resource fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this specification, such as picture, document, media.
featureCategory	A String. Collection of feature category.
featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
note	A Note. Collection of notes.

© TM Forum 2025. All Rights Reserved.

policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef. The associated party or partyRole.
validFor	A CalendarPeriod. Collection of validity information.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2025. All Rights Reserved.

PolicyRef sub-resource fields

See [PolicyRef] for details.

Feature sub-resource fields

featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Feature can be instantiated as * GeographicSiteFeature
attachment	This property is present in subclasses
featureCategory	This property is present in subclasses
note	This property is present in subclasses
relatedParty	This property is present in subclasses
validFor	This property is present in subclasses

GeographicSiteRelationship sub-resource fields

href	A String. Reference of the related geographic site.

© TM Forum 2025. All Rights Reserved.

id	A String. Unique identifier of the related site entity within the server.
relationshipType	A String. Type of relationship.
role	A String. Role of the related site in the relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.

© TM Forum 2025. All Rights Reserved.

subUnit	A GeographicSubAddressUnit. Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

subUnitNumber	A String. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

See [GeographicLocation] for details.

GeographicSite sub-resource fields

See [GeographicSite] for details.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [Characteristic] for details.

© TM Forum 2025. All Rights Reserved.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

ProductOfferingRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

© TM Forum 2025. All Rights Reserved.

version	A String. Version of the product offering.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedOrderItem sub-resource fields

orderHref	A String. Reference of the related entity.
orderId	A String. Unique identifier of a related Order.
orderItemAction	An ItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
orderItemId	A String. Id of an item of a product order.
role	A String. Role played by the Order.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductPrice sub-resource fields

description	A String. Description of the Product price.
name	A String. Name of the Product price.
price	A Price. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration.
priceAlteration	A PriceAlteration. Collection of price changes.
priceType	A String. Indicate if the price is for recurrent or no-recurrent charge.
productOfferingPrice	A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased.

© TM Forum 2025. All Rights Reserved.

recurringChargePeriod	A String. Used for recurring charge to indicate period (month, week, etc..).
unitOfMeasure	A String. Unit of Measure if price depending on it (Gb, SMS volume, etc..).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Price sub-resource fields

dutyFreeAmount	A Money. A base / value business entity used to represent money.
percentage	A Float. Percentage to apply for ProdOfferPriceAlteration.
taxIncludedAmount	A Money. A base / value business entity used to represent money.
taxRate	A Float. Tax rate.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PriceAlteration sub-resource fields

applicationDuration	An Integer. Duration during which the alteration applies on the order item price (for instance 2 months free of charge for the recurring charge).
description	A String. A narrative that explains in detail the semantics of this order item price alteration.
name	A String. Name of the order item price alteration.
price	A Price. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration.
priceType	A String. A category that describes the price such as recurring, one time and usage.
priority	An Integer. Priority level for applying this alteration among all the defined alterations on the order item price.
productOfferingPrice	A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased.
recurringChargePeriod	A String. Could be month, week…

© TM Forum 2025. All Rights Reserved.

unitOfMeasure	A String. Could be minutes, GB…
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingPriceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering price.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingPriceRef sub-resource fields

See [ProductOfferingPriceRef] for details.

ProductRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Relationship type as relies on, bundles, etc…
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.

© TM Forum 2025. All Rights Reserved.

name	A String. Name of the referred entity.
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
version	A String. Version of the product specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetProductSchema sub-resource fields

See [TargetProductSchema] for details.

ProductTerm sub-resource fields

description	A String. Description of the productTerm.
duration	A Duration. A time interval in a given unit of time.
name	A String. Name of the productTerm.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2025. All Rights Reserved.

ServiceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyOrPartyRole sub-resource fields

See [RelatedPartyOrPartyRole] for details.

BusinessPartner sub-resource fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.

© TM Forum 2025. All Rights Reserved.

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	BusinessPartner can be instantiated as * LoyaltyProgramPartner
externalIdentifier	This property is present in subclasses
partnerCatalogCategory	This property is present in subclasses
partnerCatalogOffer	This property is present in subclasses

AccountRef sub-resource fields

See [AccountRef] for details.

AgreementRef sub-resource fields

See [AgreementRef] for details.

CreditProfile sub-resource fields

See [CreditProfile] for details.

PartyRoleSpecificationRef sub-resource fields

© TM Forum 2025. All Rights Reserved.

See [PartyRoleSpecificationRef] for details.

PaymentMethodRef sub-resource fields

See [PaymentMethodRef] for details.

CategoryRef sub-resource fields

See [CategoryRef] for details.

Consumer sub-resource fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.

© TM Forum 2025. All Rights Reserved.

paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Individual sub-resource fields

aristocraticTitle	A String. E.g. Baron, Graf, Earl.
birthDate	A DateTime. Birth date.
contactMedium	A ContactMedium. List of means for contacting the party, e.g. mobile phone, email address. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
countryOfBirth	A String. Country where the individual was born.
creditRating	A PartyCreditProfile. List of credit profiles and scores for the party, typically received from an external credit broker.
deathDate	A DateTime. Date of death.
disability	A Disability. List of disabilities suffered by the individual.
externalReference	An ExternalIdentifier. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system.
familyName	A String. Contains the non-chosen or inherited name. Also known as last name in the Western context.
familyNamePrefix	A String. Family name prefix.

© TM Forum 2025. All Rights Reserved.

formattedName	A String. A fully formatted name in one string with all of its pieces in their proper place and all of the necessary punctuation. Useful for specific contexts (Chinese, Japanese, Korean).
gender	A String. Gender.
generation	A String. E.g.. Sr, Jr, III (the third).
givenName	A String. First name of the individual.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
individualIdentification	An IndividualIdentification. List of official identifications issued to the individual, such as passport, driving licence, social security number.
languageAbility	A LanguageAbility. List of national languages known by the individual.
legalName	A String. Legal name or birth name (name one has for official purposes).
location	A String. Temporary current location of the individual (may be used if the individual has approved its sharing).
maritalStatus	A String. Marital status (married, divorced, widow …).
middleName	A String. Middles name or initial.
name	A String. Full name flatten (first, middle, and last names) - this is the name that is expected to be presented in reference data types such as PartyRef, RelatedParty, etc. that refer to Individual.
nationality	A String. Nationality.
otherName	An OtherNameIndividual. List of other names by which this individual is known.
partyCharacteristic	A Characteristic. List of additional characteristics that a Party can take on. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
placeOfBirth	A String. Reference to the place where the individual was born.

© TM Forum 2025. All Rights Reserved.

preferredGivenName	A String. Contains the chosen name by which the individual prefers to be addressed. Note: This name may be a name other than a given name, such as a nickname.
relatedParty	A RelatedPartyOrPartyRole. List of parties and/or party roles related to this party.
skill	A Skill. List of skills exhibited by the individual.
status	An IndividualStateType. Valid values for the lifecycle state of the individual. ENUMERATED with values: * initialized * validated * deceased
taxExemptionCertificate	A TaxExemptionCertificate. List of tax exemptions granted to the party. For example, a war veteran might have partial exemption from state tax and a full exemption from federal tax.
title	A String. Useful for titles (aristocratic, social,…) Pr, Dr, Sir, …
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyCreditProfile sub-resource fields

creditAgencyName	A String. Name of the credit agency giving the score.
creditAgencyType	A String. Type of the credit agency giving the score.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
ratingReference	A String. Reference corresponding to the credit rating.
ratingScore	An Integer32. A measure of a party's creditworthiness calculated on the basis of a combination of factors such as their income and credit history.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Disability sub-resource fields

disabilityCode	A String. Code of the disability.
disabilityName	A String. Name of the disability.

© TM Forum 2025. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

IndividualIdentification sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
identificationId	A String. Identifier.
identificationType	A String. Identification type (passport, national identity card, drivers license, social security number, birth certificate).
issuingAuthority	A String. Authority which has issued the identifier, such as: social security, town hall.
issuingDate	A DateTime. Date at which the identifier was issued.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LanguageAbility sub-resource fields

isFavouriteLanguage	A Boolean. A ?true? value specifies whether the language is considered by the individual as his favourite one.
languageCode	A String. Language code (RFC 5646).
languageName	A String. Language name.
listeningProficiency	A String. Listening proficiency evaluated for this language.
readingProficiency	A String. Reading proficiency evaluated for this language.
speakingProficiency	A String. Speaking proficiency evaluated for this language.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
writingProficiency	A String. Writing proficiency evaluated for this language.

OtherNameIndividual sub-resource fields

aristocraticTitle	A String. E.g. Baron, Graf, Earl, etc.
familyName	A String. Contains the non-chosen or inherited name. Also known as last name in the Western context.
familyNamePrefix	A String. Family name prefix.

© TM Forum 2025. All Rights Reserved.

formattedName	A String. . A fully formatted name in one string with all of its pieces in their proper place and all of the necessary punctuation. Useful for specific contexts (Chinese, Japanese, Korean, etc.).
fullName	A String. Full name flatten (first, middle, and last names).
generation	A String. E.g. Sr, Jr, etc.
givenName	A String. First name.
legalName	A String. Legal name or birth name (name one has for official purposes).
middleName	A String. Middle name or initial.
preferredGivenName	A String. Contains the chosen name by which the person prefers to be addressed. Note: This name may be a name other than a given name, such as a nickname.
title	A String. Use for titles (aristocatic, social, …): Pr, Dr, Sir,….
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

Skill sub-resource fields

comment	A String. A free text comment linked to the evaluation done.
evaluatedLevel	A String. Level of expertise in a skill evaluated for an individual.
skillCode	A String. Code of the skill.
skillName	A String. Name of the skill, such as Java language.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

TaxExemptionCertificate sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
certificateNumber	A String. Identifier of a document that shows proof of exemption from taxes for the taxing jurisdiction.
id	A String. Identifier of the tax exemption within list of the exemptions.
issuingJurisdiction	A String. Name of the jurisdiction that issued the exemption.
reason	A String. Reason for the tax exemption.
taxDefinition	A TaxDefinition. A list of taxes that are covered by the exemption, e.g. City Tax, State Tax. The definition would include the exemption (e.g. for a rate exemption 0% would be a full exemption, 5% could be a partial exemption if the actual rate was 10%).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

© TM Forum 2025. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TaxDefinition sub-resource fields

id	A String. Unique identifier of the tax.
jurisdictionLevel	A String. Level of the jurisdiction that levies the tax.
jurisdictionName	A String. Name of the jurisdiction that levies the tax.
name	A String. Tax name.
taxType	A String. Type of the tax.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LoyaltyProgramMember sub-resource fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic

© TM Forum 2025. All Rights Reserved.

contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
tier	A String. The member tier based on tiered loyalty program rules.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LoyaltyProgramPartner sub-resource fields

account	An AccountRef. Collection of accounts.

© TM Forum 2025. All Rights Reserved.

agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partnerCatalogCategory	A CategoryRef. The partner catalog categories containing partner product offers made available for purchasing by Loyalty Program Members using loyalty rewards.
partnerCatalogOffer	A ProductOfferingRef. The partner catalog offers exposed to Loyalty Program Members for purchasing using Loyalty Rewards.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.

© TM Forum 2025. All Rights Reserved.

role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Organization sub-resource fields

contactMedium	A ContactMedium. List of means for contacting the party, e.g. mobile phone, email address. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditRating	A PartyCreditProfile. List of credit profiles and scores for the party, typically received from an external credit broker.
existsDuring	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
externalReference	An ExternalIdentifier. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isHeadOffice	A Boolean. If value is true, the organization is the head office.
isLegalEntity	A Boolean. If value is true, the organization is a legal entity known by a national referential.
name	A String. Organization name (department name for example).
nameType	A String. Type of the name : Co, Inc, Ltd, etc.

© TM Forum 2025. All Rights Reserved.

organizationChildRelationship	An OrganizationChildRelationship. List of organizations that are contained within this organization. For example if this organization is the Legal Department, the child organizations might include Claims, Courts, Contracts.
organizationIdentification	An OrganizationIdentification. List of official identifiers given to the organization, for example company number in the registry of companies.
organizationParentRelationship	An OrganizationParentRelationship. Parent references of an organization in a structure of organizations.
organizationType	A String. Type of Organization (company, department…).
otherName	An OtherNameOrganization. List of additional names by which the organization is known.
partyCharacteristic	A Characteristic. List of additional characteristics that a Party can take on. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
relatedParty	A RelatedPartyOrPartyRole. List of parties and/or party roles related to this party.
status	An OrganizationStateType. Valid values for the lifecycle state of the organization. ENUMERATED with values: * initialized * validated * closed
taxExemptionCertificate	A TaxExemptionCertificate. List of tax exemptions granted to the party. For example, a war veteran might have partial exemption from state tax and a full exemption from federal tax.
tradingName	A String. Name that the organization (unit) trades under.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OrganizationChildRelationship sub-resource fields

organization	An OrganizationRef.

© TM Forum 2025. All Rights Reserved.

relationshipType	A String. Type of the relationship. Could be juridical, hierarchical, geographical, functional for example.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OrganizationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OrganizationIdentification sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
identificationId	A String. Identifier.
identificationType	A String. Type of identification information used to identify the company in a country or internationally.
issuingAuthority	A String. Authority which has issued the identifier (chamber of commerce…).
issuingDate	A DateTime. Date at which the identifier was issued.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OrganizationParentRelationship sub-resource fields

organization	An OrganizationRef.

© TM Forum 2025. All Rights Reserved.

relationshipType	A String. Type of the relationship. Could be juridical, hierarchical, geographical, functional for example.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OtherNameOrganization sub-resource fields

name	A String. Organization name (department name for example).
nameType	A String. Co. , Inc. , Ltd. , Pty Ltd. , Plc; , Gmbh.
tradingName	A String. The name that the organization trades under.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRole sub-resource fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic

© TM Forum 2025. All Rights Reserved.

contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	PartyRole can be instantiated as * BusinessPartner * Consumer * LoyaltyProgramMember * Producer * Supplier
externalIdentifier	This property is present in subclasses

© TM Forum 2025. All Rights Reserved.

tier	This property is present in subclasses

Producer sub-resource fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.

© TM Forum 2025. All Rights Reserved.

status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Supplier sub-resource fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

© TM Forum 2025. All Rights Reserved.

name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyProgramProduct' resource object.


```
{
    "id": "1765",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/1765",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "creationDate": "2024-05-01T20:26:25.122Z",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": true,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
```


© TM Forum 2025. All Rights Reserved.

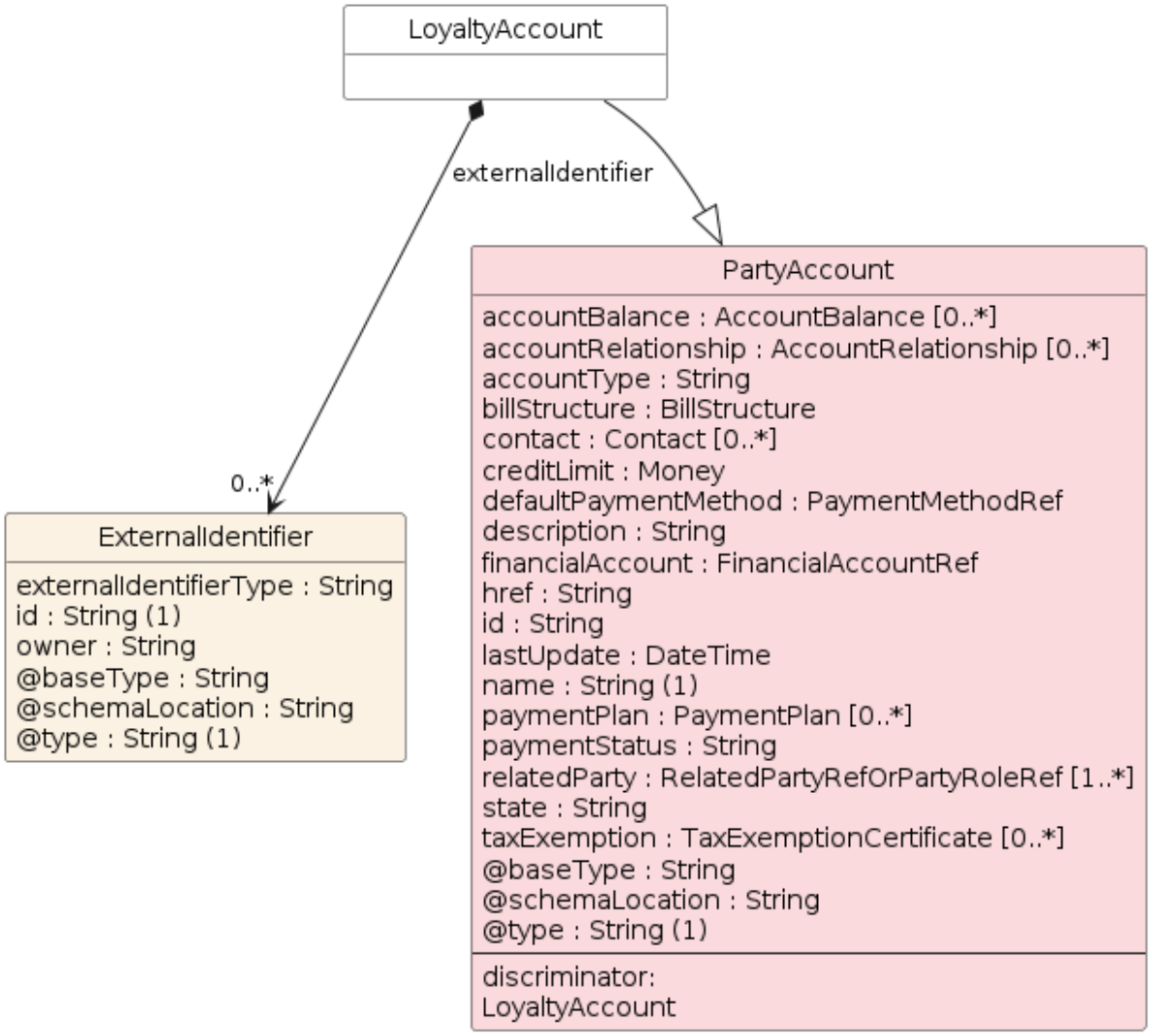
```
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "status": "created",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "9664",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
        "@type": "LoyaltyProgramMemberRef",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "1155",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
        "@type": "LoyaltyAccountRef",
        "name": "John Doe's loyalty account",
        "@referredType": "LoyaltyAccount"
    }
}
```


© TM Forum 2025. All Rights Reserved.

LoyaltyAccount resource

A LoyaltyAccount corresponds to a set of balances to carry loyalty movement according to different validity period and unit. The LoyaltyAccount is a container for one of more LoyaltyBalances. A LoyaltyAccount may be required to realize a LoyaltyProgramProduct according to the rules carried by the corresponding LoyaltyProgramProductSpecification. A LoyaltyAccount may collect results from one or more LoyaltyProgramProduct resources.

Resource model



Color	Description
	Resource (entry point)
	Sub-resource
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 17 - LoyaltyAccount

© TM Forum 2025. All Rights Reserved.

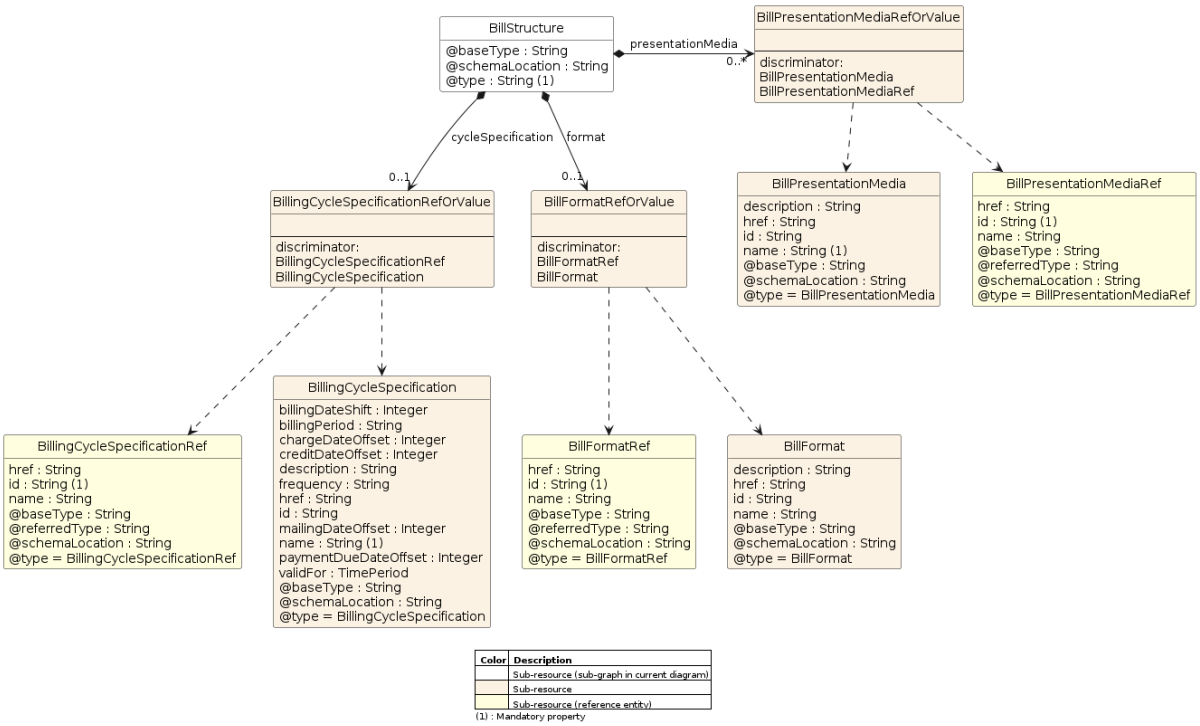


Figure 18 - BillStructure

© TM Forum 2025. All Rights Reserved.

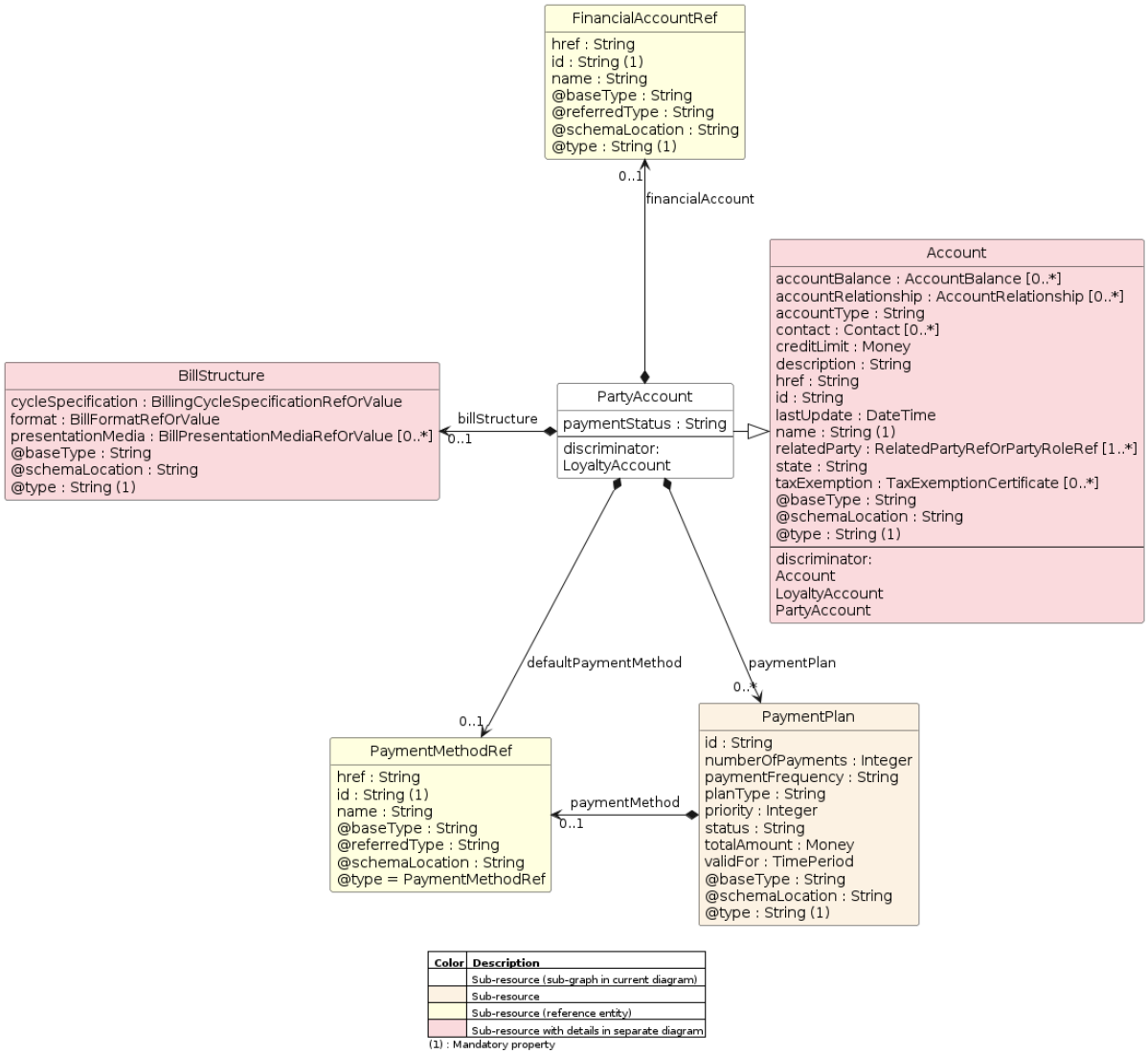


Figure 19 - PartyAccount

© TM Forum 2025. All Rights Reserved.

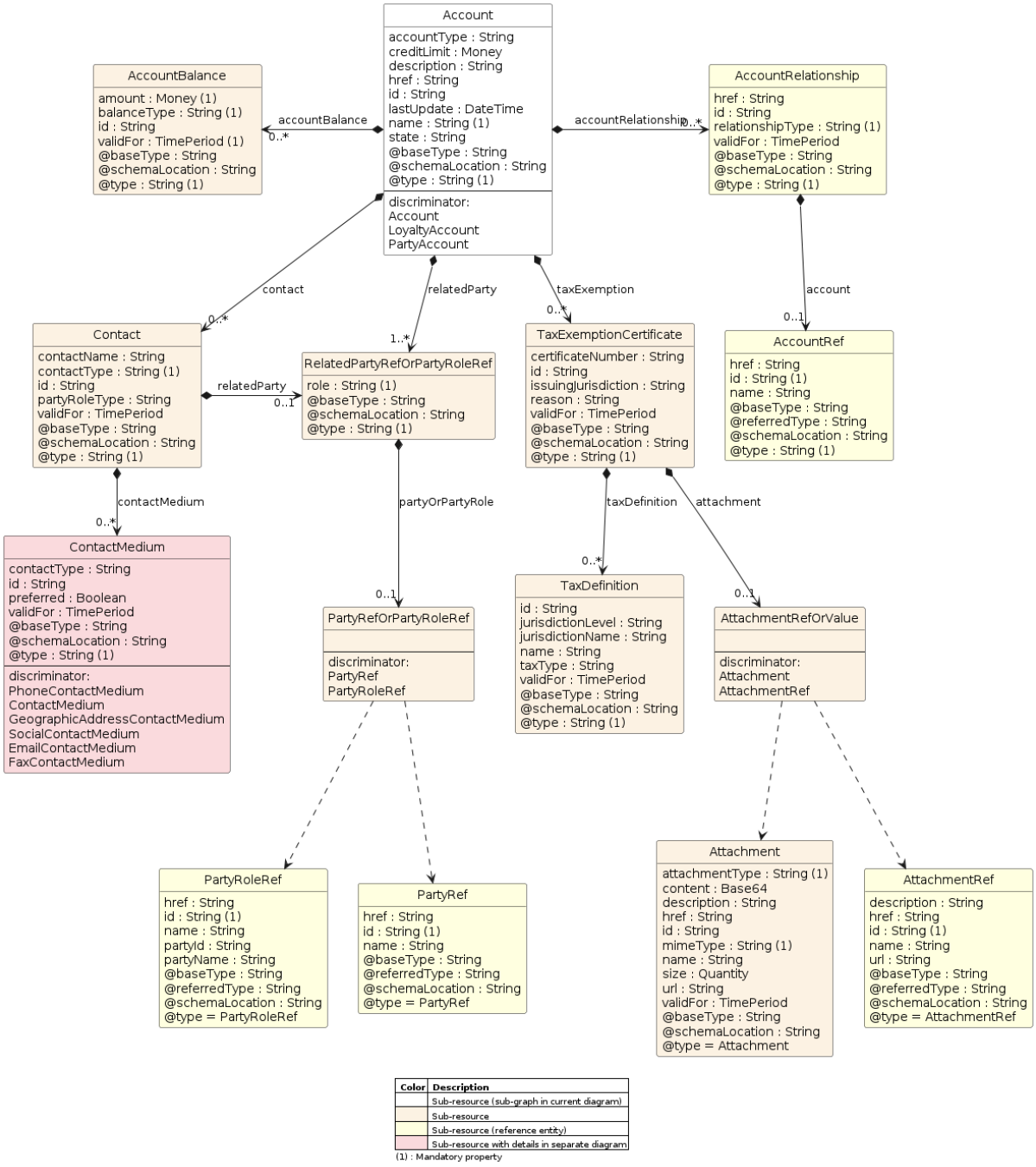


Figure 20 - Account

See Figure 5 for ContactMedium diagrams.

Field descriptions

LoyaltyAccount fields

accountBalance	An AccountBalance. List of balances for the account, for example regular postpaid balance, deposit balance, write-off balance.

© TM Forum 2025. All Rights Reserved.

accountRelationship	An AccountRelationship. List of balances related to the account. For example a list of billing accounts that contribute to a financial account.
accountType	A String. A categorization of an account, such as individual, joint, and so forth, whose instances share some of the same characteristics. Note: for flexibility we use a String here but an implementation may use an enumeration with a limited list of valid values.
billStructure	A BillStructure. The structure of the bill for party accounts (billing or settlement).
contact	A Contact. List of people who could be contacted regarding the account, for example the accountant who authorizes payments.
creditLimit	A Money. A base / value business entity used to represent money.
defaultPaymentMethod	A PaymentMethodRef. PaymentMethod reference. A payment method defines a specific mean of payment (e.g direct debit).
description	A String. Detailed description of the account.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
financialAccount	A FinancialAccountRef. AccountReceivable reference. An account of money owed by a party to another entity in exchange for goods or services that have been delivered or used. An account receivable aggregates the amounts of one or more party accounts (billing or settlement) owned by a given party.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. The date and time that the account was last updated.
name	A String. Name of the account.
paymentPlan	A PaymentPlan. A list of payment plans that are active or expired for the account, for example if the customer had difficulty in paying.
paymentStatus	A String. The condition of the account, such as due, paid, in arrears.
relatedParty	A RelatedPartyRefOrPartyRoleRef. List of parties that have some relationship with the account, for example the customer to whom the account belongs.
state	A String. Contains the lifecycle state such as: Active, Closed, Suspended and so on.
taxExemption	A TaxExemptionCertificate. List of tax exemptions that can be applied when calculating charges levied to the account.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

AccountBalance sub-resource fields

amount	A Money. A base / value business entity used to represent money.
balanceType	A String. Type of the balance : deposit balance, disputed balance, loyalty balance, receivable balance…
id	A String. Identifier of the balance within list of the balances (for entities with list).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AccountRelationship sub-resource fields

account	An AccountRef. Account reference. A account may be a party account or a financial account.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String. Type of relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AccountRef sub-resource fields

See [AccountRef] for details.

BillStructure sub-resource fields

cycleSpecification	A BillingCycleSpecificationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the BillingCycleSpecification entity and not the BillingCycleSpecificationRefOrValue class itself.

© TM Forum 2025. All Rights Reserved.

format	A BillFormatRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the BillFormat entity and not the BillFormatRefOrValue class itself.
presentationMedia	A BillPresentationMediaRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the BillPresentationMedia entity and not the BillPresentationMediaRefOrValue class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillingCycleSpecification sub-resource fields

billingDateShift	An Integer. An offset of a billing/settlement date. The offset is expressed as number of days with regard to the start of the billing/settlement period.
billingPeriod	A String. A billing time period. It can be recurring, for example: week, month, quarter of year, year .
chargeDateOffset	An Integer. An offset of a date through which charges previously received by the billing system will appear on the bill. The offset is expressed as number of days with regard to the start of the BillingPeriod.
creditDateOffset	An Integer. An offset of a date through which credits previously received by the billing system will appear on the bill. The offset is expressed as number of days with regard to the start of the BillingPeriod.
description	A String. An explanation regarding this billing cycle specification.
frequency	A String. Frequency of the billing cycle (monthly for instance).
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mailingDateOffset	An Integer. An offset of a customer bill mailing date. The offset is expressed as number of days with regard to the start of the BillingPeriod.
name	A String. A short descriptive name.
paymentDueDateOffset	An Integer. An offset of a payment due date. The offset is expressed as number of days with regard to the start of the BillingPeriod.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2025. All Rights Reserved.

@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillingCycleSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillFormat sub-resource fields

description	A String. An explanatory text for this bill format.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A short descriptive name.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillFormatRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillPresentationMedia sub-resource fields

© TM Forum 2025. All Rights Reserved.

description	A String. An explanatory text describing this bill presentation media.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A short descriptive name.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BillPresentationMediaRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Contact sub-resource fields

contactMedium	A ContactMedium. List of contact mediums for the contact, such as mobile phone number, email addreess, etc. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
contactName	A String. A displayable name for that contact.
contactType	A String. Type of contact (primary, secondary…).
id	A String. Identifier of the contact within list of the contacts (for entities with list).
partyRoleType	A String. Identifies what kind of party role type is linked to the contact (a account manager…).
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

© TM Forum 2025. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

See [ContactMedium] for details.

GeographicAddressRef sub-resource fields

See [GeographicAddressRef] for details.

EmailContactMedium sub-resource fields

See [EmailContactMedium] for details.

FaxContactMedium sub-resource fields

See [FaxContactMedium] for details.

GeographicAddressContactMedium sub-resource fields

See [GeographicAddressContactMedium] for details.

PhoneContactMedium sub-resource fields

See [PhoneContactMedium] for details.

SocialContactMedium sub-resource fields

See [SocialContactMedium] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

PaymentMethodRef sub-resource fields

See [PaymentMethodRef] for details.

© TM Forum 2025. All Rights Reserved.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

FinancialAccountRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PaymentPlan sub-resource fields

id	A String. Identifier of the plan within list of the plans (for entities with list).
numberOfPayments	An Integer. Number of payments used to spread the global payment.
paymentFrequency	A String. Frequency of the payments, such as monthly and bimonthly.
paymentMethod	A PaymentMethodRef. PaymentMethod reference. A payment method defines a specific mean of payment (e.g direct debit).
planType	A String. Type of payment plan.
priority	An Integer. Priority of the payment plan.
status	A String. Status of the payment plan (effective, ineffective).
totalAmount	A Money. A base / value business entity used to represent money.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

© TM Forum 2025. All Rights Reserved.

TaxExemptionCertificate sub-resource fields

See [TaxExemptionCertificate] for details.

Attachment sub-resource fields

See [Attachment] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

TaxDefinition sub-resource fields

See [TaxDefinition] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyAccount' resource object.


```
{
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/lovaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "name": "Loyalty Account",
    "state": "Active",
    "accountType": "Loyalty Account",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "@type": "AccountBalance",
            "amount": {
                "value": 20.00,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        },
```


© TM Forum 2025. All Rights Reserved.

```
        {
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "@type": "AccountBalance",
            "amount": {
                "value": 0.00,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```


LoyaltyProgramPartner resource

A partner engaged in the Loyalty Program.

Resource model

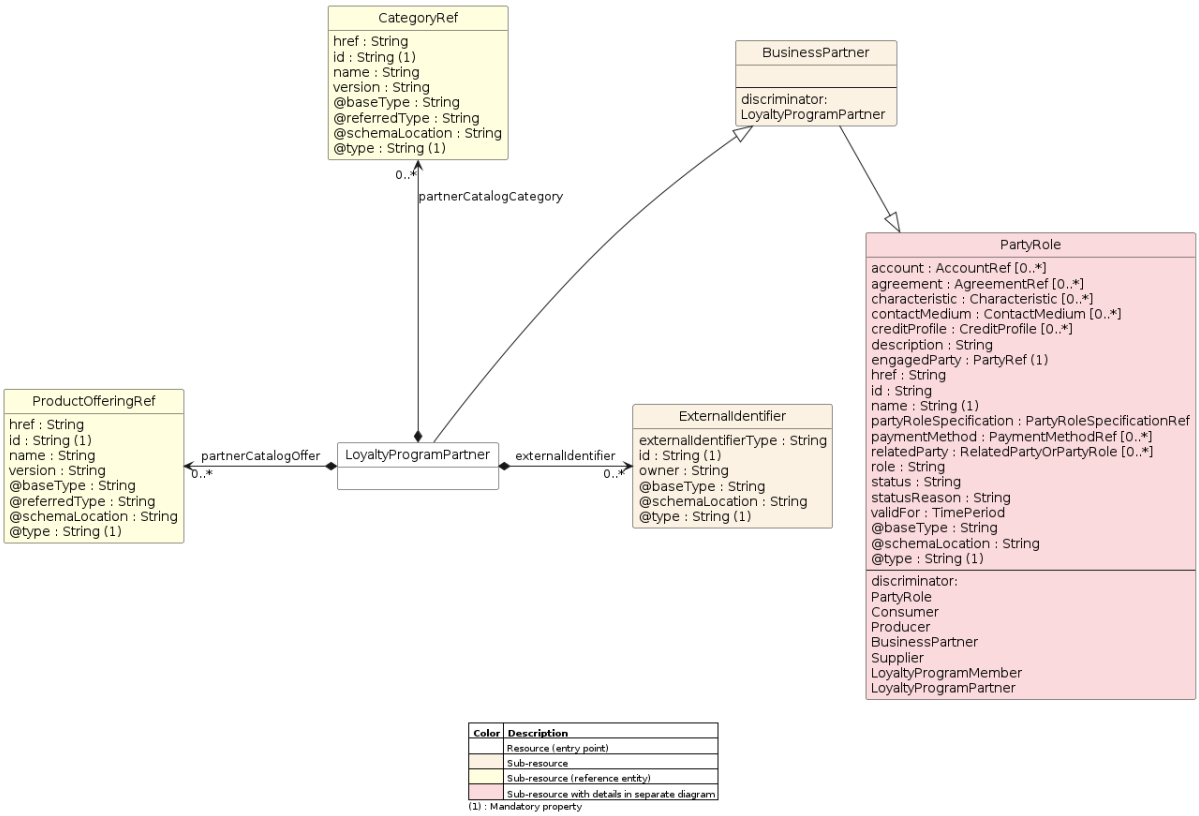


Figure 21 - LoyaltyProgramPartner

See Figure 5 for ContactMedium diagrams.

See Figure 6 for PartyRole diagrams.

See Figure 7 for Characteristic diagrams.

© TM Forum 2025. All Rights Reserved.

Field descriptions

LoyaltyProgramPartner fields

account	An AccountRef. Collection of accounts.
agreement	An AgreementRef. Collection of agreements.
characteristic	A Characteristic. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium. Collection of contact mediums. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creditProfile	A CreditProfile. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef. A Party reference.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partnerCatalogCategory	A CategoryRef. The partner catalog categories containing partner product offers made available for purchasing by Loyalty Program Members using loyalty rewards.
partnerCatalogOffer	A ProductOfferingRef. The partner catalog offers exposed to Loyalty Program Members for purchasing using Loyalty Rewards.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.

© TM Forum 2025. All Rights Reserved.

paymentMethod	A PaymentMethodRef. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by partyRoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AccountRef sub-resource fields

See [AccountRef] for details.

AgreementRef sub-resource fields

See [AgreementRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

© TM Forum 2025. All Rights Reserved.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

ContactMedium sub-resource fields

See [ContactMedium] for details.

GeographicAddressRef sub-resource fields

See [GeographicAddressRef] for details.

EmailContactMedium sub-resource fields

See [EmailContactMedium] for details.

FaxContactMedium sub-resource fields

See [FaxContactMedium] for details.

GeographicAddressContactMedium sub-resource fields

See [GeographicAddressContactMedium] for details.

PhoneContactMedium sub-resource fields

© TM Forum 2025. All Rights Reserved.

See [PhoneContactMedium] for details.

SocialContactMedium sub-resource fields

See [SocialContactMedium] for details.

CreditProfile sub-resource fields

See [CreditProfile] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

CategoryRef sub-resource fields

See [CategoryRef] for details.

ProductOfferingRef sub-resource fields

See [ProductOfferingRef] for details.

PartyRoleSpecificationRef sub-resource fields

See [PartyRoleSpecificationRef] for details.

PaymentMethodRef sub-resource fields

See [PaymentMethodRef] for details.

RelatedPartyOrPartyRole sub-resource fields

See [RelatedPartyOrPartyRole] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyProgramPartner' resource object.


```
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "active",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
```


© TM Forum 2025. All Rights Reserved.

```
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
```


© TM Forum 2025. All Rights Reserved.

```
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```


LoyaltyEarn resource

A specialised task to process direct loyalty points earnings.

Resource model

© TM Forum 2025. All Rights Reserved.

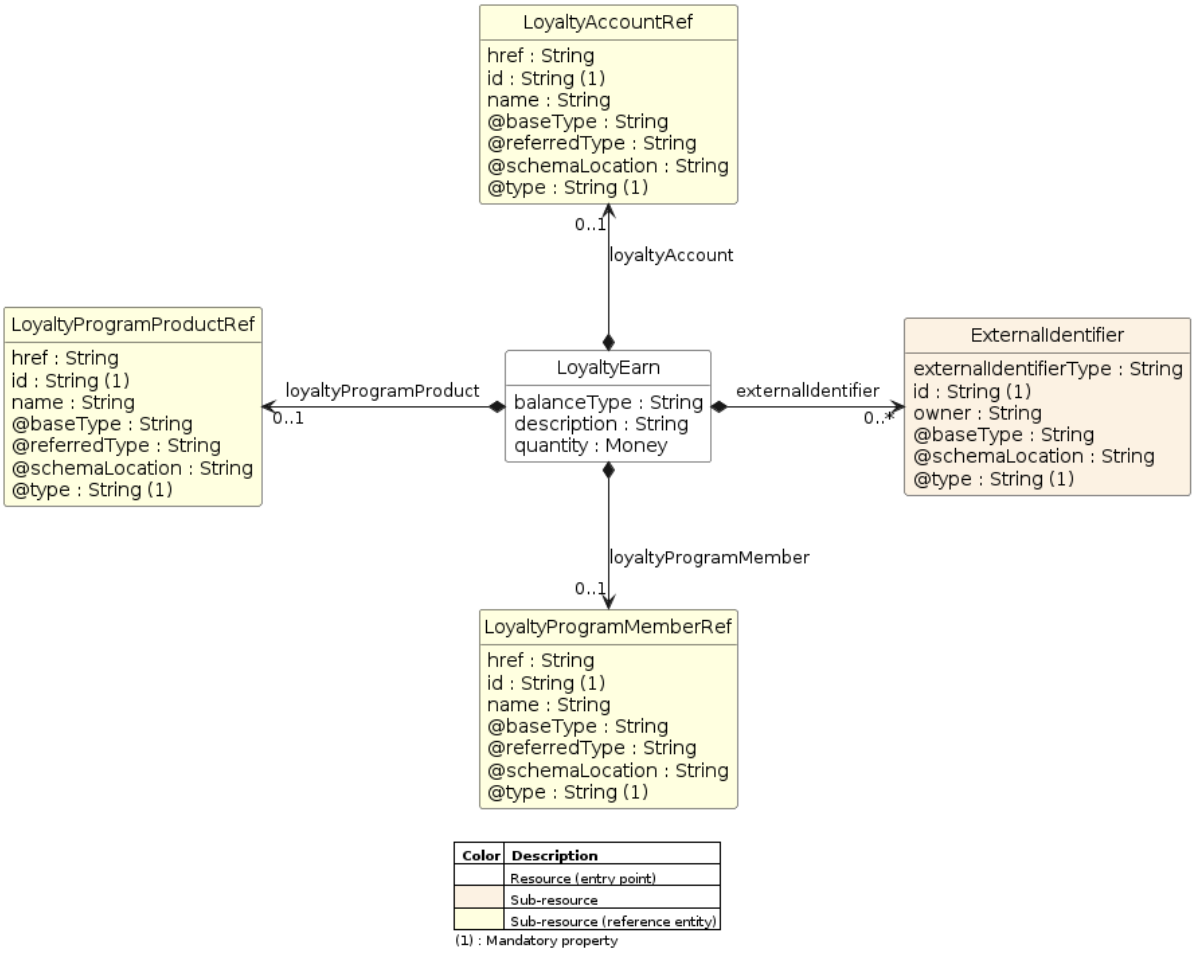


Figure 22 - LoyaltyEarn

Field descriptions

LoyaltyEarn fields

balanceType	A String. Type of the loyalty balance. Specifies the category of a loyalty account balance, such as earned, pending, banked, redeemed, expired, to define its purpose and status within the loyalty program. In the context of a loyalty burn, only the banked balance or in certain circumstances the bonus points wallet may be used.
description	A String. A description detailing the earn event.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
loyaltyAccount	A LoyaltyAccountRef. Reference to a loyalty account.
loyaltyProgramMember	A LoyaltyProgramMemberRef. Reference to a loyalty program member.
loyaltyProgramProduct	A LoyaltyProgramProductRef. A reference to a loyalty product.
quantity	A Money. A base / value business entity used to represent money.

ExternalIdentifier sub-resource fields

© TM Forum 2025. All Rights Reserved.

See [ExternalIdentifier] for details.

LoyaltyAccountRef sub-resource fields

See [LoyaltyAccountRef] for details.

LoyaltyProgramMemberRef sub-resource fields

See [LoyaltyProgramMemberRef] for details.

LoyaltyProgramProductRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyEarn' resource object.


```
{
    "id": "6651",
    "href": "https://mycsp.com:8080/tmf-api/loyaltyManagement/v5/loyaltyEarn/6651",
    "@type": "LoyaltyEarn",
    "@baseType": "ProductUsage",
    "usageDate": "2024-04-21T09:13:16-07:00",
    "description": "Partner brand interaction earning",
    "status": "received",
    "usageType": "Partner interaction",
    "balanceType": "LoyaltyEarningsWallet",
    "quantity": {
        "value": 25,
        "unit": "POINTS"
    },
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Business Partner",
            "partyOrPartyRole": {
                "id": "8121",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/8121",
                "@type": "PartyRoleRef",
                "name": "Lifestyle Goods",
                "@referredType": "LoyaltyProgramPartner"
            }
```


© TM Forum 2025. All Rights Reserved.

```
        }
    ],
    "loyaltyProgramMember": {
        "id": "9664",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
        "@type": "LoyaltyProgramMemberRef",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "1765",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/1765",
        "@type": "LoyaltyProgramProductRef",
        "name": "John Doe's Loyalty Program",
        "@referredType": "LoyaltyProgramProduct"
    },
    "loyaltyAccount": {
        "id": "1155",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
        "@type": "LoyaltyAccountRef",
        "name": "John Doe's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    }
}
```


LoyaltyBurn resource

A specialised task to process direct loyalty points redemption for purchases.

Resource model

© TM Forum 2025. All Rights Reserved.

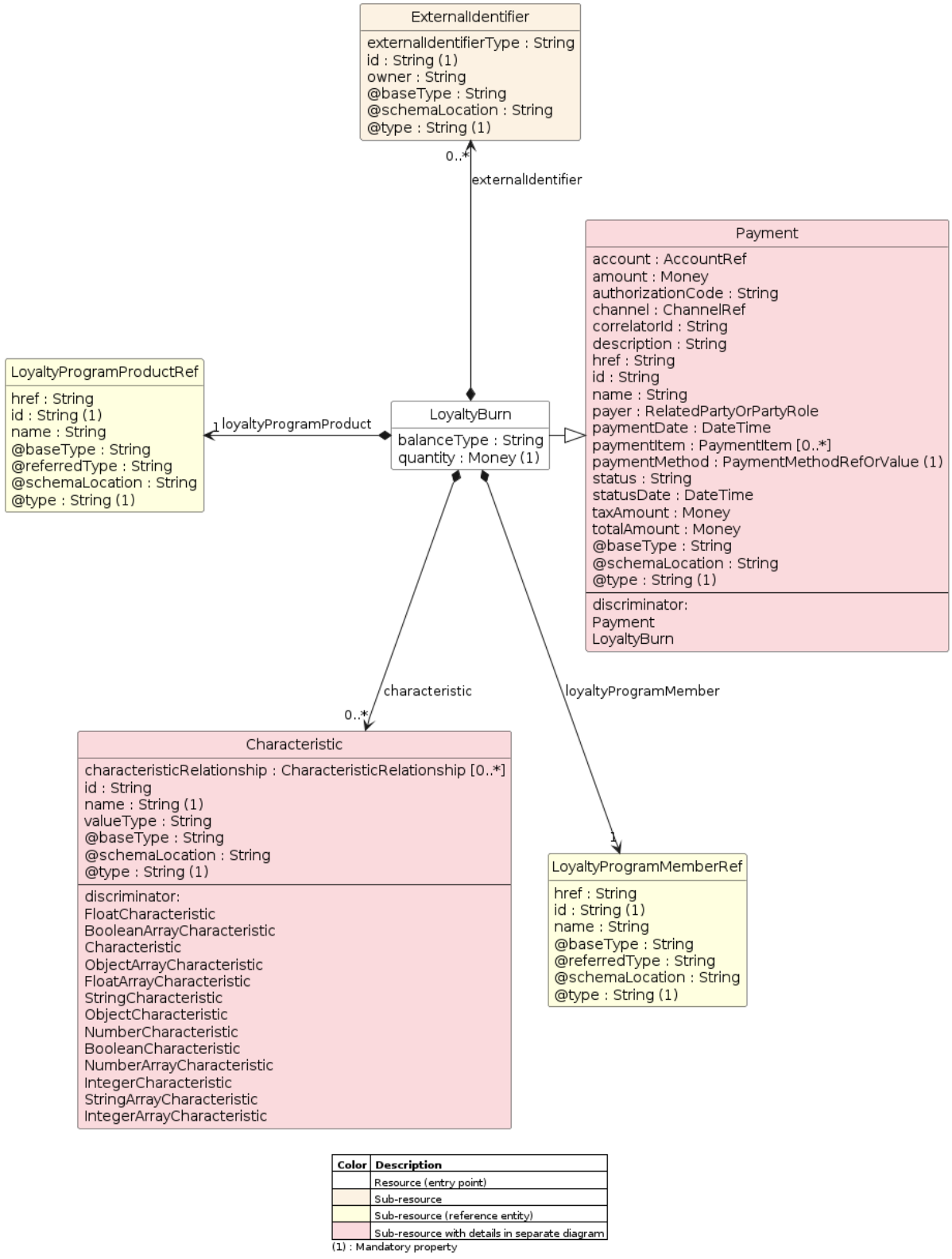


Figure 23 - LoyaltyBurn

© TM Forum 2025. All Rights Reserved.

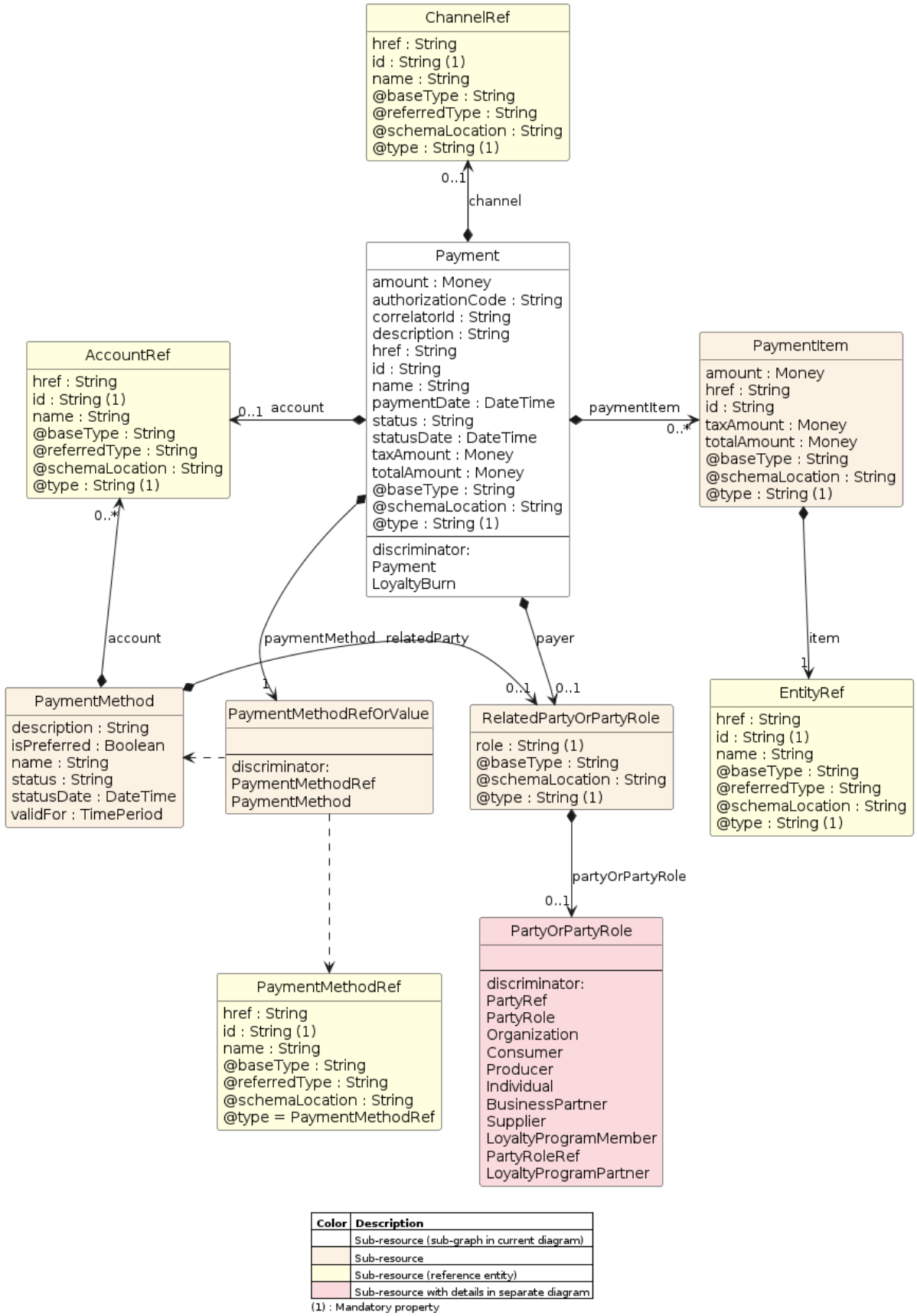


Figure 24 - Payment

© TM Forum 2025. All Rights Reserved.

See Figure 5 for ContactMedium diagrams.

See Figure 7 for Characteristic diagrams.

See Figure 12 for Organization diagrams.

See Figure 14 for Individual diagrams.

See Figure 15 for Party diagrams.

See Figure 16 for PartyOrPartyRole diagrams.

See Figure 6 for PartyRole diagrams.

Field descriptions

LoyaltyBurn fields

account	An AccountRef. Account reference. A account may be a party account or a financial account.
amount	A Money. A base / value business entity used to represent money.
authorizationCode	A String. Authorization code retrieved from an external payment gateway that could be used for conciliation.
balanceType	A String. Type of the loyalty balance. Specifies the category of a loyalty account balance, such as earned, pending, banked, redeemed, expired, to define its purpose and status within the loyalty program. In the context of a loyalty burn, only the banked balance or in certain circumstances the bonus points wallet may be used.
channel	A ChannelRef. The channel to which the resource reference to. e.g. channel for selling product offerings, channel for opening a trouble ticket etc..
characteristic	A Characteristic. Collection of characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
correlatorId	A String. Unique identifier in the client for the payment in case it is needed to correlate.
description	A String. A description detailing the burn event.

© TM Forum 2025. All Rights Reserved.

externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
loyaltyProgramMember	A LoyaltyProgramMemberRef. Reference to a loyalty program member.
loyaltyProgramProduct	A LoyaltyProgramProductRef. A reference to a loyalty product.
name	A String. Screen name of the payment.
payer	A RelatedPartyOrPartyRole. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
paymentDate	A DateTime. Date when the payment was performed.
paymentItem	A PaymentItem. The paymentItem is the result of lettering process. It enables to assign automatically or manually part of incoming payment amount to a bill.
paymentMethod	A PaymentMethodRefOrValue. Link to the resource that holds information about the payment mean used to complete the operation.
quantity	A Money. A base / value business entity used to represent money.
status	A String. Status of the payment.
statusDate	A DateTime. Date when the status was recorded.
taxAmount	A Money. A base / value business entity used to represent money.
totalAmount	A Money. A base / value business entity used to represent money.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AccountRef sub-resource fields

See [AccountRef] for details.

ChannelRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2025. All Rights Reserved.

@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

© TM Forum 2025. All Rights Reserved.

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

LoyaltyProgramMemberRef sub-resource fields

See [LoyaltyProgramMemberRef] for details.

LoyaltyProgramProductRef sub-resource fields

See [LoyaltyProgramProductRef] for details.

RelatedPartyOrPartyRole sub-resource fields

See [RelatedPartyOrPartyRole] for details.

BusinessPartner sub-resource fields

See [BusinessPartner] for details.

AgreementRef sub-resource fields

See [AgreementRef] for details.

ContactMedium sub-resource fields

See [ContactMedium] for details.

GeographicAddressRef sub-resource fields

See [GeographicAddressRef] for details.

EmailContactMedium sub-resource fields

See [EmailContactMedium] for details.

FaxContactMedium sub-resource fields

See [FaxContactMedium] for details.

GeographicAddressContactMedium sub-resource fields

See [GeographicAddressContactMedium] for details.

© TM Forum 2025. All Rights Reserved.

PhoneContactMedium sub-resource fields

See [PhoneContactMedium] for details.

SocialContactMedium sub-resource fields

See [SocialContactMedium] for details.

CreditProfile sub-resource fields

See [CreditProfile] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleSpecificationRef sub-resource fields

See [PartyRoleSpecificationRef] for details.

PaymentMethodRef sub-resource fields

See [PaymentMethodRef] for details.

CategoryRef sub-resource fields

See [CategoryRef] for details.

ProductOfferingRef sub-resource fields

See [ProductOfferingRef] for details.

Consumer sub-resource fields

See [Consumer] for details.

Individual sub-resource fields

See [Individual] for details.

PartyCreditProfile sub-resource fields

See [PartyCreditProfile] for details.

Disability sub-resource fields

See [Disability] for details.

IndividualIdentification sub-resource fields

See [IndividualIdentification] for details.

Attachment sub-resource fields

© TM Forum 2025. All Rights Reserved.

See [Attachment] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

LanguageAbility sub-resource fields

See [LanguageAbility] for details.

OtherNameIndividual sub-resource fields

See [OtherNameIndividual] for details.

Skill sub-resource fields

See [Skill] for details.

TaxExemptionCertificate sub-resource fields

See [TaxExemptionCertificate] for details.

TaxDefinition sub-resource fields

See [TaxDefinition] for details.

LoyaltyProgramMember sub-resource fields

See [LoyaltyProgramMember] for details.

LoyaltyProgramPartner sub-resource fields

See [LoyaltyProgramPartner] for details.

Organization sub-resource fields

See [Organization] for details.

OrganizationChildRelationship sub-resource fields

See [OrganizationChildRelationship] for details.

OrganizationRef sub-resource fields

See [OrganizationRef] for details.

OrganizationIdentification sub-resource fields

See [OrganizationIdentification] for details.

OrganizationParentRelationship sub-resource fields

See [OrganizationParentRelationship] for details.

© TM Forum 2025. All Rights Reserved.

OtherNameOrganization sub-resource fields

See [OtherNameOrganization] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRole sub-resource fields

See [PartyRole] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

Producer sub-resource fields

See [Producer] for details.

Supplier sub-resource fields

See [Supplier] for details.

PaymentItem sub-resource fields

amount	A Money. A base / value business entity used to represent money.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
item	An EntityRef. Entity reference schema to be use for all entityRef class.
taxAmount	A Money. A base / value business entity used to represent money.
totalAmount	A Money. A base / value business entity used to represent money.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

See [EntityRef] for details.

PaymentMethod sub-resource fields

© TM Forum 2025. All Rights Reserved.

account	An AccountRef. Account reference. A account may be a party account or a financial account.
description	A String. The contents of the payment method.
isPreferred	A Boolean. If the method is the preferred one by the owner. Typically used when querying for the payment methods of a specific customer or account.
name	A String. Friendly name assigned to the payment method.
relatedParty	A RelatedPartyOrPartyRole. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
status	A String.
statusDate	A DateTime.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

PaymentMethodRef sub-resource fields

See [PaymentMethodRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyBurn' resource object.


```
{
    "id": "542F9866-97E2-4841-AC60-804D0A1CA810",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyBurn/542F9866-97E2-4841-AC60-804D0A1CA810",
    "@type": "LoyaltyBurn",
    "@baseType": "Payment",
    "correlatorId": "542F9866-97E2-4841-AC60-804D0A1CA810",
    "paymentDate": "2024-05-01T20:39:38.832Z",
    "name": "Part payment for partner catalog purchases",
    "description": "Payment in full or in part with loyalty points / virtual
currency earnings",
    "authorizationCode": "542F9866-97E2-4841-AC60-804D0A1CA810",
    "status": "received",
    "statusDate": "2024-05-01T20:39:38.832Z",
    "quantity": {
        "value": 2000,
        "unit": "POINTS"
    },
    "taxAmount": {
        "unit": "USD",
        "value": 0
    },
    "channel": {
        "id": "45",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/45",
        "@type": "ChannelRef",
        "name": "Online purchase channels",
        "@referredType": "SalesChannel"
    },
    "payer": {
```


© TM Forum 2025. All Rights Reserved.

```
        "@type": "RelatedPartyOrPartyRole",
        "role": "Loyalty Program Member",
        "partyOrPartyRole": {
            "id": "9664",
            "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
            "@type": "PartyRoleRef",
            "name": "John Doe",
            "@referredType": "LoyaltyProgramMember"
        }
    },
    "paymentMethod": {
        "@type": "LoyaltyPaymentMethod",
        "@baseType": "PaymentMethod",
        "name": "Joseph's Loyalty Points",
        "status": "Active",
        "statusDate": "2024-11-21T11:21:11.445Z"
    },
    "paymentItem": [
        {
            "id": "1",
            "amount": {
                "value": 450.0,
                "unit": "USD"
            },
            "item": {
                "id": "1201",
                "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/1201",
                "name": "Smart Phone",
                "@type": "EntityRef",
                "@referredType": "ProductOrder"
            },
            "taxAmount": {
                "value": 0.0,
                "unit": "USD"
            },
            "totalAmount": {
                "value": 450.0,
                "unit": "USD"
            },
            "@type": "PaymentItem"
        }
    ],
    "totalAmount": {
        "value": 450.0,
        "unit": "USD"
    },
    "loyaltyProgramMember": {
        "id": "9664",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
        "@type": "LoyaltyProgramMemberRef",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "1745",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/1745",
        "@type": "LoyaltyProgramProductRef",
        "name": "John Doe's Loyalty Program",
        "@referredType": "LoyaltyProgramProduct"
```


© TM Forum 2025. All Rights Reserved.

```
    },
    "account": {
        "id": "1155",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
        "@type": "AccountRef",
        "name": "John Doe's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    }
}
```


LoyaltyExecutionPoint resource

A LoyaltyExecutionPoint is a type of Policy Execution Point. A LoyaltyExecutionPoint tracks the application of a Loyalty Policy Action and serves as a record of an applied Loyalty Policy Action.

Resource model

© TM Forum 2025. All Rights Reserved.

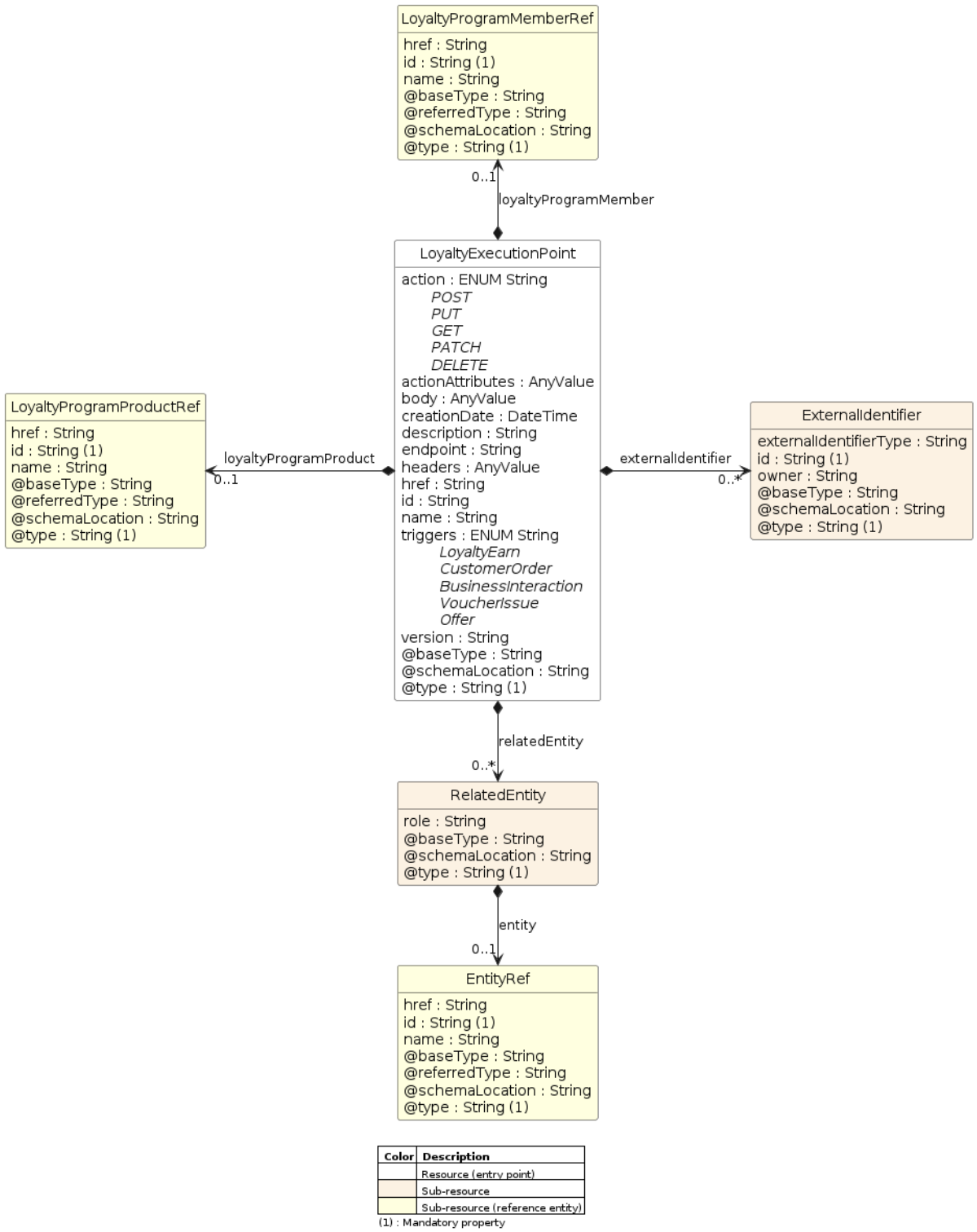


Figure 25 - LoyaltyExecutionPoint

Field descriptions

LoyaltyExecutionPoint fields

© TM Forum 2025. All Rights Reserved.

action	A String. The HTTP operation to be used when calling the endpoint. ENUMERATED with values: * POST * PUT * GET * PATCH * DELETE
actionAttributes	An AnyValue.
body	An AnyValue.
creationDate	A DateTime. The timestamp that the action was applied.
description	A String. A free-form description of the loyalty execution point.
endpoint	A String. The endpoint called to trigger the Loyalty Program reward.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
headers	An AnyValue.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
loyaltyProgramMember	A LoyaltyProgramMemberRef. Reference to a loyalty program member.
loyaltyProgramProduct	A LoyaltyProgramProductRef. A reference to a loyalty product.
name	A String. A user-friendly identifier of the loyalty execution point.
relatedEntity	A RelatedEntity. A reference to an entity, where the type of the entity is not known in advance.
triggers	A String. The type of loyalty action. ENUMERATED with values: * LoyaltyEarn * CustomerOrder * BusinessInteraction * VoucherIssue * Offer
version	A String. A string that identifies the version of the loyalty action.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

LoyaltyProgramMemberRef sub-resource fields

See [LoyaltyProgramMemberRef] for details.

© TM Forum 2025. All Rights Reserved.

LoyaltyProgramProductRef sub-resource fields

See [LoyaltyProgramProductRef] for details.

RelatedEntity sub-resource fields

entity	An EntityRef. Entity reference schema to be use for all entityRef class.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

See [EntityRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'LoyaltyExecutionPoint' resource object.


```
{
    "id": "7756",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyExecutionPoint/7756",
    "@type": "LoyaltyExecutionPoint",
    "name": "John Doe's Birthday Gift Loyalty Execution Point",
    "triggers": "CustomerOrder",
    "loyaltyProgramMember": {
        "id": "9664",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
        "@type": "LoyaltyProgramMemberRef",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "1745",
        "href": "https://mycsp.com:8080/tmf-
api/loyalty/v5/loyaltyProgramProduct/1745",
        "@type": "LoyaltyProgramProductRef",
        "name": "John Doe's Loyalty Program",
        "@referredType": "LoyaltyProgramProduct"
    },
    "body": {
        "category": "Birthday Gift product order",
        "priority": "1",
        "channel": [
            {
                "role": "Used channel for order capture",
                "@type": "RelatedChannel",
                "channel": {
```


© TM Forum 2025. All Rights Reserved.

```
                    "id": "1",
                    "name": "Online channel",
                    "@type": "ChannelRef"
                }
            }
        ],
        "productOrderItem": [
            {
                "id": "100",
                "quantity": 1,
                "action": "add",
                "productOffering": {
                    "id": "14277",
                    "href":
"https://host:port/productCatalogManagement/v5/productOffering/14277",
                    "name": "TMF25",
                    "@type": "ProductOfferingRef"
                },
                "@type": "ProductOrderItem"
            }
        ],
        "relatedParty": [
            {
                "role": "Customer",
                "partyOrPartyRole": {
                    "id": "{{customer.id}}",
                    "href":
"https://host:port/customerManagement/v5/customer/{{customer.id}}",
                    "name": "{{customer.name}}",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                },
                "@type": "RelatedPartyRefOrPartyRoleRef"
            }
        ],
        "@type": "ProductOrder"
    },
    "version": "1.0",
    "action": "POST",
    "endpoint": "https://mycsp.com:8080/tmf-api/productOrdering/v5/productOrder",
    "creationDate": "2024-05-01T20:36:13.313Z"
}
```


Notification Resource Models

27 notifications are defined for this API.

Notifications related to LoyaltyProgramProductSpecification:

- Update Event
- Create Event
- Delete Event
- State Change Event

Notifications related to LoyaltyProgramMember:

- Update Event

© TM Forum 2025. All Rights Reserved.

- Create Event
- Delete Event
- State Change Event

Notifications related to LoyaltyProgramProduct:

- Update Event
- Create Event
- Delete Event
- State Change Event

Notifications related to LoyaltyAccount:

- Update Event
- Create Event
- Delete Event
- State Change Event

Notifications related to LoyaltyProgramPartner:

- Update Event
- Create Event
- Delete Event
- State Change Event

Notifications related to LoyaltyEarn:

- Create Event
- State Change Event

Notifications related to LoyaltyBurn:

- Create Event
- State Change Event

Notifications related to LoyaltyExecutionPoint:

- Update Event
- Create Event
- Delete Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the

© TM Forum 2025. All Rights Reserved.

resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

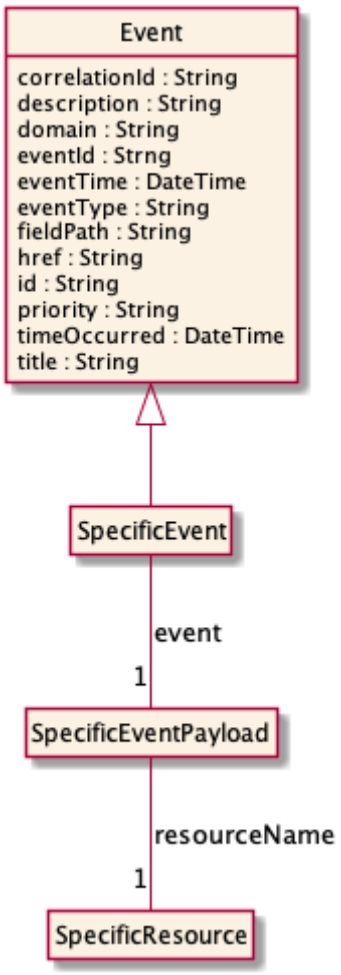


Figure 26 Notification Pattern

LoyaltyProgramProductSpecification

Update Event

Message example for LoyaltyProgramProductSpecificationUpdateEvent

```
Content-Type: application/json
{
    "correlationId": "4eeb06e2-8210",
    "description": "LoyaltyProgramProductSpecificationUpdateEvent illustration",
    "domain": "Commercial",
    "eventId": "46c2-8e2b-0182de193a9c",
    "eventTime": "2022-08-25T12:17:59.613Z",
    "eventType": "LoyaltyProgramProductSpecificationUpdateEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:17:57.413Z",
    "title": "LoyaltyProgramProductSpecificationUpdateEvent",
    "event": {
        "loyaltyProgramProductSpecification": {
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/9881",
```

© TM Forum 2025. All Rights Reserved.

```
            "id": "9881",
            "@type": "LoyaltyProgramProductSpecification",
            "relatedParty": [
                {
                    "role": "Owner",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                        "id": "777",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/777",
                        "name": "John Doe",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                    }
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "868",
        "name": "APP-394",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "372",
        "name": "APP-92",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductSpecificationUpdateEvent"
}
```


Create Event

Message example for LoyaltyProgramProductSpecificationCreateEvent


```
Content-Type: application/json
{
    "correlationId": "be58c421-12cf",
    "description": "LoyaltyProgramProductSpecificationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4a0a-9a52-86f728494b4e",
    "eventTime": "2022-08-25T12:17:59.603Z",
    "eventType": "LoyaltyProgramProductSpecificationCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:17:52.237Z",
    "title": "LoyaltyProgramProductSpecificationCreateEvent",
    "event": {
        "loyaltyProgramProductSpecification": {
            "id": "100",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
            "@type": "LoyaltyProgramProductSpecification",
            "@baseType": "ProductSpecification",
            "brand": "My MVNO",
            "description": "Rewards Program to incentivise onboarding with
benefits",
            "isBundle": false,
```


© TM Forum 2025. All Rights Reserved.

```
            "productNumber": "LOYALTY-01",
            "category": [
                {
                    "id": "1",
                    "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/category/1",
                    "@type": "CategoryRef",
                    "name": "Smart City Loyalty Incentive",
                    "@referredType": "Category",
                    "version": "1.0"
                }
            ],
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2030-04-12T23:20:50.52Z"
            },
            "version": "1.0",
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "Catalog Vendor",
                    "partyOrPartyRole": {
                        "id": "501",
                        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/501",
                        "@type": "PartyRoleRef",
                        "name": "Loyalty Benefit Partner",
                        "@referredType": "LoyaltyProgramPartner"
                    }
                }
            ],
            "lastUpdate": "2024-05-01T18:15:13.414Z",
            "lifecycleStatus": "Active",
            "name": "Inner City Transportation Incentive Program",
            "needsLoyaltyAccount": true,
            "productSpecCharacteristic": [
                {
                    "name": "Bicycle Model",
                    "description": "The bicycle model that the citizen rents",
                    "valueType": "string",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "isUnique": false,
                    "@type": "CharacteristicSpecification",
                    "characteristicValueSpecification": [
                        {
                            "isDefault": true,
                            "validFor": {
                                "startDateTime": "2024-09-23T00:00:00Z",
                                "endDateTime": "2028-11-24T00:00:00Z"
                            },
                            "value": "Trek Domane",
                            "@type": "StringCharacteristicValueSpecification"
                        },
                        {
                            "isDefault": false,
                            "validFor": {
                                "startDateTime": "2022-09-23T00:00:00Z",
                                "endDateTime": "2026-11-24T00:00:00Z"
                            },
                            "value": "Giant Trance",
                            "@type": "StringCharacteristicValueSpecification"
```


© TM Forum 2025. All Rights Reserved.

```
                        },
                        {
                            "isDefault": false,
                            "validFor": {
                                "startDateTime": "2025-09-23T00:00:00Z",
                                "endDateTime": "2030-11-24T00:00:00Z"
                            },
                            "value": "Yeti SB130",
                            "@type": "StringCharacteristicValueSpecification"
                        }
                    ],
                    "validFor": {
                        "startDateTime": "2024-09-23T16:42:23Z"
                    }
                },
                {
                    "name": "Color",
                    "description": "Color of the bicycle",
                    "valueType": "string",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "extensible": true,
                    "isUnique": true,
                    "@type": "CharacteristicSpecification",
                    "characteristicValueSpecification": [
                        {
                            "isDefault": true,
                            "valueType": "string",
                            "value": "Black",
                            "@type": "StringCharacteristicValueSpecification"
                        },
                        {
                            "isDefault": false,
                            "valueType": "string",
                            "value": "White",
                            "@type": "StringCharacteristicValueSpecification"
                        }
                    ],
                    "validFor": {
                        "startDateTime": "2024-09-23T16:42:23Z"
                    }
                }
            ],
            "policy": [
                {
                    "id": "88987",
                    "href": "https://mycsp.com:8080/tmf-
api/policyManagement/v5/policyRule/88987",
                    "@type": "PolicyRef",
                    "name": "High LSM Rule",
                    "@referredType": "PolicyRule"
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "868",
        "name": "APP-394",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
```


© TM Forum 2025. All Rights Reserved.

```
        "id": "372",
        "name": "APP-92",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductSpecificationCreateEvent"
}
```

Delete Event

Message example for LoyaltyProgramProductSpecificationDeleteEvent

```
Content-Type: application/json
{
    "correlationId": "f3a34f89-0046",
    "description": "LoyaltyProgramProductSpecificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4996-b9b1-e533c09a0186",
    "eventTime": "2022-08-25T12:17:59.636Z",
    "eventType": "LoyaltyProgramProductSpecificationDeleteEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:17:55.035Z",
    "title": "LoyaltyProgramProductSpecificationDeleteEvent",
    "event": {
        "loyaltyProgramProductSpecification": {
            "id": "9881",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/9881",
            "@type": "LoyaltyProgramProductSpecification"
        }
    },
    "reportingSystem": {
        "id": "868",
        "name": "APP-394",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "372",
        "name": "APP-92",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductSpecificationDeleteEvent"
}
```

State Change Event

Message example for LoyaltyProgramProductSpecificationStateChangeEvent

```
Content-Type: application/json
{
    "correlationId": "d991dd5c-cb42",
    "description": "LoyaltyProgramProductSpecificationStateChangeEvent
illustration",
```

© TM Forum 2025. All Rights Reserved.

```
    "domain": "Commercial",
    "eventId": "4768-a3c5-c47b06010b06",
    "eventTime": "2022-08-25T12:17:59.625Z",
    "eventType": "LoyaltyProgramProductSpecificationStateChangeEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:17:58.436Z",
    "title": "LoyaltyProgramProductSpecificationStateChangeEvent",
    "event": {
        "loyaltyProgramProductSpecification": {
            "href": "http://servername/loyaltyProgramProductSpecification/9881",
            "id": "9881",
            "@type": "LoyaltyProgramProductSpecification",
            "lifecycleStatus": "Active"
        }
    },
    "reportingSystem": {
        "id": "868",
        "name": "APP-394",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "372",
        "name": "APP-92",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductSpecificationStateChangeEvent"
}
```


LoyaltyProgramMember

Update Event

Message example for LoyaltyProgramMemberUpdateEvent


```
Content-Type: application/json
{
    "correlationId": "5030abc8-0b95",
    "description": "LoyaltyProgramMemberUpdateEvent illustration",
    "domain": "Commercial",
    "eventId": "4e5c-a999-008f4029d6d9",
    "eventTime": "2022-08-25T12:21:50.030Z",
    "eventType": "LoyaltyProgramMemberUpdateEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:21:49.819Z",
    "title": "LoyaltyProgramMemberUpdateEvent",
    "event": {
        "loyaltyProgramMember": {
            "id": "9664",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramMember/9664",
            "@type": "LoyaltyProgramMember",
            "relatedParty": [
                {
                    "role": "administrator",
                    "@type": "RelatedPartyOrPartyRole",
                    "partyOrPartyRole": {
```


© TM Forum 2025. All Rights Reserved.

```
                        "id": "777",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/777",
                        "name": "John Smith",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                    }
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramMemberUpdateEvent"
}
```

Create Event

Message example for LoyaltyProgramMemberCreateEvent

```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyProgramMemberCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
    "eventType": "LoyaltyProgramMemberCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyProgramMemberCreateEvent",
    "event": {
        "loyaltyProgramMember": {
            "id": "9664",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
            "@type": "LoyaltyProgramMember",
            "@baseType": "PartyRole",
            "name": "John Doe",
            "description": "Loyalty Program Member Profile",
            "engagedParty": {
                "id": "7872",
                "href": "https:/host:port/tmf-
api/partyManagement/v2/individual/7872",
                "@type": "PartyRef",
                "name": "John Doe",
                "@referredType": "Individual"
            },
            "account": [
```

© TM Forum 2025. All Rights Reserved.

```
                {
                    "id": "1155",
                    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
                    "@type": "AccountRef",
                    "name": "John Doe's Loyalty Account",
                    "@referredType": "LoyaltyAccount"
                }
            ],
            "contactMedium": [
                {
                    "preferred": true,
                    "validFor": {
                        "startDateTime": "2025-01-15T07:49:25.246Z"
                    },
                    "@type": "EmailContactMedium",
                    "emailAddress": "john.doe@gmail.com"
                },
                {
                    "preferred": false,
                    "validFor": {
                        "startDateTime": "2025-01-15T07:49:25.246Z"
                    },
                    "@type": "GeographicAddressContactMedium",
                    "city": "Morristown",
                    "country": "USA",
                    "postCode": "07960",
                    "stateOrProvince": "New Jersey",
                    "street1": "240 Headquarters Plazza",
                    "street2": "East Tower - 10th Floor"
                }
            ],
            "agreement": [
                {
                    "id": "4721",
                    "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
                    "@type": "AgreementRef",
                    "name": "John Doe's Loyalty Program Membership Agreement"
                }
            ],
            "characteristic": [
                {
                    "name": "Last Active",
                    "value": "2025-01-15T07:49:25.246Z",
                    "@type": "StringCharacteristic"
                }
            ],
            "status": "active",
            "statusReason": "Loyalty Member Enrolment",
            "validFor": {
                "startDateTime": "2025-01-12T23:20:50.52Z",
                "endDateTime": "2030-04-12T23:20:50.52Z"
            },
            "tier": "Gold"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
```


© TM Forum 2025. All Rights Reserved.

```
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyProgramMemberCreateEvent",
    "@baseType": "Event"
}
```

Delete Event

Message example for LoyaltyProgramMemberDeleteEvent

```
Content-Type: application/json
{
    "correlationId": "17eabb55-f69c",
    "description": "LoyaltyProgramMemberDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "46c6-86cd-47b3fa4d39e3",
    "eventTime": "2025-01-25T12:21:50.052Z",
    "eventType": "LoyaltyProgramMemberDeleteEvent",
    "priority": "2",
    "timeOccurred": "2025-01-25T12:21:47.370Z",
    "title": "LoyaltyProgramMemberDeleteEvent",
    "event": {
        "loyaltyProgramMember": {
            "id": "9664",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramMember/9664",
            "@type": "LoyaltyProgramMember"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramMemberDeleteEvent"
}
```

State Change Event

Message example for LoyaltyProgramMemberStateChangeEvent

```
Content-Type: application/json
{
    "correlationId": "9bd57948-24b6",
    "description": "LoyaltyProgramMemberStateChangeEvent illustration",
```

© TM Forum 2025. All Rights Reserved.

```
    "domain": "Commercial",
    "eventId": "4d28-8a7a-cf4c696bc7ec",
    "eventTime": "2025-01-25T12:21:50.041Z",
    "eventType": "LoyaltyProgramMemberStateChangeEvent",
    "priority": "2",
    "timeOccurred": "2025-01-25T12:21:43.703Z",
    "title": "LoyaltyProgramMemberStateChangeEvent",
    "event": {
        "loyaltyProgramMember": {
            "id": "9664",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramMember/9664",
            "@type": "LoyaltyProgramMember",
            "status": "validated"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramMemberStateChangeEvent"
}
```


LoyaltyProgramProduct

Update Event

Message example for LoyaltyProgramProductUpdateEvent


```
Content-Type: application/json
{
    "correlationId": "5030abc8-0b95",
    "description": "LoyaltyProgramProductUpdateEvent illustration",
    "domain": "Commercial",
    "eventId": "4e5c-a999-008f4029d6d9",
    "eventTime": "2022-08-25T12:21:50.030Z",
    "eventType": "LoyaltyProgramProductUpdateEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:21:49.819Z",
    "title": "LoyaltyProgramProductUpdateEvent",
    "event": {
        "loyaltyProgramProduct": {
            "id": "55",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/55",
            "@type": "LoyaltyProgramProduct",
            "relatedParty": [
                {
                    "role": "administrator",
                    "@type": "RelatedPartyOrPartyRole",
```


© TM Forum 2025. All Rights Reserved.

```
                    "partyOrPartyRole": {
                        "id": "777",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/777",
                        "name": "John Smith",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                    }
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductUpdateEvent"
}
```

Create Event

Message example for LoyaltyProgramProductCreateEvent

```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyProgramProductCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
    "eventType": "LoyaltyProgramProductCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyProgramProductCreateEvent",
    "event": {
        "loyaltyProgramProduct": {
            "id": "1765",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/1765",
            "@type": "LoyaltyProgramProduct",
            "@baseType": "Product",
            "creationDate": "2024-05-01T20:26:25.122Z",
            "description": "Program enrolment for Inner City transportation
incentives",
            "isBundle": true,
            "isCustomerVisible": true,
            "name": " Inner City transportation incentives enrolment",
            "orderDate": "2024-05-01T20:26:25.122Z",
            "productSerialNumber": "LOYALTY-14434394",
            "productTerm": [
                {
```

© TM Forum 2025. All Rights Reserved.

```
                    "@type": "ProductTerm",
                    "description": "Loyalty enrolment subscription term",
                    "duration": {
                        "amount": 24,
                        "units": "Month"
                    },
                    "validFor": {
                        "startDateTime": "2024-04-12T23:20:50.52Z",
                        "endDateTime": "2026-04-12T23:20:50.52Z"
                    },
                    "name": "24 month enrolment"
                }
            ],
            "startDate": "2024-04-12T20:26:25.122Z",
            "status": "created",
            "terminationDate": "2026-04-12T20:26:25.122Z",
            "productCharacteristic": [
                {
                    "name": "Bicycle Model",
                    "valueType": "string",
                    "value": "Trek Domane",
                    "@type": "StringCharacteristic"
                },
                {
                    "name": "Color",
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristic"
                }
            ],
            "productOffering": {
                "id": "7121",
                "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
                "@type": "ProductOfferingRef",
                "name": "Inner City Transportation Incentive Program",
                "@referredType": "ProductOffering"
            },
            "productSpecification": {
                "id": "100",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
                "@type": "ProductSpecificationRef",
                "name": "Inner City Transportation Incentive Program",
                "@referredType": "LoyaltyProgramProductSpecification"
            },
            "loyaltyProgramMember": {
                "id": "9664",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
                "@type": "LoyaltyProgramMemberRef",
                "name": "John Doe",
                "@referredType": "LoyaltyProgramMember"
            },
            "loyaltyAccount": {
                "id": "1155",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
                "@type": "LoyaltyAccountRef",
                "name": "John Doe's loyalty account",
                "@referredType": "LoyaltyAccount"
            }
        }
```


© TM Forum 2025. All Rights Reserved.

```
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyProgramProductCreateEvent",
    "@baseType": "Event"
}
```


Delete Event

Message example for LoyaltyProgramProductDeleteEvent


```
Content-Type: application/json
{
    "correlationId": "17eabb55-f69c",
    "description": "LoyaltyProgramProductDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "46c6-86cd-47b3fa4d39e3",
    "eventTime": "2022-08-25T12:21:50.052Z",
    "eventType": "LoyaltyProgramProductDeleteEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:47.370Z",
    "title": "LoyaltyProgramProductDeleteEvent",
    "event": {
        "loyaltyProgramProduct": {
            "id": "55",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/55",
            "@type": "LoyaltyProgramProduct"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductDeleteEvent"
}
```


© TM Forum 2025. All Rights Reserved.

State Change Event

Message example for LoyaltyProgramProductStateChangeEvent

```
Content-Type: application/json
{
    "correlationId": "9bd57948-24b6",
    "description": "LoyaltyProgramProductStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d28-8a7a-cf4c696bc7ec",
    "eventTime": "2022-08-25T12:21:50.041Z",
    "eventType": "LoyaltyProgramProductStateChangeEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:43.703Z",
    "title": "LoyaltyProgramProductStateChangeEvent",
    "event": {
        "loyaltyProgramProduct": {
            "id": "55",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/55",
            "@type": "LoyaltyProgramProduct",
            "status": "suspended"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramProductStateChangeEvent"
}
```

LoyaltyAccount

Update Event

Message example for LoyaltyAccountUpdateEvent

```
Content-Type: application/json
{
    "correlationId": "5030abc8-0b95",
    "description": "LoyaltyAccountUpdateEvent illustration",
    "domain": "Commercial",
    "eventId": "4e5c-a999-008f4029d6d9",
    "eventTime": "2022-08-25T12:21:50.030Z",
    "eventType": "LoyaltyAccountUpdateEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:21:49.819Z",
    "title": "LoyaltyAccountUpdateEvent",
    "event": {
```

© TM Forum 2025. All Rights Reserved.

```
        "loyaltyAccount": {
            "id": "55",
            "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyAccount/55",
            "@type": "LoyaltyAccount",
            "relatedParty": [
                {
                    "role": "administrator",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                        "id": "777",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/777",
                        "name": "John Smith",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                    }
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyAccountUpdateEvent"
}
```


Create Event

Message example for LoyaltyAccountCreateEvent


```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyAccountCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
    "eventType": "LoyaltyAccountCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyAccountCreateEvent",
    "event": {
        "loyaltyAccount": {
            "id": "501",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/lovaltyAccount/501",
            "description": "Loyalty account for loyalty program member",
            "lastUpdate": "2024-05-01T20:30:52.370Z",
            "name": "Loyalty Account",
            "state": "Active",
```


© TM Forum 2025. All Rights Reserved.

```
            "accountType": "Loyalty Account",
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "Loyalty Program Member",
                    "partyOrPartyRole": {
                        "id": "3035235555",
                        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                        "@type": "PartyRoleRef",
                        "name": "Joseph",
                        "@referredType": "LoyaltyProgramMember"
                    }
                }
            ],
            "accountBalance": [
                {
                    "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
                    "@type": "AccountBalance",
                    "amount": {
                        "value": 20.0,
                        "unit": "POINTS"
                    },
                    "balanceType": "LoyaltyEarningsWallet",
                    "validFor": {
                        "startDateTime": "2024-04-12T23:20:50.52Z",
                        "endDateTime": "2026-04-12T23:20:50.52Z"
                    }
                },
                {
                    "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
                    "@type": "AccountBalance",
                    "amount": {
                        "value": 0.0,
                        "unit": "POINTS"
                    },
                    "balanceType": "LoyaltyBankedWallet",
                    "validFor": {
                        "startDateTime": "2024-04-12T23:20:50.52Z",
                        "endDateTime": "2026-04-12T23:20:50.52Z"
                    }
                }
            ],
            "@type": "LoyaltyAccount",
            "@baseType": "PartyAccount"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyAccountCreateEvent",
    "@baseType": "Event"
}
```


© TM Forum 2025. All Rights Reserved.

Delete Event

Message example for LoyaltyAccountDeleteEvent

```
Content-Type: application/json
{
    "correlationId": "17eabb55-f69c",
    "description": "LoyaltyAccountDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "46c6-86cd-47b3fa4d39e3",
    "eventTime": "2022-08-25T12:21:50.052Z",
    "eventType": "LoyaltyAccountDeleteEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:47.370Z",
    "title": "LoyaltyAccountDeleteEvent",
    "event": {
        "loyaltyAccount": {
            "id": "55",
            "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyAccount/55",
            "@type": "LoyaltyAccount"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyAccountDeleteEvent"
}
```

State Change Event

Message example for LoyaltyAccountStateChangeEvent

```
Content-Type: application/json
{
    "correlationId": "9bd57948-24b6",
    "description": "LoyaltyAccountStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d28-8a7a-cf4c696bc7ec",
    "eventTime": "2022-08-25T12:21:50.041Z",
    "eventType": "LoyaltyAccountStateChangeEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:43.703Z",
    "title": "LoyaltyAccountStateChangeEvent",
    "event": {
        "loyaltyAccount": {
            "id": "55",
            "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyAccount/55",
            "@type": "LoyaltyAccount",
```

© TM Forum 2025. All Rights Reserved.

```
            "state": "confirmed"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyAccountStateChangeEvent"
}
```

LoyaltyProgramPartner

Update Event

Message example for LoyaltyProgramPartnerUpdateEvent

```
Content-Type: application/json
{
    "correlationId": "5030abc8-0b95",
    "description": "LoyaltyProgramPartnerUpdateEvent illustration",
    "domain": "Commercial",
    "eventId": "4e5c-a999-008f4029d6d9",
    "eventTime": "2022-08-25T12:21:50.030Z",
    "eventType": "LoyaltyProgramPartnerUpdateEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:21:49.819Z",
    "title": "LoyaltyProgramPartnerUpdateEvent",
    "event": {
        "loyaltyProgramPartner": {
            "id": "1",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramPartner/1",
            "@type": "LoyaltyProgramPartner",
            "relatedParty": [
                {
                    "role": "administrator",
                    "@type": "RelatedPartyOrPartyRole",
                    "partyOrPartyRole": {
                        "id": "777",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/777",
                        "name": "John Smith",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                    }
                }
            ]
        }
    },
    "reportingSystem": {
```

© TM Forum 2025. All Rights Reserved.

```
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramPartnerUpdateEvent"
}
```


Create Event

Message example for LoyaltyProgramPartnerCreateEvent


```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyProgramPartnerCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
    "eventType": "LoyaltyProgramPartnerCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyProgramPartnerCreateEvent",
    "event": {
        "loyaltyProgramPartner": {
            "id": "1",
            "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
            "@type": "LoyaltyProgramPartner",
            "@baseType": "BusinessPartner",
            "name": "Lifestyle Market",
            "status": "active",
            "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "engagedParty": {
                "id": "1131",
                "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
                "@type": "PartyRef",
                "name": "Leisure Market",
                "@referredType": "Organization"
            },
            "contactMedium": [
                {
                    "preferred": true,
                    "validFor": {
                        "startDateTime": "2025-01-15T07:49:25.246Z"
                    },
                    "@type": "EmailContactMedium",
                    "emailAddress": "settlements@lifestylemarket.com"
```


© TM Forum 2025. All Rights Reserved.

```
                },
                {
                    "preferred": false,
                    "validFor": {
                        "startDateTime": "2023-01-15T07:49:25.246Z"
                    },
                    "@type": "GeographicAddressContactMedium",
                    "city": "Morristown",
                    "country": "USA",
                    "postCode": "07960",
                    "stateOrProvince": "New Jersey",
                    "street1": "777 Headquarters Plazza",
                    "street2": "West Tower - 12th Floor"
                }
            ],
            "account": [
                {
                    "id": "7723",
                    "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
                    "@type": "AccountRef",
                    "name": "Lifestyle Market's Settlement Account",
                    "@referredType": "SettlementAccount"
                }
            ],
            "agreement": [
                {
                    "id": "442",
                    "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
                    "@type": "AgreementRef",
                    "name": "Lifestyle Market's Partnership Agreement"
                }
            ],
            "characteristic": [
                {
                    "name": "Partner sector",
                    "value": "Entertainment",
                    "@type": "StringCharacteristic"
                }
            ],
            "partnerCatalogCategory": [
                {
                    "id": "35",
                    "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
                    "@type": "CategoryRef",
                    "name": "Leisure Goods",
                    "version": "1.1"
                }
            ],
            "partnerCatalogOffer": [
                {
                    "id": "6654",
                    "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
                    "@type": "ProductOfferingRef",
                    "name": "World Cup Football",
                    "version": "1.1"
                },
                {
                    "id": "6712",
                    "href": "https://myscp.com:8080/tmf-
```


© TM Forum 2025. All Rights Reserved.

```
api/productCatalogManagement/v5/productOffering/6712",
                    "@type": "ProductOfferingRef",
                    "name": "Tennis Racket",
                    "version": "1.0"
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyProgramPartnerCreateEvent",
    "@baseType": "Event"
}
```

Delete Event

Message example for LoyaltyProgramPartnerDeleteEvent

```
Content-Type: application/json
{
    "correlationId": "17eabb55-f69c",
    "description": "LoyaltyProgramPartnerDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "46c6-86cd-47b3fa4d39e3",
    "eventTime": "2025-01-25T12:21:50.052Z",
    "eventType": "LoyaltyProgramPartnerDeleteEvent",
    "priority": "2",
    "timeOccurred": "2025-01-25T12:21:47.370Z",
    "title": "LoyaltyProgramPartnerDeleteEvent",
    "event": {
        "loyaltyProgramPartner": {
            "id": "1",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramPartner/1",
            "@type": "LoyaltyProgramPartner"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
```

© TM Forum 2025. All Rights Reserved.

```
    "@type": "LoyaltyProgramPartnerDeleteEvent"
}
```

State Change Event

Message example for LoyaltyProgramPartnerStateChangeEvent

```
Content-Type: application/json
{
    "correlationId": "9bd57948-24b6",
    "description": "LoyaltyProgramPartnerStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d28-8a7a-cf4c696bc7ec",
    "eventTime": "2025-01-25T12:21:50.041Z",
    "eventType": "LoyaltyProgramPartnerStateChangeEvent",
    "priority": "2",
    "timeOccurred": "2025-01-25T12:21:43.703Z",
    "title": "LoyaltyProgramPartnerStateChangeEvent",
    "event": {
        "loyaltyProgramPartner": {
            "id": "1",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramPartner/1",
            "@type": "LoyaltyProgramPartner",
            "status": "validated"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyProgramPartnerStateChangeEvent"
}
```

LoyaltyEarn

Create Event

Message example for LoyaltyEarnCreateEvent

```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyEarnCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
```

© TM Forum 2025. All Rights Reserved.

```
    "eventType": "LoyaltyEarnCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyEarnCreateEvent",
    "event": {
        "loyaltyEarn": {
            "id": "6651",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyEarn/6651",
            "@type": "LoyaltyEarn",
            "@baseType": "ProductUsage",
            "usageDate": "2024-04-21T09:13:16-07:00",
            "description": "Partner brand interaction earning",
            "status": "received",
            "usageType": "Partner interaction",
            "balanceType": "LoyaltyEarningsWallet",
            "quantity": {
                "value": 25,
                "unit": "POINTS"
            },
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "Business Partner",
                    "partyOrPartyRole": {
                        "id": "8121",
                        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/8121",
                        "@type": "PartyRoleRef",
                        "name": "Lifestyle Goods",
                        "@referredType": "LoyaltyProgramPartner"
                    }
                }
            ],
            "loyaltyProgramMember": {
                "id": "9664",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
                "@type": "LoyaltyProgramMemberRef",
                "name": "John Doe",
                "@referredType": "LoyaltyProgramMember"
            },
            "loyaltyProgramProduct": {
                "id": "1765",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/1765",
                "@type": "LoyaltyProgramProductRef",
                "name": "John Doe's Loyalty Program",
                "@referredType": "LoyaltyProgramProduct"
            },
            "loyaltyAccount": {
                "id": "1155",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
                "@type": "LoyaltyAccountRef",
                "name": "John Doe's Loyalty Account",
                "@referredType": "LoyaltyAccount"
            }
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
```


© TM Forum 2025. All Rights Reserved.

```
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyEarnCreateEvent",
    "@baseType": "Event"
}
```


State Change Event

Message example for LoyaltyEarnStateChangeEvent


```
Content-Type: application/json
{
    "correlationId": "9bd57948-24b6",
    "description": "LoyaltyEarnStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d28-8a7a-cf4c696bc7ec",
    "eventTime": "2022-08-25T12:21:50.041Z",
    "eventType": "LoyaltyEarnStateChangeEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:43.703Z",
    "title": "LoyaltyEarnStateChangeEvent",
    "event": {
        "loyaltyEarn": {
            "id": "55",
            "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyEarn/55",
            "status": "received",
            "@type": "LoyaltyEarn"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyEarnStateChangeEvent"
}
```


LoyaltyBurn

Create Event

Message example for LoyaltyBurnCreateEvent

© TM Forum 2025. All Rights Reserved.

```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyBurnCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
    "eventType": "LoyaltyBurnCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyBurnCreateEvent",
    "event": {
        "loyaltyBurn": {
            "id": "542F9866-97E2-4841-AC60-804D0A1CA810",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyBurn/542F9866-97E2-4841-AC60-804D0A1CA810",
            "@type": "LoyaltyBurn",
            "@baseType": "Payment",
            "correlatorId": "542F9866-97E2-4841-AC60-804D0A1CA810",
            "paymentDate": "2024-05-01T20:39:38.832Z",
            "name": "Part payment for partner catalog purchases",
            "description": "Payment in full or in part with loyalty points / virtual
currency earnings",
            "authorizationCode": "542F9866-97E2-4841-AC60-804D0A1CA810",
            "status": "received",
            "statusDate": "2024-05-01T20:39:38.832Z",
            "quantity": {
                "value": 2000,
                "unit": "POINTS"
            },
            "taxAmount": {
                "unit": "USD",
                "value": 0
            },
            "channel": {
                "id": "45",
                "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/45",
                "@type": "ChannelRef",
                "name": "Online purchase channels",
                "@referredType": "SalesChannel"
            },
            "payer": {
                "@type": "RelatedPartyOrPartyRole",
                "role": "Loyalty Program Member",
                "partyOrPartyRole": {
                    "id": "9664",
                    "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
                    "@type": "PartyRoleRef",
                    "name": "John Doe",
                    "@referredType": "LoyaltyProgramMember"
                }
            },
            "paymentMethod": {
                "@type": "LoyaltyPaymentMethod",
                "@baseType": "PaymentMethod",
                "name": "Joseph's Loyalty Points",
                "status": "Active",
                "statusDate": "2024-11-21T11:21:11.445Z"
            },
            "paymentItem": [
```

© TM Forum 2025. All Rights Reserved.

```
                {
                    "id": "1",
                    "amount": {
                        "value": 450.0,
                        "unit": "USD"
                    },
                    "item": {
                        "id": "1201",
                        "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/1201",
                        "name": "Smart Phone",
                        "@type": "EntityRef",
                        "@referredType": "ProductOrder"
                    },
                    "taxAmount": {
                        "value": 0.0,
                        "unit": "USD"
                    },
                    "totalAmount": {
                        "value": 450.0,
                        "unit": "USD"
                    },
                    "@type": "PaymentItem"
                }
            ],
            "totalAmount": {
                "value": 450.0,
                "unit": "USD"
            },
            "loyaltyProgramMember": {
                "id": "9664",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
                "@type": "LoyaltyProgramMemberRef",
                "name": "John Doe",
                "@referredType": "LoyaltyProgramMember"
            },
            "loyaltyProgramProduct": {
                "id": "1745",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/1745",
                "@type": "LoyaltyProgramProductRef",
                "name": "John Doe's Loyalty Program",
                "@referredType": "LoyaltyProgramProduct"
            },
            "account": {
                "id": "1155",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
                "@type": "AccountRef",
                "name": "John Doe's Loyalty Account",
                "@referredType": "LoyaltyAccount"
            }
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
```


© TM Forum 2025. All Rights Reserved.

```
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyBurnCreateEvent",
    "@baseType": "Event"
}
```


State Change Event

Message example for LoyaltyBurnStateChangeEvent


```
Content-Type: application/json
{
    "correlationId": "9bd57948-24b6",
    "description": "LoyaltyBurnStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d28-8a7a-cf4c696bc7ec",
    "eventTime": "2022-08-25T12:21:50.041Z",
    "eventType": "LoyaltyBurnStateChangeEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:43.703Z",
    "title": "LoyaltyBurnStateChangeEvent",
    "event": {
        "loyaltyBurn": {
            "id": "55",
            "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyBurn/55",
            "status": "confirmed",
            "@type": "LoyaltyBurn"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyBurnStateChangeEvent"
}
```


LoyaltyExecutionPoint

Update Event

Message example for LoyaltyExecutionPointUpdateEvent


```
Content-Type: application/json
{
    "correlationId": "5030abc8-0b95",
```


© TM Forum 2025. All Rights Reserved.

```
    "description": "LoyaltyExectionPointUpdateEvent illustration",
    "domain": "Commercial",
    "eventId": "4e5c-a999-008f4029d6d9",
    "eventTime": "2022-08-25T12:21:50.030Z",
    "eventType": "LoyaltyExectionPointUpdateEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:21:49.819Z",
    "title": "LoyaltyExectionPointUpdateEvent",
    "event": {
        "loyaltyExecutionPoint": {
            "id": "55",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyExectionPoint/55",
            "@type": "LoyaltyExecutionPoint",
            "loyaltyProgramMember":
                {
                        "id": "777",
                        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/777",
                        "name": "John Smith",
                        "@type": "LoyaltyProgramMemberRef",
                        "@referredType": "LoyaltyProgramMember"
                }
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "LoyaltyExecutionPointUpdateEvent"
}
```


Create Event

Message example for LoyaltyExecutionPointCreateEvent


```
Content-Type: application/json
{
    "correlationId": "c78a9aa0-d7e4",
    "description": "LoyaltyExectionPointCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4fc8-a266-fa6506329869",
    "eventTime": "2022-08-25T12:21:50.000Z",
    "eventType": "LoyaltyExectionPointCreateEvent",
    "priority": "4",
    "timeOccurred": "2022-08-25T12:21:44.821Z",
    "title": "LoyaltyExectionPointCreateEvent",
    "event": {
        "loyaltyExecutionPoint": {
            "id": "7756",
            "href": "https://mycsp.com:8080/tmf-
```


© TM Forum 2025. All Rights Reserved.

```
api/loyaltyManagement/v5/loyaltyExecutionPoint/7756",
            "@type": "LoyaltyExecutionPoint",
            "name": "John Doe's Birthday Gift Loyalty Execution Point",
            "triggers": "CustomerOrder",
            "loyaltyProgramMember": {
                "id": "9664",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
                "@type": "LoyaltyProgramMemberRef",
                "name": "John Doe",
                "@referredType": "LoyaltyProgramMember"
            },
            "loyaltyProgramProduct": {
                "id": "1745",
                "href": "https://mycsp.com:8080/tmf-
api/loyalty/v5/loyaltyProgramProduct/1745",
                "@type": "LoyaltyProgramProductRef",
                "name": "John Doe's Loyalty Program",
                "@referredType": "LoyaltyProgramProduct"
            },
            "body": {
                "category": "Birthday Gift product order",
                "priority": "1",
                "channel": [
                    {
                        "role": "Used channel for order capture",
                        "@type": "RelatedChannel",
                        "channel": {
                            "id": "1",
                            "name": "Online channel",
                            "@type": "ChannelRef"
                        }
                    }
                ],
                "productOrderItem": [
                    {
                        "id": "100",
                        "quantity": 1,
                        "action": "add",
                        "productOffering": {
                            "id": "14277",
                            "href":
"https://host:port/productCatalogManagement/v5/productOffering/14277",
                            "name": "TMF25",
                            "@type": "ProductOfferingRef"
                        },
                        "@type": "ProductOrderItem"
                    }
                ],
                "relatedParty": [
                    {
                        "role": "Customer",
                        "partyOrPartyRole": {
                            "id": "{{customer.id}}",
                            "href":
"https://host:port/customerManagement/v5/customer/{{customer.id}}",
                            "name": "{{customer.name}}",
                            "@type": "PartyRoleRef",
                            "@referredType": "Customer"
                        },
                        "@type": "RelatedPartyRefOrPartyRoleRef"
                    }
                ],
```


© TM Forum 2025. All Rights Reserved.

```
                "@type": "ProductOrder"
            },
            "version": "1.0",
            "action": "POST",
            "endpoint": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder",
            "creationDate": "2024-05-01T20:36:13.313Z"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "LoyaltyExecutionPointCreateEvent",
    "@baseType": "Event"
}
```


Delete Event

Message example for LoyaltyExecutionPointDeleteEvent


```
Content-Type: application/json
{
    "correlationId": "17eabb55-f69c",
    "description": "LoyaltyExectionPointDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "46c6-86cd-47b3fa4d39e3",
    "eventTime": "2022-08-25T12:21:50.052Z",
    "eventType": "LoyaltyExectionPointDeleteEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:21:47.370Z",
    "title": "LoyaltyExecutionPointDeleteEvent",
    "event": {
        "loyaltyExecutionPoint": {
            "id": "55",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyExectionPoint/55",
            "@type": "LoyaltyExecutionPoint"
        }
    },
    "reportingSystem": {
        "id": "426",
        "name": "APP-49",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "487",
        "name": "APP-329",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
```


© TM Forum 2025. All Rights Reserved.

```
    "@baseType": "Event",
    "@type": "LoyaltyExecutionPointDeleteEvent"
}
```

© TM Forum 2025. All Rights Reserved.

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Complete Update of an Entity	PUT Resource	PUT must be used to completely update a resource identified by its resource URI
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on LoyaltyProgramProductSpecification

List or find LoyaltyProgramProductSpecification objects

`GET /loyaltyProgramProductSpecification?fields=…`

Description

This operation list LoyaltyProgramProductSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Program Product Spec GET operation result

Request

```
GET /loyaltyProgramProductSpecification?lifecycleStatus=Active
Content-Type: application/json
```

Response

© TM Forum 2025. All Rights Reserved.

```
200
[
    {
        "id": "1151",
        "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
        "@type": "LoyaltyProgramProductSpecification",
        "@baseType": "ProductSpecification",
        "brand": "Nice",
        "description": "This is the program for the rewarding of citizens making use
of inner city public transportation services.",
        "isBundle": false,
        "productNumber": "Nice-CC-01",
        "validFor": {
            "startDateTime": "2024-01-01T00:00:00Z",
            "endDateTime": "2028-12-31T23:59:59Z"
        },
        "version": "1.0",
        "lastUpdate": "2023-12-01T12:01:01Z",
        "lifecycleStatus": "Active",
        "name": "City of Nice Connected Citizens Rewards Programme",
        "externalIdentifier": [
            {
                "id": "7761",
                "@type": "ExternalIdentifier",
                "owner": "MagentoCommerce",
                "externalIdentifierType": "ProductSpecification"
            }
        ],
        "needsLoyaltyAccount": true,
        "productSpecCharacteristic": [
            {
                "name": "Bicycle Model",
                "description": "The bicycle model that the citizen rents",
                "valueType": "string",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1,
                "isUnique": false,
                "@type": "CharacteristicSpecification",
                "characteristicValueSpecification": [
                    {
                        "isDefault": true,
                        "validFor": {
                            "startDateTime": "2024-09-23T00:00:00Z",
                            "endDateTime": "2028-11-24T00:00:00Z"
                        },
                        "value": "Trek Domane",
                        "@type": "StringCharacteristicValueSpecification"
                    },
                    {
                        "isDefault": false,
                        "validFor": {
                            "startDateTime": "2022-09-23T00:00:00Z",
                            "endDateTime": "2026-11-24T00:00:00Z"
                        },
                        "value": "Giant Trance",
                        "@type": "StringCharacteristicValueSpecification"
                    },
                    {
                        "isDefault": false,
                        "validFor": {
```


© TM Forum 2025. All Rights Reserved.

```
                            "startDateTime": "2025-09-23T00:00:00Z",
                            "endDateTime": "2030-11-24T00:00:00Z"
                        },
                        "value": "Yeti SB130",
                        "@type": "StringCharacteristicValueSpecification"
                    }
                ],
                "validFor": {
                    "startDateTime": "2024-09-23T16:42:23Z"
                }
            },
            {
                "name": "Color",
                "description": "Color of the bicycle",
                "valueType": "string",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1,
                "extensible": true,
                "isUnique": true,
                "@type": "CharacteristicSpecification",
                "characteristicValueSpecification": [
                    {
                        "isDefault": true,
                        "valueType": "string",
                        "value": "Black",
                        "@type": "StringCharacteristicValueSpecification"
                    },
                    {
                        "isDefault": false,
                        "valueType": "string",
                        "value": "White",
                        "@type": "StringCharacteristicValueSpecification"
                    }
                ],
                "validFor": {
                    "startDateTime": "2024-09-23T16:42:23Z"
                }
            }
        ],
        "policy": [
            {
                "id": "5565",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
                "@type": "PolicyRef",
                "name": "CityOfNiceRewardsRule1",
                "@referredType": "PolicyRule",
                "version": "1.0"
            }
        ]
    },
    {
        "id": "1712",
        "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1712",
        "@type": "LoyaltyProgramProductSpecification",
        "@baseType": "ProductSpecification",
        "brand": "Telco1",
        "description": "This is the program for the rewarding of prepaid subscriber
behavioural changes.",
        "isBundle": false,
        "productNumber": "Telco-Rewards-01",
```


© TM Forum 2025. All Rights Reserved.

```
        "validFor": {
            "startDateTime": "2024-01-01T00:00:00Z",
            "endDateTime": "2028-12-31T23:59:59Z"
        },
        "version": "1.0",
        "lastUpdate": "2023-12-01T12:01:01Z",
        "lifecycleStatus": "Active",
        "name": "City of Nice Connected Citizens Rewards Programme",
        "externalIdentifier": [
            {
                "id": "7761",
                "@type": "ExternalIdentifier",
                "owner": "MagentoCommerce",
                "externalIdentifierType": "ProductSpecification"
            }
        ],
        "needsLoyaltyAccount": true,
        "productSpecCharacteristic": [
            {
                "name": "Bicycle Model",
                "description": "The bicycle model that the citizen rents",
                "valueType": "string",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1,
                "isUnique": false,
                "@type": "CharacteristicSpecification",
                "characteristicValueSpecification": [
                    {
                        "isDefault": true,
                        "validFor": {
                            "startDateTime": "2024-09-23T00:00:00Z",
                            "endDateTime": "2028-11-24T00:00:00Z"
                        },
                        "value": "Trek Domane",
                        "@type": "StringCharacteristicValueSpecification"
                    },
                    {
                        "isDefault": false,
                        "validFor": {
                            "startDateTime": "2022-09-23T00:00:00Z",
                            "endDateTime": "2026-11-24T00:00:00Z"
                        },
                        "value": "Giant Trance",
                        "@type": "StringCharacteristicValueSpecification"
                    },
                    {
                        "isDefault": false,
                        "validFor": {
                            "startDateTime": "2025-09-23T00:00:00Z",
                            "endDateTime": "2030-11-24T00:00:00Z"
                        },
                        "value": "Yeti SB130",
                        "@type": "StringCharacteristicValueSpecification"
                    }
                ],
                "validFor": {
                    "startDateTime": "2024-09-23T16:42:23Z"
                }
            },
            {
                "name": "Color",
                "description": "Color of the bicycle",
```


© TM Forum 2025. All Rights Reserved.

```
                "valueType": "string",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1,
                "extensible": true,
                "isUnique": true,
                "@type": "CharacteristicSpecification",
                "characteristicValueSpecification": [
                    {
                        "isDefault": true,
                        "valueType": "string",
                        "value": "Black",
                        "@type": "StringCharacteristicValueSpecification"
                    },
                    {
                        "isDefault": false,
                        "valueType": "string",
                        "value": "White",
                        "@type": "StringCharacteristicValueSpecification"
                    }
                ],
                "validFor": {
                    "startDateTime": "2024-09-23T16:42:23Z"
                }
            }
        ],
        "policy": [
            {
                "id": "6121",
                "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/6121",
                "@type": "PolicyRef",
                "name": "VoucherRechargeBehaviourRule",
                "@referredType": "PolicyRule",
                "version": "1.0"
            }
        ]
    }
]
```


Retrieves a LoyaltyProgramProductSpecification by ID

`GET /loyaltyProgramProductSpecification/{id}?fields=…`

Description

This operation retrieves a LoyaltyProgramProductSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Program Product Spec GET operation result

Request

```
GET /loyaltyProgramProductSpecification/1151
```

© TM Forum 2025. All Rights Reserved.

```
Content-Type: application/json
```

Response

```
200
{
    "id": "1151",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "Nice-CC-01",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "1.0",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
```

© TM Forum 2025. All Rights Reserved.

```
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```


© TM Forum 2025. All Rights Reserved.

Creates a LoyaltyProgramProductSpecification

POST /loyaltyProgramProductSpecification?fields=…

Description

This operation creates a LoyaltyProgramProductSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
name	
policy	M
@type	M

Usage samples

This is an example of creating a Loyalty Program Product Spec - UC1 Inner City Spec

Request

```
POST /loyaltyProgramProductSpecification
Content-Type: application/json
{
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "Nice-CC-01",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "1.0",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
```

© TM Forum 2025. All Rights Reserved.

```
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
```


© TM Forum 2025. All Rights Reserved.

```
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```

Response

```
201
{
    "id": "1151",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "Nice-CC-01",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "1.0",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
```

© TM Forum 2025. All Rights Reserved.

```
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
```


© TM Forum 2025. All Rights Reserved.

```
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```

This is an example of creating a Loyalty Program Product Spec - UC2 Prepaid Spec

Request

```
POST /loyaltyProgramProductSpecification
Content-Type: application/json
{
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Telco1",
    "description": "This is the program for the rewarding of prepaid subscriber
behavioural changes.",
    "isBundle": false,
    "productNumber": "Telco-Rewards-01",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "1.0",
    "lifecycleStatus": "Active",
    "name": "Prepaid Topup Off-peak Reward",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Topup Period",
            "description": "The time of the day for rewarding topup",
            "valueType": "object",
            "configurable": false,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "value": {
                        "startTime": "06:00:00",
                        "endTime": "08:00:00"
                    },
                    "@type": "ObjectCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
```

© TM Forum 2025. All Rights Reserved.

```
            "id": "6121",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/6121",
            "@type": "PolicyRef",
            "name": "VoucherRechargeBehaviourRule",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```

Response

```
201
{
    "id": "1712",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1712",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Telco1",
    "description": "This is the program for the rewarding of prepaid subscriber
behavioural changes.",
    "isBundle": false,
    "productNumber": "Telco-Rewards-01",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "1.0",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "Prepaid Topup Off-peak Reward",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Topup Period",
            "description": "The time of the day for rewarding topup",
            "valueType": "object",
            "configurable": false,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "value": {
                        "startTime": "06:00:00",
                        "endTime": "08:00:00"
                    },
                    "@type": "ObjectCharacteristicValueSpecification"
                }
            ],
            "validFor": {
```

© TM Forum 2025. All Rights Reserved.

```
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "6121",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/6121",
            "@type": "PolicyRef",
            "name": "VoucherRechargeBehaviourRule",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```


Updates partially a LoyaltyProgramProductSpecification

`PATCH /loyaltyProgramProductSpecification/{id}?fields=…`

Description

This operation allows partial updates of a LoyaltyProgramProductSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
lastUpdate	
needsLoyaltyAccount	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
brand	
bundledProductSpecification	
category	
description	

© TM Forum 2025. All Rights Reserved.

Patchable Attributes	Rule
eventTopic	
externalIdentifier	
intentSpecification	
isBundle	
lifecycleStatus	
name	
policy	
productNumber	
productSpecCharacteristic	
productSpecificationRelationship	
relatedParty	
resourceSpecification	
serviceSpecification	
targetProductSchema	
validFor	
version	

Usage samples

Here is an example of a request for patching a specification. In this example, a new lifecycle status and version is set.

Request

```
PATCH /loyaltyProgramProductSpecification/1151
Content-Type: application/merge-patch+json
{
    "productNumber": "loyalty-123",
    "lifecycleStatus": "Active",
    "version": "2.0",
    "@type": "LoyaltyProgramProductSpecification"
}
```

Response

```
200
{
    "id": "1151",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
```

© TM Forum 2025. All Rights Reserved.

```
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "loyalty-123",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "2.0",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
```


© TM Forum 2025. All Rights Reserved.

```
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```

Here is an example of a request for patching a specification. In this example, a new version is set, together with validity period.

Request

```
PATCH /loyaltyProgramProductSpecification/1151
Content-Type: application/json
{
    "version": "3.1",
    "validFor": {
        "startDateTime": "2024-11-24T00:00:00Z",
        "endDateTime": "2027-11-24T00:00:00Z"
    },
    "@type": "LoyaltyProgramProductSpecification"
}
```

© TM Forum 2025. All Rights Reserved.

Response


```
200
{
    "id": "1151",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "loyalty-123",
    "validFor": {
        "startDateTime": "2024-11-24T00:00:00Z",
        "endDateTime": "2027-11-24T00:00:00Z"
    },
    "version": "3.1",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
```


© TM Forum 2025. All Rights Reserved.

```
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```


Update a Loyalty Program Product Specification using json-patch

Request

```
PATCH /loyaltyProgramProductSpecification/1151
```

© TM Forum 2025. All Rights Reserved.

```
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "/relatedParty",
    "value": {
        "role": "Technical Expert",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
            "id": "4321",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/partyRole/4321",
            "name": "Jean Jones",
            "@type": "PartyRefOrPartyRoleRef",
            "@referredType": "PartyRole"
        }
    }
  }
]
```

Response

```
200
{
    "id": "1151",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "loyalty-123",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "relatedParty": [
        {
            "role": "Technical Expert",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
                "id": "4321",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/partyRole/4321",
                "name": "Jean Jones",
                "@type": "PartyRefOrPartyRoleRef",
                "@referredType": "PartyRole"
            }
        }
    ],
    "version": "2.0",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
```

© TM Forum 2025. All Rights Reserved.

```
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "Black",
```


© TM Forum 2025. All Rights Reserved.

```
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```

Update a Loyalty Program Product Specification using json-patch query

Request

```
PATCH /loyaltyProgramProductSpecification/1151
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path":
"/productSpecCharacteristic/characteristicValueSpecification/isDefault?value=/produc
tSpecCharacteristic/characteristicValueSpecification/Black",
    "value": false
  },
  {
    "op": "replace",
    "path":
"/productSpecCharacteristic/characteristicValueSpecification/isDefault?value=/produc
tSpecCharacteristic/characteristicValueSpecification/White",
    "value": true
  }
]
```

Response

```
200
{
    "id": "1151",
    "href": "https://serverRoot/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/1151",
```

© TM Forum 2025. All Rights Reserved.

```
    "@type": "LoyaltyProgramProductSpecification",
    "@baseType": "ProductSpecification",
    "brand": "Nice",
    "description": "This is the program for the rewarding of citizens making use of
inner city public transportation services.",
    "isBundle": false,
    "productNumber": "loyalty-123",
    "validFor": {
        "startDateTime": "2024-01-01T00:00:00Z",
        "endDateTime": "2028-12-31T23:59:59Z"
    },
    "version": "2.0",
    "lastUpdate": "2023-12-01T12:01:01Z",
    "lifecycleStatus": "Active",
    "name": "City of Nice Connected Citizens Rewards Programme",
    "externalIdentifier": [
        {
            "id": "7761",
            "@type": "ExternalIdentifier",
            "owner": "MagentoCommerce",
            "externalIdentifierType": "ProductSpecification"
        }
    ],
    "needsLoyaltyAccount": true,
    "productSpecCharacteristic": [
        {
            "name": "Bicycle Model",
            "description": "The bicycle model that the citizen rents",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": false,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "validFor": {
                        "startDateTime": "2024-09-23T00:00:00Z",
                        "endDateTime": "2028-11-24T00:00:00Z"
                    },
                    "value": "Trek Domane",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2022-09-23T00:00:00Z",
                        "endDateTime": "2026-11-24T00:00:00Z"
                    },
                    "value": "Giant Trance",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": false,
                    "validFor": {
                        "startDateTime": "2025-09-23T00:00:00Z",
                        "endDateTime": "2030-11-24T00:00:00Z"
                    },
                    "value": "Yeti SB130",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
```


© TM Forum 2025. All Rights Reserved.

```
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        },
        {
            "name": "Color",
            "description": "Color of the bicycle",
            "valueType": "string",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "extensible": true,
            "isUnique": true,
            "@type": "CharacteristicSpecification",
            "characteristicValueSpecification": [
                {
                    "isDefault": false,
                    "valueType": "string",
                    "value": "Black",
                    "@type": "StringCharacteristicValueSpecification"
                },
                {
                    "isDefault": true,
                    "valueType": "string",
                    "value": "White",
                    "@type": "StringCharacteristicValueSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2024-09-23T16:42:23Z"
            }
        }
    ],
    "policy": [
        {
            "id": "5565",
            "href": "https://serverRoot/tmf-
api/policyManagement/v5/policyRule/5565",
            "@type": "PolicyRef",
            "name": "CityOfNiceRewardsRule1",
            "@referredType": "PolicyRule",
            "version": "1.0"
        }
    ]
}
```


Deletes a LoyaltyProgramProductSpecification

`DELETE /loyaltyProgramProductSpecification/{id}`

Description

This operation deletes a LoyaltyProgramProductSpecification entity.

Operations on LoyaltyProgramMember

© TM Forum 2025. All Rights Reserved.

List or find LoyaltyProgramMember objects

GET /loyaltyProgramMember?fields=…

Description

This operation list LoyaltyProgramMember entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example request for retrieving multiple Loyalty Program Member

Request

```
GET /loyaltyProgramMember
Content-Type: application/json
```

Response

```
200
[
    {
        "id": "9664",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
        "@type": "LoyaltyProgramMember",
        "@baseType": "PartyRole",
        "name": "John Doe",
        "description": "Loyalty Program Member Profile",
        "engagedParty": {
            "id": "7872",
            "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
            "@type": "PartyRef",
            "name": "John Doe",
            "@referredType": "Individual"
        },
        "account": [
            {
                "id": "1155",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
                "@type": "AccountRef",
                "name": "John Doe's Loyalty Account",
                "@referredType": "LoyaltyAccount"
            }
        ],
        "contactMedium": [
            {
                "preferred": true,
                "validFor": {
                    "startDateTime": "2025-01-15T07:49:25.246Z"
                },
                "@type": "EmailContactMedium",
                "emailAddress": "john.doe@gmail.com"
            },
```

© TM Forum 2025. All Rights Reserved.

```
            {
                "preferred": false,
                "validFor": {
                    "startDateTime": "2025-01-15T07:49:25.246Z"
                },
                "@type": "GeographicAddressContactMedium",
                "city": "Morristown",
                "country": "USA",
                "postCode": "07960",
                "stateOrProvince": "New Jersey",
                "street1": "240 Headquarters Plazza",
                "street2": "East Tower - 10th Floor"
            }
        ],
        "agreement": [
            {
                "id": "4721",
                "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
                "@type": "AgreementRef",
                "name": "John Doe's Loyalty Program Membership Agreement"
            }
        ],
        "characteristic": [
            {
                "name": "Last Active",
                "value": "2025-01-15T07:49:25.246Z",
                "@type": "StringCharacteristic"
            }
        ],
        "status": "active",
        "statusReason": "Loyalty Member Enrolment",
        "validFor": {
            "startDateTime": "2025-01-12T23:20:50.52Z",
            "endDateTime": "2030-04-12T23:20:50.52Z"
        },
        "tier": "Gold"
    },
    {
        "id": "6649",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/6649",
        "@type": "LoyaltyProgramMember",
        "@baseType": "PartyRole",
        "name": "Jane Doe",
        "description": "Loyalty Program Member Profile",
        "engagedParty": {
            "id": "441",
            "href": "https:/host:port/tmf-api/partyManagement/v2/individual/441",
            "@type": "PartyRef",
            "name": "Jane Doe",
            "@referredType": "Individual"
        },
        "account": [
            {
                "id": "525",
                "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/525",
                "@type": "AccountRef",
                "name": "Jane Doe's Loyalty Account",
                "@referredType": "LoyaltyAccount"
            }
        ],
```


© TM Forum 2025. All Rights Reserved.

```
        "contactMedium": [
            {
                "preferred": true,
                "validFor": {
                    "startDateTime": "2024-01-15T07:49:25.246Z"
                },
                "@type": "EmailContactMedium",
                "emailAddress": "jane.doe@gmail.com"
            },
            {
                "preferred": false,
                "validFor": {
                    "startDateTime": "2024-01-15T07:49:25.246Z"
                },
                "@type": "GeographicAddressContactMedium",
                "city": "Morristown",
                "country": "USA",
                "postCode": "07960",
                "stateOrProvince": "New Jersey",
                "street1": "240 Headquarters Plazza",
                "street2": "East Tower - 10th Floor"
            }
        ],
        "agreement": [
            {
                "id": "1474",
                "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/1474",
                "@type": "AgreementRef",
                "name": "Jane Doe's Loyalty Program Membership Agreement"
            }
        ],
        "characteristic": [
            {
                "name": "Last Active",
                "value": "2025-01-22T13:03:12.144Z",
                "@type": "StringCharacteristic"
            }
        ],
        "status": "active",
        "statusReason": "Loyalty Member Enrolment",
        "validFor": {
            "startDateTime": "2024-01-12T23:20:50.52Z",
            "endDateTime": "2030-04-12T23:20:50.52Z"
        },
        "tier": "Platinum"
    }
]
```


Retrieves a LoyaltyProgramMember by ID

`GET /loyaltyProgramMember/{id}?fields=…`

Description

This operation retrieves a LoyaltyProgramMember entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

© TM Forum 2025. All Rights Reserved.

Usage samples

This is an example request for retrieving a specific Loyalty Program Member

Request

```
GET /loyaltyProgramMember/9664
Content-Type: application/json
```

Response

```
200
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
```

© TM Forum 2025. All Rights Reserved.

```
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "active",
    "statusReason": "Loyalty Member Enrolment",
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
}
```

Creates a LoyaltyProgramMember

POST /loyaltyProgramMember?fields=…

Description

This operation creates a LoyaltyProgramMember entity.

Mandatory Attributes

Mandatory Attributes	Rule
engagedParty	M
name	M
@type	M

Usage samples

This is an example request for creating a Loyalty Program Member

Request

```
POST /loyaltyProgramMember
Content-Type: application/json
{
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
```

© TM Forum 2025. All Rights Reserved.

```
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "active",
    "statusReason": "Loyalty Member Enrolment",
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
```


© TM Forum 2025. All Rights Reserved.

```
}
```

Response

```
201
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
```

© TM Forum 2025. All Rights Reserved.

```
            }
        ],
        "characteristic": [
            {
                "name": "Last Active",
                "value": "2025-01-15T07:49:25.246Z",
                "@type": "StringCharacteristic"
            }
        ],
        "status": "active",
        "statusReason": "Loyalty Member Enrolment",
        "validFor": {
            "startDateTime": "2025-01-12T23:20:50.52Z",
            "endDateTime": "2030-04-12T23:20:50.52Z"
        },
        "tier": "Gold"
    }
```


Updates partially a LoyaltyProgramMember

`PATCH /loyaltyProgramMember/{id}?fields=…`

Description

This operation allows partial updates of a LoyaltyProgramMember entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
account	
agreement	
characteristic	
contactMedium	
creditProfile	
description	
engagedParty	
externalIdentifier	
href	
id	
name	
partyRoleSpecification	
paymentMethod	

© TM Forum 2025. All Rights Reserved.

Non Patchable Attributes	Rule
relatedParty	
role	
status	
statusReason	
tier	
validFor	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule

Usage samples

Here's an example of a request for updating a LoyaltyProgramMember resource - set status to validated. This example illustrates patch using application/json.

Request

```
PATCH /loyaltyProgramMember/9664
Content-Type: application/json
{
    "@type": "LoyaltyProgramMember",
    "status": "validated"
}
```

Response

```
200
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
```

© TM Forum 2025. All Rights Reserved.

```
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "validated",
    "statusReason": "Loyalty Member Enrolment",
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
}
```

Here's an example of a request for updating a LoyaltyProgramMember resource - set status to validated. This example illustrates patch using application/merge-patch+json.

Request

```
PATCH /loyaltyProgramMember/9664
```

© TM Forum 2025. All Rights Reserved.

```
Content-Type: application/merge-patch+json
{
    "@type": "LoyaltyProgramMember",
    "status": "validated"
}
```

Response

```
200
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
```

© TM Forum 2025. All Rights Reserved.

```
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "validated",
    "statusReason": "Loyalty Member Enrolment",
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
}
```

Here's an example of a request for updating a LoyaltyProgramMember resource - set status to validated. This example illustrates patch using application/json-patch+json.

Request

```
PATCH /loyaltyProgramMember/9664
Content-Type: application/json-patch+json
[
    {
        "op": "replace",
        "path": "/status",
        "value": "validated"
    }
]
```

Response

```
200
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
```

© TM Forum 2025. All Rights Reserved.

```
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "preferred": true,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "john.doe@gmail.com"
        },
        {
            "preferred": false,
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "validated",
    "statusReason": "Loyalty Member Enrolment",
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
}
```


Here's an example of a request for updating a LoyaltyProgramMember resource - change phone number. This example illustrates patch using application/json-patch-query+json.

Request

© TM Forum 2025. All Rights Reserved.

```
PATCH /loyaltyProgramMember/9664
Content-Type: application/json-patch-query+json
[
    {
        "op": "replace",
        "path": "/contactMedium?id=1",
        "value": {
            "@type": "PhoneContactMedium",
            "phoneNumber": "+3312345678"
        }
    }
]
```

Response

```
200
{
    "id": "9664",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/9664",
    "@type": "LoyaltyProgramMember",
    "@baseType": "PartyRole",
    "name": "John Doe",
    "description": "Loyalty Program Member Profile",
    "engagedParty": {
        "id": "7872",
        "href": "https:/host:port/tmf-api/partyManagement/v2/individual/7872",
        "@type": "PartyRef",
        "name": "John Doe",
        "@referredType": "Individual"
    },
    "account": [
        {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "AccountRef",
            "name": "John Doe's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        }
    ],
    "contactMedium": [
        {
            "@type": "PhoneContactMedium",
            "id": "1",
            "phoneNumber": "+3312345678"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "240 Headquarters Plazza",
```

© TM Forum 2025. All Rights Reserved.

```
            "street2": "East Tower - 10th Floor"
        }
    ],
    "agreement": [
        {
            "id": "4721",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/4721",
            "@type": "AgreementRef",
            "name": "John Doe's Loyalty Program Membership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Last Active",
            "value": "2025-01-15T07:49:25.246Z",
            "@type": "StringCharacteristic"
        }
    ],
    "status": "validated",
    "statusReason": "Loyalty Member Enrolment",
    "validFor": {
        "startDateTime": "2025-01-12T23:20:50.52Z",
        "endDateTime": "2030-04-12T23:20:50.52Z"
    },
    "tier": "Gold"
}
```


Deletes a LoyaltyProgramMember

`DELETE /loyaltyProgramMember/{id}`

Description

This operation deletes a LoyaltyProgramMember entity.

Operations on LoyaltyProgramProduct

Retrieves a LoyaltyProgramProduct by ID

`GET /loyaltyProgramProduct/{id}?fields=…`

Description

This operation retrieves a LoyaltyProgramProduct entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Program Product GET operation result

Request

```
GET /loyaltyProgramProduct/100
```

© TM Forum 2025. All Rights Reserved.

```
Content-Type: application/json
```

Response

```
200
{
    "id": "100",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/100",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "creationDate": "2024-05-01T20:26:25.122Z",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
```

© TM Forum 2025. All Rights Reserved.

```
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    }
}
```


List or find LoyaltyProgramProduct objects

`GET /loyaltyProgramProduct?fields=…`

Description

This operation list LoyaltyProgramProduct entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Program Product GET operation result

Request

```
GET /loyaltyProgramProduct?status=active
Content-Type: application/json
```

Response


```
200
[
    {
        "id": "100",
        "href": "/tmf-api/loyaltyManagement/v5/loyaltyProgramProduct/100",
        "@type": "LoyaltyProgramProduct",
        "@baseType": "Product",
        "creationDate": "2024-05-01T20:26:25.122Z",
        "description": "Program enrolment for Inner City transportation incentives",
        "isBundle": false,
```


© TM Forum 2025. All Rights Reserved.

```
        "isCustomerVisible": true,
        "name": " Inner City transportation incentives enrolment",
        "orderDate": "2024-05-01T20:26:25.122Z",
        "productSerialNumber": "LOYALTY-14434394",
        "productTerm": [
            {
                "@type": "ProductTerm",
                "description": "Loyalty enrolment subscription term",
                "duration": {
                    "amount": 24,
                    "units": "Month"
                },
                "validFor": {
                    "startDateTime": "2024-04-12T23:20:50.52Z",
                    "endDateTime": "2026-04-12T23:20:50.52Z"
                },
                "name": "24 month enrolment"
            }
        ],
        "startDate": "2024-04-12T20:26:25.122Z",
        "terminationDate": "2026-04-12T20:26:25.122Z",
        "status": "active",
        "productCharacteristic": [
            {
                "name": "Bicycle Model",
                "valueType": "string",
                "value": "Trek Domane",
                "@type": "StringCharacteristic"
            },
            {
                "name": "Color",
                "valueType": "string",
                "value": "Black",
                "@type": "StringCharacteristic"
            }
        ],
        "productOffering": {
            "id": "7121",
            "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
            "@type": "ProductOfferingRef",
            "name": "Inner City Transportation Incentive Program",
            "@referredType": "ProductOffering"
        },
        "productSpecification": {
            "id": "100",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
            "@type": "ProductSpecificationRef",
            "name": "Inner City Transportation Incentive Program",
            "@referredType": "LoyaltyProgramProductSpecification"
        },
        "loyaltyProgramMember": {
            "id": "3035235555",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
            "@type": "LoyaltyProgramMemberRef",
            "name": "Joseph",
            "@referredType": "LoyaltyProgramMember"
        },
        "loyaltyAccount": {
            "id": "3035235555",
            "href": "https://mycsp.com:8080/tmf-
```


© TM Forum 2025. All Rights Reserved.

```
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
            "@type": "LoyaltyAccountRef",
            "name": "Joseph loyalty account",
            "@referredType": "LoyaltyAccount"
        }
    },
    {
        "id": "200",
        "href": "/tmf-api/loyaltyManagement/v5/loyaltyProgramProduct/200",
        "@type": "LoyaltyProgramProduct",
        "@baseType": "Product",
        "creationDate": "2024-05-01T20:26:25.122Z",
        "description": "Program enrolment for balance topup behavior change
incentives",
        "isBundle": false,
        "isCustomerVisible": true,
        "name": "Prepaid subscriber behavioural changes enrolment",
        "orderDate": "2024-05-01T20:26:25.122Z",
        "productSerialNumber": "LOYALTY-14434394",
        "productTerm": [
            {
                "@type": "ProductTerm",
                "@baseType": "string",
                "description": "Loyalty enrolment subscription term",
                "duration": {
                    "amount": 24,
                    "units": "Month"
                },
                "validFor": {
                    "startDateTime": "2024-04-12T23:20:50.52Z",
                    "endDateTime": "2026-04-12T23:20:50.52Z"
                },
                "name": "24 month enrolment"
            }
        ],
        "startDate": "2024-04-12T20:26:25.122Z",
        "terminationDate": "2026-04-12T20:26:25.122Z",
        "status": "active",
        "productCharacteristic": [
            {
                "name": "Bicycle Model",
                "valueType": "string",
                "value": "Trek Domane",
                "@type": "StringCharacteristic"
            },
            {
                "name": "Color",
                "valueType": "string",
                "value": "White",
                "@type": "StringCharacteristic"
            }
        ],
        "productSpecification": {
            "id": "100",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
            "@type": "ProductSpecificationRef",
            "name": "Prepaid Subscriber Behavior Change Incentive Program",
            "@referredType": "LoyaltyProgramProductSpecification"
        },
        "loyaltyProgramMember": {
            "id": "3035235555",
            "href": "https://mycsp.com:8080/tmf-
```


© TM Forum 2025. All Rights Reserved.

```
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
            "@type": "LoyaltyProgramMemberRef",
            "name": "Joseph",
            "@referredType": "LoyaltyProgramMember"
        },
        "loyaltyAccount": {
            "id": "3035235555",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
            "@type": "LoyaltyAccountRef",
            "name": "Joseph loyalty account",
            "@referredType": "LoyaltyAccount"
        }
    }
]
```

Creates a LoyaltyProgramProduct

POST /loyaltyProgramProduct?fields=…

Description

This operation creates a LoyaltyProgramProduct entity.

Mandatory Attributes

Mandatory Attributes	Rule
loyaltyProgramMember	M
@type	M

Usage samples

This is an example of creating a Loyalty Program Product - UC1 Inner City Spec

Request

```
POST /loyaltyProgramProduct
Content-Type: application/json
{
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
```

© TM Forum 2025. All Rights Reserved.

```
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    }
}
```


Response

© TM Forum 2025. All Rights Reserved.

```
201
{
    "id": "100",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/100",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
```


© TM Forum 2025. All Rights Reserved.

```
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    }
}
```

This is an example of creating a Loyalty Program Product - UC2 Prepaid Spec

Request

```
POST /loyaltyProgramProduct
Content-Type: application/json
{
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "description": "Program enrolment for balance topup behavior change incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": "Prepaid subscriber behavioural changes enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "@baseType": "string",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Prepaid Subscriber Behavior Change Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
```

© TM Forum 2025. All Rights Reserved.

```
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    }
}
```


Response


```
201
{
    "id": "200",
    "href": "/tmf-api/loyaltyManagement/v5/loyaltyProgramProduct/200",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "description": "Program enrolment for balance topup behavior change incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": "Prepaid subscriber behavioural changes enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "@baseType": "string",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Prepaid Subscriber Behavior Change Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
```


© TM Forum 2025. All Rights Reserved.

```
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    }
}
```


Updates partially a LoyaltyProgramProduct

`PATCH /loyaltyProgramProduct/{id}?fields=…`

Description

This operation allows partial updates of a LoyaltyProgramProduct entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
agreementItem	
billingAccount	
description	
externalIdentifier	
intent	
isBundle	
isCustomerVisible	

© TM Forum 2025. All Rights Reserved.

Patchable Attributes	Rule
loyaltyAccount	
loyaltyProgramMember	
name	
orderDate	
place	
product	
productCharacteristic	
productOffering	
productOrderItem	
productPrice	
productRelationship	
productSerialNumber	
productSpecification	
productTerm	
realizingResource	
realizingService	
relatedParty	
startDate	
status	
terminationDate	

Usage samples

Here's an example of a request for updating a LoyaltyProgramProduct resource - set status to active. This example illustrates patch using application/json.

Request

```
PATCH /loyaltyProgramProduct/100
Content-Type: application/json
{
    "@type": "LoyaltyProgramProduct",
    "status": "active"
}
```

Response

```
200
{
    "id": "100",
```

© TM Forum 2025. All Rights Reserved.

```
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/100",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "creationDate": "2024-05-01T20:26:25.122Z",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
```


© TM Forum 2025. All Rights Reserved.

```
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    },
    "relatedParty": [
        {
            "role": "User",
            "partyOrPartyRole": {
                "id": "777",
                "href": "https://host:port/partyManagement/v5/individual/777",
                "name": "Pierre",
                "@type": "PartyRef",
                "@referredType": "Individual"
            },
            "@type": "RelatedPartyOrPartyRole"
        }
    ]
}
```

Here's an example of a request for updating a LoyaltyProgramProduct resource - set status to active. This example illustrates patch using application/merge-patch+json.

Request

```
PATCH /loyaltyProgramProduct/100
Content-Type: application/merge-patch+json
{
    "@type": "LoyaltyProgramProduct",
    "status": "active"
}
```

Response

```
200
{
    "id": "100",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/100",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "creationDate": "2024-05-01T20:26:25.122Z",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
```

© TM Forum 2025. All Rights Reserved.

```
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    },
    "relatedParty": [
```


© TM Forum 2025. All Rights Reserved.

```
        {
            "role": "User",
            "partyOrPartyRole": {
                "id": "777",
                "href": "https://host:port/partyManagement/v5/individual/777",
                "name": "Pierre",
                "@type": "PartyRef",
                "@referredType": "Individual"
            },
            "@type": "RelatedPartyOrPartyRole"
        }
    ]
}
```

Here's an example of a request for updating a LoyaltyProgramProduct resource - set status to active. This example illustrates patch using application/json-patch+json.

Request

```
PATCH /loyaltyProgramProduct/100
Content-Type: application/json-patch+json
[
    {
        "op": "replace",
        "path": "/status",
        "value": "active"
    }
]
```

Response

```
200
{
    "id": "100",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/100",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "creationDate": "2024-05-01T20:26:25.122Z",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            },
```

© TM Forum 2025. All Rights Reserved.

```
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    },
    "relatedParty": [
        {
            "role": "User",
            "partyOrPartyRole": {
                "id": "777",
                "href": "https://host:port/partyManagement/v5/individual/777",
                "name": "Pierre",
                "@type": "PartyRef",
                "@referredType": "Individual"
            },
            "@type": "RelatedPartyOrPartyRole"
```


© TM Forum 2025. All Rights Reserved.

```
        }
    ]
}
```

Here's an example of a request for updating a LoyaltyProgramProduct resource - change relatedParty and color characteristic. This example illustrates patch using application/json-patch-query+json.

Request

```
PATCH /loyaltyProgramProduct/100
Content-Type: application/json-patch-query+json
[
    {
        "op": "replace",
        "path": "/relatedParty/partyOrPartyRole/id?relatedParty.role=user",
        "value": "777"
    },
    {
        "op": "replace",
        "path": "/productCharacteristic/value?name=Color",
        "value": {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    }
]
```

Response

```
200
{
    "id": "100",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProduct/100",
    "@type": "LoyaltyProgramProduct",
    "@baseType": "Product",
    "creationDate": "2024-05-01T20:26:25.122Z",
    "description": "Program enrolment for Inner City transportation incentives",
    "isBundle": false,
    "isCustomerVisible": true,
    "name": " Inner City transportation incentives enrolment",
    "orderDate": "2024-05-01T20:26:25.122Z",
    "productSerialNumber": "LOYALTY-14434394",
    "productTerm": [
        {
            "@type": "ProductTerm",
            "description": "Loyalty enrolment subscription term",
            "duration": {
                "amount": 24,
                "units": "Month"
            },
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
```

© TM Forum 2025. All Rights Reserved.

```
            },
            "name": "24 month enrolment"
        }
    ],
    "startDate": "2024-04-12T20:26:25.122Z",
    "terminationDate": "2026-04-12T20:26:25.122Z",
    "status": "active",
    "productCharacteristic": [
        {
            "name": "Bicycle Model",
            "valueType": "string",
            "value": "Trek Domane",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Color",
            "valueType": "string",
            "value": "Black",
            "@type": "StringCharacteristic"
        }
    ],
    "productOffering": {
        "id": "7121",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7121",
        "@type": "ProductOfferingRef",
        "name": "Inner City Transportation Incentive Program",
        "@referredType": "ProductOffering"
    },
    "productSpecification": {
        "id": "100",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramProductSpecification/100",
        "@type": "ProductSpecificationRef",
        "name": "Transportation Incentive Program",
        "@referredType": "LoyaltyProgramProductSpecification"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyAccount": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/3035235555",
        "@type": "LoyaltyAccountRef",
        "name": "Joseph loyalty account",
        "@referredType": "LoyaltyAccount"
    },
    "relatedParty": [
        {
            "role": "User",
            "partyOrPartyRole": {
                "id": "777",
                "href": "https://host:port/partyManagement/v5/individual/777",
                "name": "Pierre",
                "@type": "PartyRef",
                "@referredType": "Individual"
            },
```


© TM Forum 2025. All Rights Reserved.

```
            "@type": "RelatedPartyOrPartyRole"
        }
    ]
}
```

Deletes a LoyaltyProgramProduct

DELETE /loyaltyProgramProduct/{id}

Description

This operation deletes a LoyaltyProgramProduct entity.

Operations on LoyaltyAccount

List or find LoyaltyAccount objects

GET /loyaltyAccount?fields=…

Description

This operation list LoyaltyAccount entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Account GET operation result

Request

```
GET /loyaltyAccount?state=active
Content-Type: application/json
```

Response

```
200
[
    {
        "id": "501",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/lovaltyAccount/501",
        "description": "Loyalty account for loyalty program member",
        "lastUpdate": "2024-05-01T20:30:52.370Z",
        "name": "Loyalty Account",
        "state": "Active",
        "accountType": "Loyalty Account",
        "relatedParty": [
            {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "Loyalty Program Member",
                "partyOrPartyRole": {
                    "id": "3035235555",
```

© TM Forum 2025. All Rights Reserved.

```
                    "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                    "@type": "PartyRoleRef",
                    "name": "Joseph",
                    "@referredType": "LoyaltyProgramMember"
                }
            }
        ],
        "accountBalance": [
            {
                "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
                "@type": "AccountBalance",
                "amount": {
                    "value": 20.0,
                    "unit": "POINTS"
                },
                "balanceType": "LoyaltyEarningsWallet",
                "validFor": {
                    "startDateTime": "2024-04-12T23:20:50.52Z",
                    "endDateTime": "2026-04-12T23:20:50.52Z"
                }
            },
            {
                "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
                "@type": "AccountBalance",
                "amount": {
                    "value": 0.0,
                    "unit": "POINTS"
                },
                "balanceType": "LoyaltyBankedWallet",
                "validFor": {
                    "startDateTime": "2024-04-12T23:20:50.52Z",
                    "endDateTime": "2026-04-12T23:20:50.52Z"
                }
            }
        ],
        "@type": "LoyaltyAccount",
        "@baseType": "PartyAccount"
    },
    {
        "id": "344",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/lovaltyAccount/344",
        "description": "Loyalty account for loyalty program member",
        "lastUpdate": "2024-05-01T20:30:52.370Z",
        "name": "Loyalty Account",
        "state": "Active",
        "accountType": "Loyalty Account",
        "relatedParty": [
            {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "Loyalty Program Member",
                "partyOrPartyRole": {
                    "id": "18112",
                    "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/18112",
                    "@type": "PartyRoleRef",
                    "name": "John Doe",
                    "@referredType": "LoyaltyProgramMember"
                }
            }
        ],
        "accountBalance": [
```


© TM Forum 2025. All Rights Reserved.

```
            {
                "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
                "@type": "AccountBalance",
                "amount": {
                    "value": 45.0,
                    "unit": "POINTS"
                },
                "balanceType": "LoyaltyEarningsWallet",
                "validFor": {
                    "startDateTime": "2024-04-12T23:20:50.52Z",
                    "endDateTime": "2026-04-12T23:20:50.52Z"
                }
            },
            {
                "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
                "@type": "AccountBalance",
                "amount": {
                    "value": 12.0,
                    "unit": "POINTS"
                },
                "balanceType": "LoyaltyBankedWallet",
                "validFor": {
                    "startDateTime": "2024-04-12T23:20:50.52Z",
                    "endDateTime": "2026-04-12T23:20:50.52Z"
                }
            }
        ],
        "@type": "LoyaltyAccount",
        "@baseType": "PartyAccount"
    }
]
```


Retrieves a LoyaltyAccount by ID

`GET /loyaltyAccount/{id}?fields=…`

Description

This operation retrieves a LoyaltyAccount entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Account GET operation result

Request

```
GET /loyaltyAccount/501
Content-Type: application/json
```

Response

```
200
{
```

© TM Forum 2025. All Rights Reserved.

```
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/lovaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "name": "Loyalty Account",
    "state": "Active",
    "accountType": "Loyalty Account",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "@type": "AccountBalance",
            "amount": {
                "value": 20.00,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        },
        {
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "@type": "AccountBalance",
            "amount": {
                "value": 0.00,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```


Creates a LoyaltyAccount

`POST /loyaltyAccount?fields=…`

Description

© TM Forum 2025. All Rights Reserved.

This operation creates a LoyaltyAccount entity.

Mandatory Attributes

Mandatory Attributes	Rule
name	
relatedParty	
@type	M

Usage samples

This is an example of creating a Loyalty Account UC1 Inner City Request Example

Request

```
POST /loyaltyAccount
Content-Type: application/json
{
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "externalIdentifier": [
        {
            "id": "12e82219-d691-43f7-b12c-789deaf03a9d",
            "owner": "CCS",
            "@type": "ExternalIdentifier"
        }
    ],
    "name": "Loyalty Account",
    "state": "Active",
    "accountType": "Loyalty Account",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "@type": "AccountBalance",
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "amount": {
                "value": 20.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
```

© TM Forum 2025. All Rights Reserved.

```
        },
        {
            "@type": "AccountBalance",
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "amount": {
                "value": 0.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```


Response


```
201
{
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "externalIdentifier": [
        {
            "id": "12e82219-d691-43f7-b12c-789deaf03a9d",
            "owner": "CCS",
            "@type": "ExternalIdentifier"
        }
    ],
    "name": "Loyalty Account",
    "state": "Active",
    "accountType": "Loyalty Account",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "@type": "AccountBalance",
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "amount": {
                "value": 20.0,
                "unit": "POINTS"
            },
```


© TM Forum 2025. All Rights Reserved.

```
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        },
        {
            "@type": "AccountBalance",
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "amount": {
                "value": 0.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```


Updates partially a LoyaltyAccount

`PATCH /loyaltyAccount/{id}?fields=…`

Description

This operation allows partial updates of a LoyaltyAccount entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
accountBalance	
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
accountRelationship	

© TM Forum 2025. All Rights Reserved.

Patchable Attributes	Rule
accountType	
billStructure	
contact	
creditLimit	
defaultPaymentMethod	
description	
externalIdentifier	
financialAccount	
name	
paymentPlan	
paymentStatus	
relatedParty	
state	
taxExemption	

Usage samples

Here is an example of a request for patching an LoyaltyAccount resource with change on credit limit attribute using patch merge

Request

```
PATCH /loyaltyAccount/501
Content-Type: application/json
{
    "@type": "LoyaltyAccount",
    "creditLimit": {
        "unit": "POINTS",
        "value": 5000
    }
}
```

Response

```
200
{
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "externalIdentifier": [
        {
            "id": "12e82219-d691-43f7-b12c-789deaf03a9d",
            "owner": "CCS",
```

© TM Forum 2025. All Rights Reserved.

```
            "@type": "ExternalIdentifier"
        }
    ],
    "name": "Loyalty Account",
    "state": "Suspended",
    "accountType": "Loyalty Account",
    "creditLimit": {
        "unit": "POINTS",
        "value": 5000
    },
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "@type": "AccountBalance",
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "amount": {
                "value": 20.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        },
        {
            "@type": "AccountBalance",
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "amount": {
                "value": 0.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```


Here is an example of a request for patching an LoyaltyAccount resource with change on credit limit attribute using patch merge

Request

© TM Forum 2025. All Rights Reserved.

```
PATCH /loyaltyAccount/501
Content-Type: application/merge-patch+json
{
    "@type": "LoyaltyAccount",
    "creditLimit": {
        "unit": "POINTS",
        "value": 5000
    }
}
```

Response

```
200
{
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "externalIdentifier": [
        {
            "id": "12e82219-d691-43f7-b12c-789deaf03a9d",
            "owner": "CCS",
            "@type": "ExternalIdentifier"
        }
    ],
    "name": "Loyalty Account",
    "state": "Suspended",
    "accountType": "Loyalty Account",
    "creditLimit": {
        "unit": "POINTS",
        "value": 5000
    },
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "@type": "AccountBalance",
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "amount": {
                "value": 20.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
```

© TM Forum 2025. All Rights Reserved.

```
            }
        },
        {
            "@type": "AccountBalance",
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "amount": {
                "value": 0.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```

Here is an example of a request for patching a LoyaltyAccount by adding another related party using json-patch

Request

```
PATCH /loyaltyAccount/501
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "/relatedParty",
    "value": {
        "role": "contact",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
            "@type":"PartyRefOrPartyRoleRef",
            "@referredType" : "Individual",
            "href": "https://host:port/tmf-api/partyManagement/v5/individual/6666",
            "id": "6666",
            "name": "John Doe"
        }
    }
  }
]
```

Response

```
200
{
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "externalIdentifier": [
        {
            "id": "12e82219-d691-43f7-b12c-789deaf03a9d",
```

© TM Forum 2025. All Rights Reserved.

```
            "owner": "CCS",
            "@type": "ExternalIdentifier"
        }
    ],
    "name": "Loyalty Account",
    "state": "Suspended",
    "accountType": "Loyalty Account",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        },
        {
            "role": "contact",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
                "@type": "PartyRefOrPartyRoleRef",
                "@referredType": "Individual",
                "href": "https://host:port/tmf-
api/partyManagement/v5/individual/6666",
                "id": "6666",
                "name": "John Doe"
            }
        }
    ],
    "accountBalance": [
        {
            "@type": "AccountBalance",
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
            "amount": {
                "value": 20.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        },
        {
            "@type": "AccountBalance",
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "amount": {
                "value": 0.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
```


© TM Forum 2025. All Rights Reserved.

```
}
```

Here is an example of a request for patching a LoyaltyAccount by extending the validity period of a tax exemption using json-patch query

Request

```
PATCH /loyaltyAccount/501
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/taxExemption/validFor?/taxExemption/certificateNumber=45678889876",
    "value": {
        "startDateTime": "2018-06-13T00:00:00.000Z",
        "endDateTime": "2020-01-10T00:00:00.000Z"
    }
  }
]
```

Response

```
200
{
    "id": "501",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/501",
    "description": "Loyalty account for loyalty program member",
    "lastUpdate": "2024-05-01T20:30:52.370Z",
    "externalIdentifier": [
        {
            "id": "12e82219-d691-43f7-b12c-789deaf03a9d",
            "owner": "CCS",
            "@type": "ExternalIdentifier"
        }
    ],
    "name": "Loyalty Account",
    "state": "Suspended",
    "accountType": "Loyalty Account",
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
                "id": "3035235555",
                "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/3035235555",
                "@type": "PartyRoleRef",
                "name": "Joseph",
                "@referredType": "LoyaltyProgramMember"
            }
        }
    ],
    "accountBalance": [
        {
            "@type": "AccountBalance",
            "id": "46c54fb7-36a3-472b-a655-724d4b00a24a",
```

© TM Forum 2025. All Rights Reserved.

```
            "amount": {
                "value": 20.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        },
        {
            "@type": "AccountBalance",
            "id": "8d2f9f78-bfeb-4d6e-b463-5c2c03cdd589",
            "amount": {
                "value": 0.0,
                "unit": "POINTS"
            },
            "balanceType": "LoyaltyBankedWallet",
            "validFor": {
                "startDateTime": "2024-04-12T23:20:50.52Z",
                "endDateTime": "2026-04-12T23:20:50.52Z"
            }
        }
    ],
    "taxExemption": [
        {
            "@type": "TaxExemption",
            "certificateNumber": "45678889876",
            "issuingJurisdiction": "USA",
            "reason": "Embassy",
            "taxDefinition": [
                {
                    "@type": "TaxDefinition",
                    "id": "96",
                    "name": "Value Added Tax",
                    "taxType": "federalVat"
                }
            ],
            "attachment": {
                "@type": "Attachment",
                "id": "654321234334534552",
                "href": "https://host:port/tmf-
api/document/v5/attachment/654321234334534552",
                "attachmentType": "taxExemptionCertificate",
                "name": "TaxExemptionCertificate-654321.pdf",
                "description": "Electronic version of the tax exemption
certificate",
                "mimeType": "application/pdf",
                "url":
"http://host:port/DocumentManager/666/TaxExemptionCertificate-654321.pdf"
            },
            "validFor": {
                "startDateTime": "2018-06-13T00:00:00.000Z",
                "endDateTime": "2020-01-10T00:00:00.000Z"
            }
        }
    ],
    "@type": "LoyaltyAccount",
    "@baseType": "PartyAccount"
}
```


© TM Forum 2025. All Rights Reserved.

Deletes a LoyaltyAccount

DELETE /loyaltyAccount/{id}

Description

This operation deletes a LoyaltyAccount entity.

Operations on LoyaltyProgramPartner

List or find LoyaltyProgramPartner objects

GET /loyaltyProgramPartner?fields=…

Description

This operation list LoyaltyProgramPartner entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example request for retrieving multiple Loyalty Program Partner

Request

```
GET /loyaltyProgramPartner
Content-Type: application/json
```

Response

```
200
[
    {
        "id": "1",
        "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
        "@type": "LoyaltyProgramPartner",
        "@baseType": "BusinessPartner",
        "name": "Lifestyle Market",
        "status": "active",
        "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
        "validFor": {
            "startDateTime": "2024-04-12T23:20:50.52Z",
            "endDateTime": "2026-04-12T23:20:50.52Z"
        },
        "engagedParty": {
            "id": "1131",
            "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
            "@type": "PartyRef",
            "name": "Leisure Market",
            "@referredType": "Organization"
        },
        "contactMedium": [
```

© TM Forum 2025. All Rights Reserved.

```
            {
                "preferred": true,
                "mediumType": "email",
                "validFor": {
                    "startDateTime": "2025-01-15T07:49:25.246Z"
                },
                "@type": "EmailContactMedium",
                "emailAddress": "settlements@lifestylemarket.com"
            },
            {
                "preferred": false,
                "mediumType": "postalAddress",
                "validFor": {
                    "startDateTime": "2023-01-15T07:49:25.246Z"
                },
                "@type": "GeographicAddressContactMedium",
                "city": "Morristown",
                "country": "USA",
                "postCode": "07960",
                "stateOrProvince": "New Jersey",
                "street1": "777 Headquarters Plazza",
                "street2": "West Tower - 12th Floor"
            }
        ],
        "account": [
            {
                "id": "7723",
                "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
                "@type": "AccountRef",
                "name": "Lifestyle Market's Settlement Account",
                "@referredType": "SettlementAccount"
            }
        ],
        "agreement": [
            {
                "id": "442",
                "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
                "@type": "AgreementRef",
                "name": "Lifestyle Market's Partnership Agreement"
            }
        ],
        "characteristic": [
            {
                "name": "Partner sector",
                "value": "Entertainment",
                "@type": "StringCharacteristic"
            }
        ],
        "partnerCatalogCategory": [
            {
                "id": "35",
                "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
                "@type": "CategoryRef",
                "name": "Leisure Goods",
                "version": "1.1"
            }
        ],
        "partnerCatalogOffer": [
            {
                "id": "6654",
```


© TM Forum 2025. All Rights Reserved.

```
                "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
                "@type": "ProductOfferingRef",
                "name": "World Cup Football",
                "version": "1.1"
            },
            {
                "id": "6712",
                "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
                "@type": "ProductOfferingRef",
                "name": "Tennis Racket",
                "version": "1.0"
            }
        ]
    },
    {
        "id": "555",
        "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/555",
        "@type": "LoyaltyProgramPartner",
        "@baseType": "BusinessPartner",
        "name": "Sip & Savour Co.",
        "status": "Active",
        "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
        "validFor": {
            "startDateTime": "2023-08-11T10:12:50.45Z",
            "endDateTime": "2027-01-12T23:20:44.52Z"
        },
        "engagedParty": {
            "id": "776",
            "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/776",
            "@type": "PartyRef",
            "name": "Sip & Savour Co.",
            "@referredType": "Organization"
        },
        "contactMedium": [
            {
                "preferred": true,
                "mediumType": "email",
                "validFor": {
                    "startDateTime": "2023-01-15T07:49:25.246Z"
                },
                "@type": "EmailContactMedium",
                "emailAddress": "settlements@sipandsavour.com"
            },
            {
                "preferred": false,
                "mediumType": "postalAddress",
                "validFor": {
                    "startDateTime": "2023-01-15T07:49:25.246Z"
                },
                "@type": "GeographicAddressContactMedium",
                "city": "Morristown",
                "country": "USA",
                "postCode": "07960",
                "stateOrProvince": "New Jersey",
                "street1": "888 Headquarters Plazza",
                "street2": "North Tower - 22nd Floor"
            }
        ],
        "account": [
```


© TM Forum 2025. All Rights Reserved.

```
            {
                "id": "2213",
                "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/2213",
                "@type": "AccountRef",
                "name": "Sip & Savour Co's Settlement Account",
                "@referredType": "SettlementAccount"
            }
        ],
        "agreement": [
            {
                "id": "777",
                "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/777",
                "@type": "AgreementRef",
                "name": "Sip & Savour Co's Partnership Agreement"
            }
        ],
        "characteristic": [
            {
                "name": "Partner sector",
                "value": "Food and beverage",
                "@type": "StringCharacteristic"
            }
        ],
        "partnerCatalogCategory": [
            {
                "id": "443",
                "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/443",
                "@type": "CategoryRef",
                "name": "Drinks",
                "version": "1.0"
            }
        ],
        "partnerCatalogOffer": [
            {
                "id": "112",
                "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/112",
                "@type": "ProductOfferingRef",
                "name": "Pepsi",
                "version": "1.1"
            },
            {
                "id": "121",
                "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/121",
                "@type": "ProductOfferingRef",
                "name": "Coffee",
                "version": "1.0"
            }
        ]
    }
]
```


Retrieves a LoyaltyProgramPartner by ID

`GET /loyaltyProgramPartner/{id}?fields=…`

© TM Forum 2025. All Rights Reserved.

Description

This operation retrieves a LoyaltyProgramPartner entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example request for retrieving a specific Loyalty Program Partner

Request

```
GET /loyaltyProgramPartner/1
Content-Type: application/json
```

Response

```
200
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "active",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
```

© TM Forum 2025. All Rights Reserved.

```
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```


© TM Forum 2025. All Rights Reserved.

Creates a LoyaltyProgramPartner

POST /loyaltyProgramPartner?fields=…

Description

This operation creates a LoyaltyProgramPartner entity.

Mandatory Attributes

Mandatory Attributes	Rule
engagedParty	M
name	M
@type	M

Usage samples

This is an example request for creating a Loyalty Program Partner

Request

```
POST /loyaltyProgramPartner
Content-Type: application/json
{
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "active",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
```

© TM Forum 2025. All Rights Reserved.

```
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
```


© TM Forum 2025. All Rights Reserved.

```
    ]
}
```

Response

```
201
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "active",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
```

© TM Forum 2025. All Rights Reserved.

```
        }
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```


Updates partially a LoyaltyProgramPartner

`PATCH /loyaltyProgramPartner/{id}?fields=…`

Description

This operation allows partial updates of a LoyaltyProgramPartner entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH

© TM Forum 2025. All Rights Reserved.

operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
account	
agreement	
characteristic	
contactMedium	
creditProfile	
description	
engagedParty	
externalIdentifier	
href	
id	
name	
partnerCatalogCategory	
partnerCatalogOffer	
partyRoleSpecification	
paymentMethod	
relatedParty	
role	
status	
statusReason	
validFor	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule

Usage samples

Here's an example of a request for updating a LoyaltyProgramPartner resource - set status to validated. This example illustrates patch using application/json.

Request

```
PATCH /loyaltyProgramPartner/1
Content-Type: application/json
```

© TM Forum 2025. All Rights Reserved.

```
{
    "@type": "LoyaltyProgramPartner",
    "status": "validated"
}
```

Response

```
200
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "validated",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
```

© TM Forum 2025. All Rights Reserved.

```
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```


Here's an example of a request for updating a LoyaltyProgramPartner resource - set status to validated. This example illustrates patch using application/merge-patch+json.

Request

```
PATCH /loyaltyProgramPartner/1
Content-Type: application/merge-patch+json
{
    "@type": "LoyaltyProgramPartner",
    "status": "validated"
```

© TM Forum 2025. All Rights Reserved.

```
}
```

Response

```
200
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "validated",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
```

© TM Forum 2025. All Rights Reserved.

```
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```

Here's an example of a request for updating a LoyaltyProgramPartner resource - set status to validated. This example illustrates patch using application/json-patch+json.

Request

```
PATCH /loyaltyProgramPartner/1
Content-Type: application/json-patch+json
[
    {
        "op": "replace",
        "path": "/status",
        "value": "validated"
    }
```

© TM Forum 2025. All Rights Reserved.

```
]
```

Response

```
200
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "validated",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
```

© TM Forum 2025. All Rights Reserved.

```
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```

Here's an example of a request for updating a LoyaltyProgramPartner resource - change phone number. This example illustrates patch using application/json-patch-query+json.

Request

```
PATCH /loyaltyProgramPartner/1
Content-Type: application/json-patch-query+json
[
    {
        "op": "replace",
        "path": "/contactMedium?id=1",
        "value": {
            "@type": "PhoneContactMedium",
```

© TM Forum 2025. All Rights Reserved.

```
            "phoneNumber": "+3312345678"
        }
    }
]
```

Response

```
200
{
    "id": "1",
    "href": "https://myscp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramPartner/1",
    "@type": "LoyaltyProgramPartner",
    "@baseType": "BusinessPartner",
    "name": "Lifestyle Market",
    "status": "active",
    "statusReason": "Loyalty Rewards Benefit Partner Onboarded",
    "validFor": {
        "startDateTime": "2024-04-12T23:20:50.52Z",
        "endDateTime": "2026-04-12T23:20:50.52Z"
    },
    "engagedParty": {
        "id": "1131",
        "href": "https://myscp.com:8080/tmf-
api/paryManagement/v5/organization/1131",
        "@type": "PartyRef",
        "name": "Leisure Market",
        "@referredType": "Organization"
    },
    "contactMedium": [
        {
            "@type": "PhoneContactMedium",
            "id": "1",
            "phoneNumber": "+3312345678"
        },
        {
            "preferred": true,
            "mediumType": "email",
            "validFor": {
                "startDateTime": "2025-01-15T07:49:25.246Z"
            },
            "@type": "EmailContactMedium",
            "emailAddress": "settlements@lifestylemarket.com"
        },
        {
            "preferred": false,
            "mediumType": "postalAddress",
            "validFor": {
                "startDateTime": "2023-01-15T07:49:25.246Z"
            },
            "@type": "GeographicAddressContactMedium",
            "city": "Morristown",
            "country": "USA",
            "postCode": "07960",
            "stateOrProvince": "New Jersey",
            "street1": "777 Headquarters Plazza",
            "street2": "West Tower - 12th Floor"
        }
    ],
    "account": [
```

© TM Forum 2025. All Rights Reserved.

```
        {
            "id": "7723",
            "href": "https://mycsp.com:8080/tmf-
api/accountManagement/v5/settlementAccount/7723",
            "@type": "AccountRef",
            "name": "Lifestyle Market's Settlement Account",
            "@referredType": "SettlementAccount"
        }
    ],
    "agreement": [
        {
            "id": "442",
            "href": "https://myscp.com:8080/tmf-
api/agreementManagement/v5/agreement/442",
            "@type": "AgreementRef",
            "name": "Lifestyle Market's Partnership Agreement"
        }
    ],
    "characteristic": [
        {
            "name": "Partner sector",
            "value": "Entertainment",
            "@type": "StringCharacteristic"
        }
    ],
    "partnerCatalogCategory": [
        {
            "id": "35",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/category/35",
            "@type": "CategoryRef",
            "name": "Leisure Goods",
            "version": "1.1"
        }
    ],
    "partnerCatalogOffer": [
        {
            "id": "6654",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6654",
            "@type": "ProductOfferingRef",
            "name": "World Cup Football",
            "version": "1.1"
        },
        {
            "id": "6712",
            "href": "https://myscp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/6712",
            "@type": "ProductOfferingRef",
            "name": "Tennis Racket",
            "version": "1.0"
        }
    ]
}
```


Deletes a LoyaltyProgramPartner

`DELETE /loyaltyProgramPartner/{id}`

Description

© TM Forum 2025. All Rights Reserved.

This operation deletes a LoyaltyProgramPartner entity.

Operations on LoyaltyEarn

List or find LoyaltyEarn objects

`GET /loyaltyEarn?fields=…`

Description

This operation list LoyaltyEarn entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Earn GET operation result

Request

```
GET /loyaltyEarn?state=active
Content-Type: application/json
```

Response

```
200
[
    {
        "id": "9111",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/9111",
        "description": "Loyalty Earn",
        "@type": "LoyaltyEarn",
        "usageDate": "2024-05-01T20:30:52.370Z",
        "balanceType": "LoyaltyEarningsWallet",
        "loyaltyProgramMember": {
            "id": "644",
            "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/644",
            "name": "John Doe",
            "@referredType": "LoyaltyProgramMember",
            "@type": "LoyaltyProgramMemberRef"
        },
        "loyaltyProgramProduct": {
            "id": "7474",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
            "@type": "LoyaltyProgramProductRef",
            "@referredType": "LoyaltyProgramProduct"
        },
        "loyaltyAccount": {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "LoyaltyAccountRef",
```

© TM Forum 2025. All Rights Reserved.

```
            "name": "John's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        },
        "quantity": {
            "value": 50,
            "unit": "POINTS"
        },
        "usageCharacteristic": [
            {
                "name": "benefitPartnerInteractionType",
                "value": "VoucherRedeemed",
                "@type": "StringCharacteristic"
            }
        ]
    },
    {
        "id": "12652",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/12652",
        "description": "Loyalty Earn",
        "@type": "LoyaltyEarn",
        "usageDate": "2024-07-01T20:22:52.122Z",
        "balanceType": "LoyaltyEarningsWallet",
        "loyaltyProgramMember": {
            "id": "555",
            "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/555",
            "name": "Jane Doe",
            "@referredType": "LoyaltyProgramMember",
            "@type": "LoyaltyProgramMemberRef"
        },
        "loyaltyProgramProduct": {
            "id": "9944",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/9944",
            "@type": "LoyaltyProgramProductRef",
            "@referredType": "LoyaltyProgramProduct"
        },
        "loyaltyAccount": {
            "id": "1155",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
            "@type": "LoyaltyAccountRef",
            "name": "John's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        },
        "quantity": {
            "value": 100,
            "unit": "POINTS"
        },
        "usageCharacteristic": [
            {
                "name": "benefitPartnerInteractionType",
                "value": "VoucherRedeemed",
                "@type": "StringCharacteristic"
            }
        ]
    }
]
```


© TM Forum 2025. All Rights Reserved.

Retrieves a LoyaltyEarn by ID

GET /loyaltyEarn/{id}?fields=…

Description

This operation retrieves a LoyaltyEarn entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Earn GET operation result

Request

```
GET /loyaltyEarn/9111
Content-Type: application/json
```

Response

```
200
{
    "id": "9111",
    "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyEarn/9111",
    "description": "Loyalty Earn",
    "@type": "LoyaltyEarn",
    "usageDate": "2024-05-01T20:30:52.370Z",
    "balanceType": "LoyaltyEarningsWallet",
    "loyaltyProgramMember": {
        "id": "644",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/644",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember",
        "@type": "LoyaltyProgramMemberRef"
    },
    "loyaltyProgramProduct": {
        "id": "7474",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
        "@type": "LoyaltyProgramProductRef",
        "@referredType": "LoyaltyProgramProduct"
    },
    "loyaltyAccount": {
        "id": "1155",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
        "@type": "LoyaltyAccountRef",
        "name": "John's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    },
    "quantity": {
        "value": 50,
        "unit": "POINTS"
    },
    "usageCharacteristic": [
```

© TM Forum 2025. All Rights Reserved.

```
        {
            "name": "benefitPartnerInteractionType",
            "value": "VoucherRedeemed",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Creates a LoyaltyEarn

POST /loyaltyEarn?fields=…

Description

This operation creates a LoyaltyEarn entity.

Mandatory Attributes

Mandatory Attributes	Rule

Usage samples

This is an example of creating a Loyalty Earn

Request

```
POST /loyaltyEarn
Content-Type: application/json
{
    "description": "Loyalty Earn",
    "@type": "LoyaltyEarn",
    "usageDate": "2024-05-01T20:30:52.370Z",
    "balanceType": "LoyaltyEarningsWallet",
    "loyaltyProgramMember": {
        "id": "644",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/644",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember",
        "@type": "LoyaltyProgramMemberRef"
    },
    "loyaltyProgramProduct": {
        "id": "7474",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
        "@type": "LoyaltyProgramProductRef",
        "@referredType": "LoyaltyProgramProduct"
    },
    "loyaltyAccount": {
        "id": "1155",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
        "@type": "LoyaltyAccountRef",
        "name": "John Doe's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    },
    "quantity": {
```

© TM Forum 2025. All Rights Reserved.

```
        "value": 50,
        "unit": "POINTS"
    },
    "usageCharacteristic": [
        {
            "name": "benefitPartnerInteractionType",
            "value": "VoucherRedeemed",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Response

```
201
{
    "id": "9111",
    "href": "https://mycsp.com:8080/loyaltyManagement/v5/loyaltyEarn/9111",
    "description": "Loyalty Earn",
    "@type": "LoyaltyEarn",
    "usageDate": "2024-05-01T20:30:52.370Z",
    "balanceType": "LoyaltyEarningsWallet",
    "loyaltyProgramMember": {
        "id": "644",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/644",
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember",
        "@type": "LoyaltyProgramMemberRef"
    },
    "loyaltyProgramProduct": {
        "id": "7474",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
        "@type": "LoyaltyProgramProductRef",
        "@referredType": "LoyaltyProgramProduct"
    },
    "loyaltyAccount": {
        "id": "1155",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1155",
        "@type": "LoyaltyAccountRef",
        "name": "John Doe's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    },
    "quantity": {
        "value": 50,
        "unit": "POINTS"
    },
    "usageCharacteristic": [
        {
            "name": "benefitPartnerInteractionType",
            "value": "VoucherRedeemed",
            "@type": "StringCharacteristic"
        }
    ]
}
```

© TM Forum 2025. All Rights Reserved.

Operations on LoyaltyBurn

Retrieves a LoyaltyBurn by ID

GET /loyaltyBurn/{id}?fields=…

Description

This operation retrieves a LoyaltyBurn entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Burn GET operation result

Request

```
GET /loyaltyBurn/501
Content-Type: application/json
```

Response

```
200
{
    "id": "8131",
    "href": "https://mycsp.com:8080/tmf-api/loyaltyManagement/v5/loyaltyBurn/8131",
    "authorizationCode": "D73JD8885",
    "correlatorId": "6fgdsf8syf-gdf77gdfg-dsf6ds7f",
    "name": "Points Redemption",
    "description": "Points redemption for Online Coffee",
    "paymentDate": "2024-11-21T11:21:11.445Z",
    "status": "received",
    "statusDate": "2024-11-21T11:21:11.445Z",
    "amount": {
        "value": 2.5,
        "unit": "USD"
    },
    "channel": {
        "id": "64",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/64",
        "name": "My Favourite Coffee Online",
        "@type": "ChannelRef",
        "@referredType": "SalesChannel"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "7474",
        "href":
```

© TM Forum 2025. All Rights Reserved.

```
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
        "@type": "LoyaltyProgramProductRef",
        "@referredType": "LoyaltyProgramProduct"
    },
    "account": {
        "id": "1234",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1234",
        "@type": "AccountRef",
        "name": "Joseph's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    },
    "paymentMethod": {
        "@type": "LoyaltyPaymentMethod",
        "@baseType": "PaymentMethod",
        "name": "Joseph's Loyalty Points",
        "status": "Active",
        "statusDate": "2024-11-21T11:21:11.445Z"
    },
    "quantity": {
        "value": 120,
        "unit": "POINTS"
    },
    "paymentItem": [
        {
            "id": "1",
            "amount": {
                "value": 2.0,
                "unit": "USD"
            },
            "item": {
                "id": "55843",
                "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/55843",
                "name": "Coffee",
                "@type": "EntityRef",
                "@referredType": "ProductOrder"
            },
            "taxAmount": {
                "value": 0.5,
                "unit": "USD"
            },
            "totalAmount": {
                "value": 2.5,
                "unit": "USD"
            },
            "@type": "PaymentItem"
        }
    ],
    "@baseType": "Payment",
    "@type": "LoyaltyBurn"
}
```


List or find LoyaltyBurn objects

`GET /loyaltyBurn?fields=…`

Description

This operation list LoyaltyBurn entities. Attribute selection is enabled for all first level

© TM Forum 2025. All Rights Reserved.

attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a Loyalty Burn GET operation result

Request

```
GET /loyaltyBurn?state=active
Content-Type: application/json
```

Response

```
200
[
    {
        "id": "8131",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/8131",
        "authorizationCode": "D73JD8885",
        "correlatorId": "6fgdsf8syf-gdf77gdfg-dsf6ds7f",
        "name": "Points Redemption",
        "description": "Points redemption for Online Coffee",
        "paymentDate": "2024-11-21T11:21:11.445Z",
        "status": "received",
        "statusDate": "2024-11-21T11:21:11.445Z",
        "amount": {
            "value": 2.5,
            "unit": "USD"
        },
        "channel": {
            "id": "64",
            "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/64",
            "name": "My Favourite Coffee Online",
            "@type": "ChannelRef",
            "@referredType": "SalesChannel"
        },
        "loyaltyProgramMember": {
            "id": "3035235555",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
            "@type": "LoyaltyProgramMemberRef",
            "name": "Joseph",
            "@referredType": "LoyaltyProgramMember"
        },
        "loyaltyProgramProduct": {
            "id": "7474",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
            "@type": "LoyaltyProgramProductRef",
            "@referredType": "LoyaltyProgramProduct"
        },
        "account": {
            "id": "1234",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1234",
```

© TM Forum 2025. All Rights Reserved.

```
            "@type": "AccountRef",
            "name": "Joseph's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        },
        "paymentMethod": {
            "@type": "LoyaltyPaymentMethod",
            "@baseType": "PaymentMethod",
            "name": "Joseph's Loyalty Points",
            "status": "Active",
            "statusDate": "2024-11-21T11:21:11.445Z"
        },
        "quantity": {
            "value": 120,
            "unit": "POINTS"
        },
        "paymentItem": [
            {
                "id": "1",
                "amount": {
                    "value": 2.0,
                    "unit": "USD"
                },
                "item": {
                    "id": "55843",
                    "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/55843",
                    "name": "Coffee",
                    "@type": "EntityRef",
                    "@referredType": "ProductOrder"
                },
                "taxAmount": {
                    "value": 0.5,
                    "unit": "USD"
                },
                "totalAmount": {
                    "value": 2.5,
                    "unit": "USD"
                },
                "@type": "PaymentItem"
            }
        ],
        "@baseType": "Payment",
        "@type": "LoyaltyBurn"
    },
    {
        "id": "39655",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/39655",
        "authorizationCode": "D73JD8885",
        "correlatorId": "6fgdsf8syf-gdf77gdfg-dsf6ds7f",
        "name": "Points Redemption",
        "description": "Points redemption for Online Coffee",
        "paymentDate": "2024-12-21T11:21:11.445Z",
        "status": "received",
        "statusDate": "2024-12-21T11:21:11.445Z",
        "amount": {
            "value": 2.5,
            "unit": "USD"
        },
        "channel": {
            "id": "64",
            "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/64",
```


© TM Forum 2025. All Rights Reserved.

```
            "name": "My Favourite Coffee Online",
            "@type": "ChannelRef",
            "@referredType": "SalesChannel"
        },
        "loyaltyProgramMember": {
            "id": "1213",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/1213",
            "@type": "LoyaltyProgramMemberRef",
            "name": "Sally Smith",
            "@referredType": "LoyaltyProgramMember"
        },
        "loyaltyProgramProduct": {
            "id": "888",
            "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/888",
            "@type": "LoyaltyProgramProductRef",
            "@referredType": "LoyaltyProgramProduct"
        },
        "account": {
            "id": "1187",
            "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1187",
            "@type": "AccountRef",
            "name": "Sally's Loyalty Account",
            "@referredType": "LoyaltyAccount"
        },
        "paymentMethod": {
            "@type": "LoyaltyPaymentMethod",
            "@baseType": "PaymentMethod",
            "name": "Sally's Loyalty Points",
            "status": "Active",
            "statusDate": "2024-11-21T11:21:11.445Z"
        },
        "quantity": {
            "value": 120,
            "unit": "POINTS"
        },
        "paymentItem": [
            {
                "id": "1",
                "amount": {
                    "value": 2.0,
                    "unit": "USD"
                },
                "item": {
                    "id": "99777",
                    "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/99777",
                    "name": "Coffee",
                    "@type": "EntityRef",
                    "@referredType": "ProductOrder"
                },
                "taxAmount": {
                    "value": 0.5,
                    "unit": "USD"
                },
                "totalAmount": {
                    "value": 2.5,
                    "unit": "USD"
                },
                "@type": "PaymentItem"
            }
```


© TM Forum 2025. All Rights Reserved.

```
        ],
        "@baseType": "Payment",
        "@type": "LoyaltyBurn"
    }
]
```

Creates a LoyaltyBurn

`POST /loyaltyBurn?fields=…`

Description

This operation creates a LoyaltyBurn entity.

Mandatory Attributes

Mandatory Attributes	Rule
account	M
loyaltyProgramMember	M
loyaltyProgramProduct	M
paymentMethod	M
quantity	M
@type	M

Usage samples

This is an example of creating a Loyalty Burn

Request

```
POST /loyaltyBurn
Content-Type: application/json
{
    "authorizationCode": "D73JD8885",
    "correlatorId": "6fgdsf8syf-gdf77gdfg-dsf6ds7f",
    "name": "Points Redemption",
    "description": "Points redemption for Online Coffee",
    "paymentDate": "2024-11-21T11:21:11.445Z",
    "status": "received",
    "statusDate": "2024-11-21T11:21:11.445Z",
    "amount": {
        "value": 2.5,
        "unit": "USD"
    },
    "channel": {
        "id": "64",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/64",
        "name": "My Favourite Coffee Online",
        "@type": "ChannelRef",
        "@referredType": "SalesChannel"
    },
    "account": {
```

© TM Forum 2025. All Rights Reserved.

```
        "id": "1234",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1234",
        "@type": "AccountRef",
        "name": "Joseph's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    },
    "paymentMethod": {
        "@type": "LoyaltyPaymentMethod",
        "@baseType": "PaymentMethod",
        "name": "Joseph's Loyalty Points",
        "status": "Active",
        "statusDate": "2024-11-21T11:21:11.445Z"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "7474",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
        "@type": "LoyaltyProgramProductRef",
        "@referredType": "LoyaltyProgramProduct"
    },
    "quantity": {
        "value": 120,
        "unit": "POINTS"
    },
    "paymentItem": [
        {
            "id": "1",
            "amount": {
                "value": 2.0,
                "unit": "USD"
            },
            "item": {
                "id": "55843",
                "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/55843",
                "name": "Coffee",
                "@type": "EntityRef",
                "@referredType": "ProductOrder"
            },
            "taxAmount": {
                "value": 0.5,
                "unit": "USD"
            },
            "totalAmount": {
                "value": 2.5,
                "unit": "USD"
            },
            "@type": "PaymentItem"
        }
    ],
    "@baseType": "Payment",
    "@type": "LoyaltyBurn"
}
```


© TM Forum 2025. All Rights Reserved.

Response


```
201
{
    "id": "8131",
    "href": "https://mycsp.com:8080/tmf-api/loyaltyManagement/v5/loyaltyBurn/8131",
    "authorizationCode": "D73JD8885",
    "correlatorId": "6fgdsf8syf-gdf77gdfg-dsf6ds7f",
    "name": "Points Redemption",
    "description": "Points redemption for Online Coffee",
    "paymentDate": "2024-11-21T11:21:11.445Z",
    "status": "received",
    "statusDate": "2024-11-21T11:21:11.445Z",
    "amount": {
        "value": 2.5,
        "unit": "USD"
    },
    "channel": {
        "id": "64",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/64",
        "name": "My Favourite Coffee Online",
        "@type": "ChannelRef",
        "@referredType": "SalesChannel"
    },
    "account": {
        "id": "1234",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyAccount/1234",
        "@type": "AccountRef",
        "name": "Joseph's Loyalty Account",
        "@referredType": "LoyaltyAccount"
    },
    "paymentMethod": {
        "@type": "LoyaltyPaymentMethod",
        "@baseType": "PaymentMethod",
        "name": "Joseph's Loyalty Points",
        "status": "Active",
        "statusDate": "2024-11-21T11:21:11.445Z"
    },
    "loyaltyProgramMember": {
        "id": "3035235555",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyProgramMember/3035235555",
        "@type": "LoyaltyProgramMemberRef",
        "name": "Joseph",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "7474",
        "href":
"https://mycsp.com:8080/loyaltyManagement/v5/loyaltyProgramProduct/7474",
        "@type": "LoyaltyProgramProductRef",
        "@referredType": "LoyaltyProgramProduct"
    },
    "quantity": {
        "value": 120,
        "unit": "POINTS"
    },
    "paymentItem": [
        {
            "id": "1",
```


© TM Forum 2025. All Rights Reserved.

```
            "amount": {
                "value": 2.0,
                "unit": "USD"
            },
            "item": {
                "id": "55843",
                "href": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder/55843",
                "name": "Coffee",
                "@type": "EntityRef",
                "@referredType": "ProductOrder"
            },
            "taxAmount": {
                "value": 0.5,
                "unit": "USD"
            },
            "totalAmount": {
                "value": 2.5,
                "unit": "USD"
            },
            "@type": "PaymentItem"
        }
    ],
    "@baseType": "Payment",
    "@type": "LoyaltyBurn"
}
```


Operations on LoyaltyExecutionPoint

List or find LoyaltyExecutionPoint objects

`GET /loyaltyExecutionPoint?fields=…`

Description

This operation list LoyaltyExecutionPoint entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of loyalty execution Point(s).

Request

```
GET /loyaltyExecutionPoint?triggers=CustomerOrder
Content-Type: application/json
```

Response


```
200
[
    {
        "id": "7756",
        "href": "https://mycsp.com:8080/tmf-
```


© TM Forum 2025. All Rights Reserved.

```
api/loyaltyManagement/v5/loyaltyExecutionPoint/7756",
        "@type": "LoyaltyExecutionPoint",
        "name": "John Doe's Birthday Gift Loyalty Execution Point",
        "triggers": "CustomerOrder",
        "loyaltyProgramMember": {
            "id": "9664",
            "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
            "@type": "LoyaltyProgramMemberRef",
            "name": "John Doe",
            "@referredType": "LoyaltyProgramMember"
        },
        "loyaltyProgramProduct": {
            "id": "1745",
            "href": "https://mycsp.com:8080/tmf-
api/loyalty/v5/loyaltyProgramProduct/1745",
            "@type": "LoyaltyProgramProductRef",
            "name": "John Doe's Loyalty Program",
            "@referredType": "LoyaltyProgramProduct"
        },
        "body": {
            "category": "Birthday Gift product order",
            "priority": "1",
            "channel": [
                {
                    "role": "Used channel for order capture",
                    "@type": "RelatedChannel",
                    "channel": {
                        "id": "1",
                        "name": "Online channel",
                        "@type": "ChannelRef"
                    }
                }
            ],
            "productOrderItem": [
                {
                    "id": "100",
                    "quantity": 1,
                    "action": "add",
                    "productOffering": {
                        "id": "14277",
                        "href":
"https://host:port/productCatalogManagement/v5/productOffering/14277",
                        "name": "TMF25",
                        "@type": "ProductOfferingRef"
                    },
                    "@type": "ProductOrderItem"
                }
            ],
            "relatedParty": [
                {
                    "role": "Customer",
                    "partyOrPartyRole": {
                        "id": "{{customer.id}}",
                        "href":
"https://host:port/customerManagement/v5/customer/{{customer.id}}",
                        "name": "{{customer.name}}",
                        "@type": "PartyRoleRef",
                        "@referredType": "Customer"
                    },
                    "@type": "RelatedPartyRefOrPartyRoleRef"
                }
            ],
```


© TM Forum 2025. All Rights Reserved.

```
            "@type": "ProductOrder"
        },
        "version": "1.0",
        "action": "POST",
        "endpoint": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder",
        "creationDate": "2024-05-01T20:36:13.313Z"
    },
    {
        "id": "8002",
        "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyExecutionPoint/8002",
        "@type": "LoyaltyExecutionPoint",
        "name": "Sally Smith's Birthday Gift Loyalty Execution Point",
        "triggers": "CustomerOrder",
        "loyaltyProgramMember": {
            "id": "8881",
            "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/8881",
            "@type": "LoyaltyProgramMemberRef",
            "name": "Sally Smith",
            "@referredType": "LoyaltyProgramMember"
        },
        "loyaltyProgramProduct": {
            "id": "1812",
            "href": "https://mycsp.com:8080/tmf-
api/loyalty/v5/loyaltyProgramProduct/1812",
            "@type": "LoyaltyProgramProductRef",
            "name": "Sally Smith's Loyalty Program",
            "@referredType": "LoyaltyProgramProduct"
        },
        "body": {
            "category": "Birthday Gift product order",
            "priority": "1",
            "channel": [
                {
                    "role": "Used channel for order capture",
                    "@type": "RelatedChannel",
                    "channel": {
                        "id": "1",
                        "name": "Online channel",
                        "@type": "ChannelRef"
                    }
                }
            ],
            "productOrderItem": [
                {
                    "id": "100",
                    "quantity": 1,
                    "action": "add",
                    "productOffering": {
                        "id": "14277",
                        "href":
"https://host:port/productCatalogManagement/v5/productOffering/14277",
                        "name": "TMF25",
                        "@type": "ProductOfferingRef"
                    },
                    "@type": "ProductOrderItem"
                }
            ],
            "relatedParty": [
                {
                    "role": "Customer",
```


© TM Forum 2025. All Rights Reserved.

```
                    "partyOrPartyRole": {
                        "id": "{{customer.id}}",
                        "href":
"https://host:port/customerManagement/v5/customer/{{customer.id}}",
                        "name": "{{customer.name}}",
                        "@type": "PartyRoleRef",
                        "@referredType": "Customer"
                    },
                    "@type": "RelatedPartyRefOrPartyRoleRef"
                }
            ],
            "@type": "ProductOrder"
        },
        "version": "1.0",
        "action": "POST",
        "endpoint": "https://mycsp.com:8080/tmf-
api/productOrdering/v5/productOrder",
        "creationDate": "2024-06-01T20:36:13.424Z"
    }
]
```

Retrieves a LoyaltyExecutionPoint by ID

`GET /loyaltyExecutionPoint/{id}?fields=…`

Description

This operation retrieves a LoyaltyExecutionPoint entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This is an example of a LoyaltyExecutionPoint GET operation result

Request

```
GET /loyaltyExecutionPoint/501
Content-Type: application/json
```

Response

```
200
{
    "id": "7756",
    "href": "https://mycsp.com:8080/tmf-
api/loyaltyManagement/v5/loyaltyExecutionPoint/7756",
    "@type": "LoyaltyExecutionPoint",
    "name": "John Doe's Birthday Gift Loyalty Execution Point",
    "triggers": "CustomerOrder",
    "loyaltyProgramMember": {
        "id": "9664",
        "href": "https://mycsp.com:8080/tmf-
api/partyRoleManagement/v5/partyRole/9664",
        "@type": "LoyaltyProgramMemberRef",
```

© TM Forum 2025. All Rights Reserved.

```
        "name": "John Doe",
        "@referredType": "LoyaltyProgramMember"
    },
    "loyaltyProgramProduct": {
        "id": "1745",
        "href": "https://mycsp.com:8080/tmf-
api/loyalty/v5/loyaltyProgramProduct/1745",
        "@type": "LoyaltyProgramProductRef",
        "name": "John Doe's Loyalty Program",
        "@referredType": "LoyaltyProgramProduct"
    },
    "body": {
        "category": "Birthday Gift product order",
        "priority": "1",
        "channel": [
            {
                "role": "Used channel for order capture",
                "@type": "RelatedChannel",
                "channel": {
                    "id": "1",
                    "name": "Online channel",
                    "@type": "ChannelRef"
                }
            }
        ],
        "productOrderItem": [
            {
                "id": "100",
                "quantity": 1,
                "action": "add",
                "productOffering": {
                    "id": "14277",
                    "href":
"https://host:port/productCatalogManagement/v5/productOffering/14277",
                    "name": "TMF25",
                    "@type": "ProductOfferingRef"
                },
                "@type": "ProductOrderItem"
            }
        ],
        "relatedParty": [
            {
                "role": "Customer",
                "partyOrPartyRole": {
                    "id": "{{customer.id}}",
                    "href":
"https://host:port/customerManagement/v5/customer/{{customer.id}}",
                    "name": "{{customer.name}}",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                },
                "@type": "RelatedPartyRefOrPartyRoleRef"
            }
        ],
        "@type": "ProductOrder"
    },
    "version": "1.0",
    "action": "POST",
    "endpoint": "https://mycsp.com:8080/tmf-api/productOrdering/v5/productOrder",
    "creationDate": "2024-05-01T20:36:13.313Z"
}
```


© TM Forum 2025. All Rights Reserved.

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

© TM Forum 2025. All Rights Reserved.

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

© TM Forum 2025. All Rights Reserved.

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

© TM Forum 2025. All Rights Reserved.

Acknowledgements

Release History

Release	Date	Release led by	Description
3.0.0	20/11/2017	Clara van Staden Globetom clara.vstaden@globetom.com	Initial release
4.0.0	07/11/2022	Philip Stander Globetom philip.stander@globetom.com Frederic Schmeler Salesforce	Updated to V4 design guidelines
5.0.0	27/02/2025	Philip Stander Globetom philip.stander@globetom.com Rochana Macharoen AIS rochanam@ais.co.th	Gen 5 API release

Contributors to Document

Contributor	Organization
Pierre Gauthier	TM Forum
Philip Stander	Globetom
Frederic Schmeler	SalesForce
Avani Shah	SalesForce
Jonathan Goldberg	TM Forum
Dan d'Albuquerque	Entronica

© TM Forum 2025. All Rights Reserved.